DPP 2026-27 — 7th Grade Math Weekly Checks

Read this page first. Everything for the year is in this one file. Nothing here needs to be written before it is copied; the only blanks are the ones students or teachers fill in by hand during the week.

What is in this file

Part	What it is	Who writes on it	Where
Index	Every benchmark in unit order with its check ID and page number	Nobody	Next page
Scale library	All 34 learning goals and 4-level scales, for the board	Nobody	After the index
Weekly check (one per benchmark)	One sheet per student, front and back: learning goal, scale, 5 items, score box on the back. Always starts on an odd page, so copy two pages two-sided.	Students, on Thursday	See index
Answer key	Answers and worked solutions for the check, re-check and comparison items	Nobody. Read it aloud for peer scoring	Page after each check
Reteach plan	One-page planner for Tuesday's reteach; fill in Monday	Croix (copy for John)	Page after each key
Re-check half (-R)	3 items on the benchmark that was retaught; two copies per page (one per page when the items are long)	Students, on Wednesday	After each reteach plan
Comparison half (-X)	2 items on a benchmark that was taught but not retaught; two copies per page	Students, on Wednesday, with the -R half	After each -R page
Backstop checks (Q2, Q3, Q4)	Mixed review check for a week when no new check is ready	Students	Back of the file, before the forms
Tally sheet	One per teacher per week: 25 scores, trouble item, reteach decision	Croix and John	Last 3 pages
Progress strip	One per student per quarter: their scores and levels, kept in the notebook	Students	Last 3 pages
Check calendar	Week-by-week list to mark which weeks have a check	Croix, once	Last page

One week, start to finish

1. **Before Thursday (Croix, ~5 min):** look up this week's benchmark in the index. Copy its two check pages two-sided, one per student, for both sections. Copy the key once. Put the key in the binder.
2. **Thursday, last 8 minutes (both):** students take the check. They swap papers, you read the key aloud, they score it, circle their level, and copy score and level onto their progress strip. Collect the checks.
3. **Thursday or Friday (both, ~5 min):** copy the 25 scores onto a tally sheet. Write the item that gave the most trouble. Put the sheet in the binder.
4. **Monday (Croix, ~5 min):** look at both tally sheets, decide what to reteach, write it on the decision line of both sheets. Photograph them and type the numbers into the tracking sheet.
5. **Monday (Croix, ~5 min):** copy the -R half for the retaught benchmark and the -X half for the most recent benchmark that was not retaught. One of each per student, both sections.
6. **Tuesday bellwork (both):** reteach the trouble item type, 8-10 minutes.
7. **Wednesday bellwork (both):** students take the -R and -X halves, peer-score, write the two numbers on their strip. Copy the numbers onto the tally sheet.

Weeks with a unit test, a break, or only three days: no check. Mark them on the calendar now.

Index

Unit order follows Math Nation Grade 7. The check ID is the benchmark, so it does not matter which week a unit lands.

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Scale library — learning goals and performance scales

Student language, one per benchmark. Level 3 is the benchmark; level 2 is the foundational skill; level 4 is application. Printed on every check; also usable on the board.

Benchmark	Learning goal	Level 2	Level 3	Level 4
MA.7.DP.2.1 Unit 2	I can find the sample space of an experiment.	I can list some of the outcomes of a simple experiment.	I can list or count every outcome in the sample space, with no repeats or misses.	I can explain why a repeated color or letter counts as one outcome, and decide whether two experiments have the same sample space.
MA.7.NSO.1.2 Unit 2	I can rewrite rational numbers in equivalent forms.	I can convert between a fraction and a terminating decimal.	I can convert between fractions, decimals, and percents, including repeating decimals.	I can choose the best form of a number to solve a problem and explain why it is equivalent.
MA.7.DP.2.2 Unit 2	I can find and interpret the probability of a simple event.	I can tell whether an event is likely, unlikely, certain, or impossible.	I can write the probability of a simple event as a fraction, decimal, or percent and say what it means.	I can compare probabilities of events from different experiments and explain which is more likely.
MA.7.DP.2.3 Unit 2	I can find the theoretical probability of an event.	I can count the favorable outcomes and the total outcomes.	I can find the theoretical probability of a simple event and write it as a fraction, decimal, or percent.	I can find the probability of an event described by a condition (such as 'an even number' or 'not red') and explain my count.
MA.7.DP.2.4 Unit 2	I can use a simulation to find experimental probability.	I can record the results of trials in a table.	I can use results from a simulation to find an experimental probability and compare it to theoretical.	I can design a simulation for a situation and predict results for a larger number of trials.
MA.7.NSO.2.2 Unit 3	I can add, subtract, multiply, and divide rational numbers.	I can add and subtract integers and multiply and divide with the correct sign.	I can add, subtract, multiply, and divide fractions, decimals, and negatives accurately.	I can solve a multi-step computation with rational numbers and check that the sign and size make sense.
MA.7.NSO.2.3 Unit 3	I can solve real-world problems with rational numbers.	I can pick the operation a word problem needs.	I can solve a real-world problem involving rational numbers and state the answer with units.	I can solve a multi-step real-world problem with rational numbers and explain whether the answer is reasonable.
MA.7.GR.1.1 Unit 4	I can find the area of trapezoids, parallelograms, and rhombi.	I can find the area of a rectangle and a triangle.	I can use the formulas to find the area of a trapezoid, parallelogram, or rhombus.	I can find a missing dimension when the area is given, or explain where a formula comes from.
MA.7.GR.1.2 Unit 4	I can find the area of polygons and composite figures.	I can split a figure into shapes I know.	I can find the area of a polygon or composite figure by adding or subtracting areas.	I can solve a real-world problem with a composite figure, including one with a region removed.
MA.7.GR.1.3 Unit 4	I can use the relationship between circumference and diameter.	I can identify the radius, diameter, and circumference of a circle.	I can find circumference from radius or diameter, and radius or diameter from circumference.	I can explain pi as the ratio of circumference to diameter and use it in a real-world problem.
MA.7.GR.1.4 Unit 4	I can find the area of a circle.	I can find the radius from the diameter and square a number.	I can find the area of a circle from its radius or diameter.	I can solve a real-world problem using circumference and area together, including semicircles.
MA.7.DP.1.3 Unit 5	I can use a random sample to make a prediction about a population.	I can tell whether a sample is random and representative.	I can use a proportion from a sample to predict a value for the whole population.	I can judge whether a prediction is reasonable and explain how sample size or bias affects it.
MA.7.GR.1.5 Unit 5	I can solve problems with scale drawings and scale factors.	I can find a scale factor between two figures.	I can find actual lengths, perimeters, and areas from a scale drawing or scale factor.	I can explain how the scale factor changes perimeter versus area and use that in a real-world problem.
MA.7.AR.3.3 Unit 6	I can convert units using proportional reasoning.	I can set up a ratio between two units.	I can convert between measurement systems for length, area, weight or mass, volume, or money, using a conversion factor.	I can solve a multi-step conversion problem across measurement systems and check the result.
MA.7.AR.3.2 Unit 6	I can set up and solve proportions for real-world problems.	I can tell whether two ratios are equivalent.	I can write a proportion from a situation and solve it.	I can solve a multi-step problem with proportions and explain why the setup makes sense.
MA.7.AR.4.5 Unit 6	I can solve multi-step problems with proportional relationships.	I can find a unit rate.	I can solve a real-world problem with a proportional relationship in two or more steps.	I can compare two proportional situations and justify which is the better choice.
MA.7.DP.1.4 Unit 7	I can create and interpret circle graphs.	I can read a percent or category from a circle graph.	I can find the sector angle or percent for each category and interpret a circle graph.	I can construct a circle graph from data and use it to answer a comparison question.
MA.7.AR.3.1 Unit 7	I can solve percent problems: discount, markup, tax, tip, interest, percent change, percent error.	I can find a percent of a number.	I can solve a one-step percent problem such as a discount, tax, tip, or simple interest.	I can solve a multi-step percent problem, including percent change or percent error, and explain the result.
MA.7.NSO.1.1 Unit 8	I can apply the laws of exponents with whole-number exponents.	I can evaluate a power with a whole-number exponent.	I can use the product, quotient, and power rules to write an equivalent expression.	I can combine several exponent rules in one expression and explain each step.
MA.7.NSO.2.1 Unit 8	I can use order of operations with rational numbers.	I can use order of operations with whole numbers.	I can evaluate an expression with rational numbers, exponents, and grouping symbols.	I can find and explain an error in someone else's work with order of operations.
MA.7.AR.1.1 Unit 9	I can add and subtract linear expressions.	I can combine like terms with integer coefficients.	I can add and subtract linear expressions with rational coefficients.	I can write and simplify an expression from a real-world situation.
MA.7.AR.1.2 Unit 9	I can tell whether two expressions are equivalent.	I can use the distributive property on a simple expression.	I can decide whether two linear expressions are equivalent and rewrite one as the other.	I can justify equivalence using properties of operations, including factoring.
MA.7.AR.2.1 Unit 10	I can write and solve one-step inequalities and graph the solutions.	I can graph an inequality on a number line.	I can solve a one-step inequality, including with negative numbers, and graph the solution.	I can write an inequality from a real-world situation, solve it, and interpret the solution.
MA.7.AR.2.2 Unit 10	I can write and solve two-step equations.	I can solve a one-step equation.	I can solve a two-step equation, including with rational numbers or the distributive property.	I can write a two-step equation from a real-world context, solve it, and check the answer.
MA.7.AR.4.3 Unit 11	I can graph proportional relationships.	I can plot points from a table.	I can graph a proportional relationship and identify the points (0, 0) and (1, r).	I can use a graph to find the constant of proportionality and interpret it in context.
MA.7.AR.4.2 Unit 11	I can find the constant of proportionality.	I can find a unit rate from a ratio.	I can find the constant of proportionality from a table, graph, or written description.	I can explain what the constant of proportionality means in the situation.
MA.7.AR.4.1 Unit 11	I can decide whether a relationship is proportional.	I can check whether two ratios are equal.	I can decide whether a table, graph, or written description shows a proportional relationship and say why.	I can explain why a relationship is not proportional using more than one representation.
MA.7.AR.4.4 Unit 11	I can write an equation for a proportional relationship.	I can find the constant of proportionality.	I can write the equation y = kx from a table, graph, or description and use it.	I can translate between a table, a graph, an equation, and a description of the same relationship.
MA.7.GR.2.1 Unit 12	I can find the surface area of a right circular cylinder.	I can identify the parts of a cylinder's net.	I can find the surface area of a cylinder from its radius and height.	I can explain how the net gives the formula and find surface area with a missing dimension.
MA.7.GR.2.2 Unit 12	I can solve problems involving surface area of cylinders.	I can find the area of a circle and a rectangle.	I can solve a real-world problem with the surface area of a cylinder.	I can solve a multi-step surface-area problem, such as a label or an open-top can.
MA.7.GR.2.3 Unit 12	I can find the volume of a right circular cylinder.	I can find the area of the base.	I can find the volume of a cylinder from its radius or diameter and height.	I can solve a real-world volume problem, including comparing two cylinders.
MA.7.DP.1.1 Unit 13	I can choose and justify measures of center and spread.	I can find the mean, median, and range of a data set.	I can choose the best measure of center or spread for a data set and explain why.	I can explain how an outlier changes the mean, median, and range.
MA.7.DP.1.2 Unit 13	I can compare two data sets using their displays.	I can read values from a box plot or histogram.	I can compare two data sets using measures of center and spread from their displays.	I can make and justify a claim about two populations using their data displays.
MA.7.DP.1.5 Unit 13	I can choose and create an appropriate data display.	I can make a bar graph or dot plot from data.	I can choose the right display for numerical or categorical data and create it accurately.	I can explain why one display is better than another for a given question.

Thursday routine — peer scoring in 3 minutes

Same words every week so it runs itself by November.

1. **"Pencils down. Pass your paper to your scoring partner."** Partners are fixed (seat neighbor). Scorer writes their own code small in the top corner.
2. **Read the key aloud, item by item.** "Number 1, B." For a short-answer item read the answer and the one thing that must be present ("18π centimeters; the π has to be there"). Scorer marks a check or an X next to each item. No partial credit: a select-all item is right only if every box is right.
3. **"Write the score out of 5 in the box. Pass it back."**
4. **Owner circles their level on the scale** and copies score and level onto their progress strip. (Level is theirs to judge; it is not a grade.)
5. **Collect.** Checks go in the period folder. Copy the 25 numbers onto the tally sheet when you have five minutes; write the item that gave the most trouble.

Rules that keep the data clean

- Spot-check five papers a week against the key. If a class can't be trusted to score, score that section yourself for a month.
- Absent = A on the tally sheet, never 0.
- The score is never a grade. Classwork points are for doing the check and filling in the strip.
- Wednesday's re-check runs the same way, with the -R and -X halves scored as two numbers (/3 and /2).

Reteach plan page

After each answer key there is a one-page reteach planner. Fill it in on Monday when the decision is made; it is the plan for both rooms on Tuesday, and three of them per quarter go in the DPP folder as evidence of the reteach step.

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____ Calculator: yes

Learning goal: I can find the sample space of an experiment. (circle your level after scoring)

4	I can explain why a repeated color or letter counts as one outcome, and decide whether two experiments have the same sample space.
3	I can list or count every outcome in the sample space, with no repeats or misses.
2	I can list some of the outcomes of a simple experiment.
1	I need help to start this benchmark.

1. Each letter of the word DIFFERENCE is written on its own card, one card for each letter, and the cards are placed in a jar. One card is drawn at random and its letter is recorded.

How many outcomes are in the sample space for this experiment?

A. 8

B. 7

C. 10

D. 6

2. A set of cards shows these colored shapes, one on each card: red circle, yellow star, yellow triangle, yellow triangle, yellow star, yellow triangle. One card is drawn at random, and the colored shape on it is recorded.

List the sample space for this experiment.

Turn over →

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____

3. Experiment 1: A spinner has 5 equal sections colored red, green, orange, purple and red. It is spun once and the color it lands on is recorded.
Experiment 2: A bag holds 1 orange, 3 green and 2 red marbles. One marble is drawn at random and its color is recorded.

Which statement about the sample spaces of the two experiments is true?

A. The sample space for the spinner has one outcome that the sample space for the bag does not have.

B. The sample space for the spinner has 5 outcomes, and the sample space for the bag has 6 outcomes.

C. The two experiments have the same sample space, {orange, green, red, purple}.

D. The sample space for the bag has one outcome that the sample space for the spinner does not have.

4. A fair 8-sided die has these numbers on its faces: 5, 5, 1, 8, 9, 8, 9, 5. The die is rolled once and the number on the top face is recorded.

List the sample space for this experiment.

5. In each experiment below, the color is recorded.

Select **all** the experiments whose sample space is {purple, green, blue}.

Select **all** the correct answers.

A. ☐ Rolling a fair 6-sided die whose faces are blue, yellow, green, purple, green, purple

B. ☐ Spinning a spinner whose 6 equal sections are green, blue, purple, green, purple, blue

C. ☐ Drawing a marble from a bag that holds 3 purple, 5 green and 4 blue marbles

D. ☐ Spinning a spinner whose 4 equal sections are green, purple, green, purple

E. ☐ Drawing a marble from a bag that holds 3 purple, 5 green, 4 blue and 4 yellow marbles

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.2.1 · MA.7.DP.2.1 · Sample space of a simple experiment

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **B. 7**

DIFFERENCE has 10 letters, so there are 10 cards, but F is on 2 cards and E is on 3 cards.
The outcome is the letter that is recorded, and a letter that is on more than one card is still one outcome.
The sample space is $\{D, I, F, E, R, N, C\}$, which has 7 outcomes, not 10.

2. Regular **{red circle, yellow star, yellow triangle}**

There are 6 cards, but yellow star is on 2 cards and yellow triangle is on 3 cards.
The outcome is the colored shape, and cards that show the same colored shape give the same outcome, so each different colored shape is listed once.
The sample space is {red circle, yellow star, yellow triangle}.

3. Regular **A. The sample space for the spinner has one outcome that the sample space for the bag does not have.**

Each outcome is a color, and a color is listed once no matter how many sections or marbles have it.
Spinner: {orange, green, red, purple}. Bag: {orange, green, red}.
The spinner can land on purple, and the bag has no purple marble, so the sample space for the spinner has one more outcome.

4. Regular $\{1, 5, 8, 9\}$

The die has 8 faces, but some faces show the same number.
Each different number is one outcome, so a number that is on more than one face is listed only once.
The sample space is $\{1, 5, 8, 9\}$.

5. Rigor **B. Spinning a spinner whose 6 equal sections are green, blue, purple, green, purple, blue; C. Drawing a marble from a bag that holds 3 purple, 5 green and 4 blue marbles**

List the colors each experiment can give, each color once:
Drawing a marble from a bag that holds 3 purple, 5 green and 4 blue marbles: {purple, green, blue}.
Spinning a spinner whose 6 equal sections are green, blue, purple, green, purple, blue: {green, blue, purple}.
Rolling a fair 6-sided die whose faces are blue, yellow, green, purple, green, purple: {blue, yellow, green, purple}.
Drawing a marble from a bag that holds 3 purple, 5 green, 4 blue and 4 yellow marbles: {purple, green, blue, yellow}.
Spinning a spinner whose 4 equal sections are green, purple, green, purple: {green, purple}.
Only the experiments whose list is exactly {purple, green, blue} have this sample space; a color on several sections, faces or marbles is still one outcome.
Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Re-check strip (3 items, C-DP.2.1-R)

1. Regular **B. 4**

DEGREE has 6 letters, so there are 6 cards, but E is on 3 cards.
The outcome is the letter that is recorded, and a letter that is on more than one card is still one outcome.
The sample space is $\{D, E, G, R\}$, which has 4 outcomes, not 6.

2. Regular **A. {blue, red, purple, green}**

The outcome that is recorded is a color, so the sample space lists each color once: {blue, red, purple, green}.
The numbers of sections tell how many sections have each color, and the numbers 1 to 6 would name sections, not colors.

3. Regular $\{1, 2, 8\}$

There are 7 tiles, but 1 is on 2 tiles, 2 is on 2 tiles and 8 is on 3 tiles.
The outcome is the number, and a number that is on more than one tile is still one outcome, so each different number is listed once.
The sample space is $\{1, 2, 8\}$.

Comparison strip (2 items, C-DP.2.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B. 6**

ADDITION has 8 letters, so there are 8 cards, but D is on 2 cards and I is on 2 cards.
The outcome is the letter that is recorded, and a letter that is on more than one card is still one outcome.
The sample space is $\{A, D, I, T, O, N\}$, which has 6 outcomes, not 8.

2. Regular $\{1, 6, 9\}$

The die has 6 faces, but some faces show the same number.
Each different number is one outcome, so a number that is on more than one face is listed only once.
The sample space is $\{1, 6, 9\}$.

Reteach plan — C-DP.2.1

MA.7.DP.2.1 · Sample space of a simple experiment

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.2.1-R

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____ Calculator: yes

1. Each letter of the word DEGREE is written on its own card, one card for each letter, and the cards are placed in a hat. One card is drawn at random and its letter is recorded.

How many outcomes are in the sample space for this experiment?

A. 6

B. 4

C. 5

D. 3

2. A spinner has 6 equal sections: 1 is blue, 1 is red, 1 is purple and 3 are green. The spinner is spun once and the color it lands on is recorded.

The sample space for this experiment is ____.

Choose the option that correctly completes the statement.

A. {blue, red, purple, green}

B. {blue, red, purple}

C. $\{1, 1, 1, 3\}$

D. $\{1, 2, 3, 4, 5, 6\}$

3. The tiles in a bag are numbered 8, 1, 1, 8, 2, 8, 2. One tile is drawn at random and its number is recorded.

List the sample space for this experiment.

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.2.1-X

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____ Calculator: yes

1. Each letter of the word ADDITION is written on its own card, one card for each letter, and the cards are placed in a jar. One card is drawn at random and its letter is recorded.

How many outcomes are in the sample space for this experiment?

A. 8 **B.** 6

C. 5 **D.** 7

2. A fair 6-sided die has these numbers on its faces: 6, 9, 9, 9, 9, 1. The die is rolled once and the number on the top face is recorded.

List the sample space for this experiment.

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.2.1-X

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____ Calculator: yes

1. Each letter of the word ADDITION is written on its own card, one card for each letter, and the cards are placed in a jar. One card is drawn at random and its letter is recorded.

How many outcomes are in the sample space for this experiment?

A. 8 **B.** 6

C. 5 **D.** 7

2. A fair 6-sided die has these numbers on its faces: 6, 9, 9, 9, 9, 1. The die is rolled once and the number on the top face is recorded.

List the sample space for this experiment.

Score: ____ **/ 2** → **copy to your progress strip**

Benchmark: **MA.7.NSO.1.2** · Rewrite rational numbers in equivalent forms Student code: _____ Calculator: no

Learning goal: I can rewrite rational numbers in equivalent forms. (circle your level after scoring)

4	I can choose the best form of a number to solve a problem and explain why it is equivalent.
3	I can convert between fractions, decimals, and percents, including repeating decimals.
2	I can convert between a fraction and a terminating decimal.
1	I need help to start this benchmark.

1. Write the repeating decimal as a fraction in simplest form.

$0.\overline{46}$

2. Write the fraction as a repeating decimal.

$\frac{13}{33}$

Turn over →

3. Two identical bottles of sports drink are partly full. The first bottle is $\frac{10}{11}$ full and the second bottle is 95% full.

The first bottle holds ____ the second bottle.

Choose the phrase that correctly completes the sentence.

A. exactly twice as much as

B. the same amount as

C. more than

D. less than

4. Write the fraction as an exact percent.

$\frac{7}{11}$

5. Maya uses a calculator to find $\frac{190}{33}$. The calculator shows 5.757575758. Maya says that $\frac{190}{33} = 5.757575758$.

Is Maya correct? Explain, and write the exact value of $\frac{190}{33}$ as a repeating decimal.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.1.2 · MA.7.NSO.1.2 · Rewrite rational numbers in equivalent forms

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\frac{46}{99}$

Two repeating digits go over 99: $0.\overline{46} = \frac{46}{99}$.

2. Regular $0.\overline{39}$

$13 \div 33 = 0.393939\ldots$, so $\frac{13}{33} = 0.\overline{39}$: the digits 39 repeat.

3. Regular **D. less than**

Rewrite both amounts in the same form. $\frac{10}{11} = 10 \div 11$, which is about 90.91%.

90.91% is less than 95%, so the first bottle holds less than the second bottle.

4. Regular $63\frac{7}{11} = \frac{700}{11}\%(\text{or } 63.\overline{63}\%)$

A percent is the value out of 100: $7 \div 11 \times 100 = 63\frac{7}{11}\%$, which is $63.\overline{63}\%$ as a repeating decimal.

5. Rigor **No. The calculator rounded a repeating decimal, so 5.757575758 is not exact; $\frac{190}{33} = 5\frac{25}{33} = 5.\overline{75}$. Score: 1 point for "No" with the rounding reason and the exact value $5.\overline{75}$ (or an equivalent repeating form); "No" alone earns 0.**

Divide: $190 \div 33 = 5.757575\ldots$, and the digits 75 repeat forever.

A calculator shows only so many digits and rounds the last one, so 5.757575758 is close to $\frac{190}{33}$ but not equal to it.

The exact value is $5.\overline{75}$.

Re-check strip (3 items, C-NSO.1.2-R)

1. Regular **B. more than**

Rewrite both amounts in the same form. $\frac{4}{11} = 4 \div 11$, which is about 36.36%.

36.36% is more than 30%, so the first bottle holds more than the second bottle.

2. Regular **A.** $0.\overline{08}$**; C.** $\frac{16}{198}$

$8 \div 99 = \frac{8}{99} = 0.\overline{08}$ (the block 08 repeats forever).

$\frac{8}{99} = \frac{16}{198}$, so $\frac{16}{198}$ also names the quotient.

0.8 repeats only the digit 8, not 08.

$\frac{99}{8}$ is $99 \div 8$, not $8 \div 99$.

0.0808 stops after 4 decimal places; it is not the repeating decimal.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

3. Regular $77\frac{7}{9} = \frac{700}{9}\%(\text{or } 77.\overline{7}\%)$

A percent is the value out of 100: $7 \div 9 \times 100 = 77\frac{7}{9}\%$, which is $77.\overline{7}\%$ as a repeating decimal.

Comparison strip (2 items, C-NSO.1.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **A. less than**

Rewrite both amounts in the same form. $\frac{2}{15} = 2 \div 15$, which is about 13.33%.

13.33% is less than 15%, so the first bottle holds less than the second bottle.

2. Regular **B.** $\frac{3}{99}$**; C.** $0.\overline{03}$

$1 \div 33 = \frac{1}{33} = 0.\overline{03}$ (the block 03 repeats forever).

$\frac{1}{33} = \frac{3}{99}$, so $\frac{3}{99}$ also names the quotient.

0.0303 stops after 4 decimal places; it is not the repeating decimal.

$\frac{33}{1}$ is $33 \div 1$, not $1 \div 33$.

$0.\overline{3}$ repeats only the digit 3, not 03.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Reteach plan — C-NSO.1.2

MA.7.NSO.1.2 · Rewrite rational numbers in equivalent forms

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.1.2-R

Benchmark: **MA.7.NSO.1.2** · Rewrite rational numbers in equivalent forms Student code: _____ Calculator: no

1. Two identical bottles of sports drink are partly full. The first bottle is $\frac{4}{11}$ full and the second bottle is 30% full.

The first bottle holds ____ the second bottle.

Choose the phrase that correctly completes the sentence.

A. the same amount as

B. more than

C. exactly twice as much as

D. less than

2. Select **all** the values that are equal to the quotient $8 \div 99$.

Select **all** the correct answers.

A. ☐ $0.\overline{08}$

B. ☐ $0.\overline{8}$

C. ☐ $\frac{16}{198}$

D. ☐ 0.0808

E. ☐ $\frac{99}{8}$

3. Write the fraction as an exact percent.

$\frac{7}{9}$

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-NSO.1.2-X

Benchmark: **MA.7.NSO.1.2** · Rewrite rational numbers in equivalent forms Student code: _____ Calculator: no

1. Two identical bottles of sports drink are partly full. The first bottle is $\frac{2}{15}$ full and the second bottle is 15% full.

The first bottle holds ____ the second bottle.

Choose the phrase that correctly completes the sentence.

A. less than

B. more than

C. the same amount as

D. exactly twice as much as

2. Select **all** the values that are equivalent to $1 \div 33$.

Select **all** the correct answers.

A. ☐ $\frac{33}{1}$

B. ☐ $\frac{3}{99}$

C. ☐ $0.\overline{03}$

D. ☐ 0.0303

E. ☐ $0.\overline{3}$

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.1.2-X

Benchmark: **MA.7.NSO.1.2** · Rewrite rational numbers in equivalent forms Student code: _____ Calculator: no

1. Two identical bottles of sports drink are partly full. The first bottle is $\frac{2}{15}$ full and the second bottle is 15% full.

The first bottle holds ____ the second bottle.

Choose the phrase that correctly completes the sentence.

A. less than

B. more than

C. the same amount as

D. exactly twice as much as

2. Select **all** the values that are equivalent to $1 \div 33$.

Select **all** the correct answers.

A. ☐ $\frac{33}{1}$

B. ☐ $\frac{3}{99}$

C. ☐ $0.\overline{03}$

D. ☐ 0.0303

E. ☐ $0.\overline{3}$

Score: ____ **/ 2** → **copy to your progress strip**

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____ Calculator: yes

Learning goal: I can find and interpret the probability of a simple event. (circle your level after scoring)

4	I can compare probabilities of events from different experiments and explain which is more likely.
3	I can write the probability of a simple event as a fraction, decimal, or percent and say what it means.
2	I can tell whether an event is likely, unlikely, certain, or impossible.
1	I need help to start this benchmark.

1. The probability of rain tomorrow is reported as 75%.

Is this event likely or unlikely? Explain using the probability.

A. Likely, because 75% is greater than 50% and closer to 1 than to 0.

B. Unlikely, because 75% is less than 100%, so the event might not happen.

C. Certain, because 75% is so close to 100% that it must happen.

D. As likely as not, because 75% is a percentage and not a fraction.

2. Which value **cannot** be the probability of an event?

A. 0

B. 110%

C. $\frac{7}{8}$

D. 1

3. The probability that Mia makes her next free throw is $\frac{5}{8}$. The probability that Jon makes his next free throw is $\frac{1}{4}$.

Which event is more likely?

A. There is not enough information.

B. Mia makes her next free throw

C. Jon makes his next free throw

D. They are equally likely.

Turn over →

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____

4. A probability is given for each event.

Describe each event as likely, unlikely or neither.

Check one box in each row.

	Likely	Unlikely	Neither likely nor unlikely
The probability that it snows in Orlando, Florida, this winter is 2%.	☐	☐	☐
The probability that a spinner with two equal sections, one red and one blue, lands on red is $\frac{1}{2}$.	☐	☐	☐
The probability that a hurricane passes within 100 miles of the city in the next two days is 0.6.	☐	☐	☐

5. A fair 10-sided die has faces numbered 1 through 10. The probability that it rains tomorrow is 0.2.

Which is more likely: that the die shows a number greater than 3, or that it rains tomorrow? Write both probabilities in the same form to justify your answer, and find the difference between the two probabilities.

Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — C-DP.2.2 · MA.7.DP.2.2 · Interpret and compare probabilities

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. Likely, because 75% is greater than 50% and closer to 1 than to 0.**

$75\% = 0.75$, which is greater than 0.5, so it is closer to 1 than to 0.

An event with a probability less than 0.5 is unlikely and an event with a probability greater than 0.5 is likely, so this event is likely.

2. Regular **B. 110%**

Every probability is between 0 and 1, inclusive (or between 0% and 100%).

110% cannot be a probability because a probability written as a percent can never be greater than 100%.

3. Regular **B. Mia makes her next free throw**

$\frac{5}{8} \approx 0.63$ and $\frac{1}{4} = 0.25$.

The greater probability is $\frac{5}{8}$, so the event "Mia makes her next free throw" is more likely.

4. Regular **The probability that it snows in Orlando, Florida, this winter is 2%. → Unlikely; The probability that a spinner with two equal sections, one red and one blue, lands on red is $\frac{1}{2}$. → Neither likely nor unlikely; The probability that a hurricane passes within 100 miles of the city in the next two days is 0.6. → Likely**

Write each probability as a decimal and compare it with 0.5: an event with a probability greater than 0.5 is likely, less than 0.5 is unlikely, and exactly 0.5 is neither.

$2\% = 0.02$: less than 0.5, so unlikely.

$\frac{1}{2} = 0.5$: exactly 0.5, so neither.

0.6: greater than 0.5, so likely.

Scoring: 1 point only if every match is correct; anything else earns 0.

5. Rigor **The die shows a number greater than 3; $\frac{7}{10} = 0.7$ and 0.2, so $0.7 > 0.2$, by 0.5 ($\frac{1}{2}$ or 50% also accepted). Score: 1 point — the more likely event with both probabilities in one form and the difference; the event alone earns 0.**

$P(\text{the die shows a number greater than } 3) = \frac{7}{10} = 0.7$ (7 favorable of 10 equally likely outcomes).

$P(\text{it rains tomorrow}) = 0.2$.

$0.7 - 0.2 = 0.5$, so a number greater than 3 is more likely, by 0.5.

Re-check strip (3 items, C-DP.2.2-R)

1. Regular **A. 140%**

Every probability is between 0 and 1, inclusive (or between 0% and 100%).

140% cannot be a probability because a probability written as a percent can never be greater than 100%.

2. Regular **B. The seed sprouts this week**

Write both probabilities in the same form. As decimals: 0.875 and $\frac{7}{10} = 0.7$.

$0.875 > 0.7$, so "the seed sprouts this week" is more likely. A probability can be compared whatever form it is written in.

3. Regular **The probability that a hurricane passes within 100 miles of the city in the next two days is 0.65. → Likely; The probability that a spinner with two equal sections, one red and one blue, lands on red is 50%. → Neither likely nor unlikely; The probability that a thunderstorm is within 5 miles of your house tomorrow is $\frac{3}{10}$. → Unlikely**

Write each probability as a decimal and compare it with 0.5: an event with a probability greater than 0.5 is likely, less than 0.5 is unlikely, and exactly 0.5 is neither.

0.65: greater than 0.5, so likely.

$50\% = 0.5$: exactly 0.5, so neither.

$\frac{3}{10} = 0.3$: less than 0.5, so unlikely.

Scoring: 1 point only if every match is correct; anything else earns 0.

Comparison strip (2 items, C-DP.2.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **D. $\frac{6}{5} = 1\frac{1}{5}$**

Every probability is between 0 and 1, inclusive (or between 0% and 100%).

$\frac{6}{5}$ cannot be a probability because a fraction whose numerator is greater than its denominator is greater than 1.

2. Regular **B. Likely, because 85% is greater than 50% and closer to 1 than to 0.**

$85\% = 0.85$, which is greater than 0.5, so it is closer to 1 than to 0.

An event with a probability less than 0.5 is unlikely and an event with a probability greater than 0.5 is likely, so this event is likely.

Reteach plan — C-DP.2.2

MA.7.DP.2.2 · Interpret and compare probabilities

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.2.2-R

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____ Calculator: yes

1. Which value **cannot** be the probability of an event?

A. 140%

B. 1

C. $\frac{5}{8}$

D. 10%

2. The probability that the seed sprouts this week is 0.875. The probability that the Tigers win their next game is $\frac{7}{10}$.

Which event is more likely?

A. The forms cannot be compared.

B. The seed sprouts this week

C. The Tigers win their next game

D. They are equally likely.

3. A probability is given for each event.

Describe each event as likely, unlikely or neither.

Check one box in each row.

	Likely	Unlikely	Neither likely nor unlikely
The probability that a hurricane passes within 100 miles of the city in the next two days is 0.65.	☐	☐	☐
The probability that a spinner with two equal sections, one red and one blue, lands on red is 50%.	☐	☐	☐
The probability that a thunderstorm is within 5 miles of your house tomorrow is $\frac{3}{10}$.	☐	☐	☐

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.2.2-X

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____ Calculator: yes

1. Which value **cannot** be the probability of an event?

A. 0

B. $\frac{3}{8}$

C. 0.8

D. $\frac{6}{5}$

2. The probability of a bus arriving on time is reported as 85%.

Is this event likely or unlikely? Explain using the probability.

A. Unlikely, because 85% is less than 100%, so the event might not happen.

B. Likely, because 85% is greater than 50% and closer to 1 than to 0.

C. As likely as not, because 85% is a percentage and not a fraction.

D. Certain, because 85% is so close to 100% that it must happen.

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.2.2-X

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____ Calculator: yes

1. Which value **cannot** be the probability of an event?

A. 0

B. $\frac{3}{8}$

C. 0.8

D. $\frac{6}{5}$

2. The probability of a bus arriving on time is reported as 85%.

Is this event likely or unlikely? Explain using the probability.

A. Unlikely, because 85% is less than 100%, so the event might not happen.

B. Likely, because 85% is greater than 50% and closer to 1 than to 0.

C. As likely as not, because 85% is a percentage and not a fraction.

D. Certain, because 85% is so close to 100% that it must happen.

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

Learning goal: I can find the theoretical probability of an event. (circle your level after scoring)

4	I can find the probability of an event described by a condition (such as 'an even number' or 'not red') and explain my count.
3	I can find the theoretical probability of a simple event and write it as a fraction, decimal, or percent.
2	I can count the favorable outcomes and the total outcomes.
1	I need help to start this benchmark.

1. A fair 20-sided die has faces numbered 1 through 20. The die is rolled once.

What is the probability of rolling a number greater than 15? Write your answer as a fraction in simplest form.

2. A fair spinner has 25 equal sections. 6 sections are green. The spinner is spun once.

What is the probability that the spinner lands on green? Write your answer as a decimal.

Turn over →

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____

3. A bag holds 12 marbles, and 3 of them are red. One marble is drawn at random.

What is the probability of drawing a red marble?

A. $\frac{3}{11}$

B. $\frac{1}{4}$

C. $\frac{3}{4}$

D. $\frac{1}{6}$

4. Letter cards for the word MEASUREMENT are placed in a bag, one card for each letter. One card is drawn at random.

What is the probability that the card shows the letter M? Write your answer as a fraction in simplest form.

5. A standard deck has 52 cards: 13 cards (ace, 2 through 10, jack, queen and king) in each of four suits — hearts and diamonds, which are red, and clubs and spades, which are black. One card is drawn at random.

What is the probability of drawing a card that is not a black king? Write your answer as a fraction in simplest form.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.2.3 · MA.7.DP.2.3 · Find theoretical probability of an event

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\frac{1}{4}$

The numbers greater than 15 are 16 through 20: 5 favorable outcomes out of 20.

$P = \frac{5}{20} = \frac{1}{4}$.

2. Regular **0.24**

All 25 sections are equally likely, so the sample space has 25 outcomes and 6 of them are green.

$P(\text{green}) = \frac{6}{25}$.

$6 \div 25 = 0.24$.

3. Regular **B.** $\frac{1}{4}$

$P(\text{red}) = \text{ favorable outcomes} \div \text{ total outcomes} = \frac{3}{12} = \frac{1}{4}$.

4. Regular $\frac{2}{11}$

There are 11 cards, one for each letter of MEASUREMENT, and each card is equally likely to be drawn.

The letter M is on 2 of the cards: 2 favorable outcomes out of 11.

$P = \frac{2}{11}$.

5. Rigor $\frac{25}{26}$

Only 2 cards are black kings: the kings that are also black. Every other card is not a black king: $52 - 2 = 50$ favorable cards.

$P = \frac{50}{52} = \frac{25}{26}$.

Re-check strip (3 items, C-DP.2.3-R)

1. Regular **75%**

All 4 sections are equally likely, so the sample space has 4 outcomes and 3 of them are yellow.

$P(\text{yellow}) = \frac{3}{4}$.

$3 \div 4 = 0.75$, and $0.75 = 75\%$.

2. Regular $\frac{3}{8}$

The numbers greater than 5 are 6 through 8: 3 favorable outcomes out of 8.

$P = \frac{3}{8}$.

3. Regular **B.** $\frac{2}{9}$

$P(\text{red}) = \text{ favorable outcomes} \div \text{ total outcomes} = \frac{4}{18} = \frac{2}{9}$.

Comparison strip (2 items, C-DP.2.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B.** $\frac{1}{6}$

$P(\text{red}) = \text{ favorable outcomes} \div \text{ total outcomes} = \frac{2}{12} = \frac{1}{6}$.

2. Regular **0.75**

All 8 sections are equally likely, so the sample space has 8 outcomes and 6 of them are orange.

$P(\text{orange}) = \frac{6}{8} = \frac{3}{4}$.

$6 \div 8 = 0.75$.

Reteach plan — C-DP.2.3

MA.7.DP.2.3 · Find theoretical probability of an event

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.2.3-R

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

1. A fair spinner has 4 equal sections. 3 sections are yellow. The spinner is spun once.

What is the probability that the spinner lands on yellow? Write your answer as a percent.

2. A fair 8-sided die has faces numbered 1 through 8. The die is rolled once.

What is the probability of rolling a number greater than 5? Write your answer as a fraction in simplest form.

3. A bag holds 18 marbles, and 4 of them are red. One marble is drawn at random.

What is the probability of drawing a red marble?

A. $\frac{4}{17}$

B. $\frac{2}{9}$

C. $\frac{7}{9}$

D. $\frac{1}{6}$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-DP.2.3-R

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

1. A fair spinner has 4 equal sections. 3 sections are yellow. The spinner is spun once.

What is the probability that the spinner lands on yellow? Write your answer as a percent.

2. A fair 8-sided die has faces numbered 1 through 8. The die is rolled once.

What is the probability of rolling a number greater than 5? Write your answer as a fraction in simplest form.

3. A bag holds 18 marbles, and 4 of them are red. One marble is drawn at random.

What is the probability of drawing a red marble?

A. $\frac{4}{17}$

B. $\frac{2}{9}$

C. $\frac{7}{9}$

D. $\frac{1}{6}$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.2.3-X

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

1. A bag holds 12 marbles, and 2 of them are red. One marble is drawn at random.

What is the probability of drawing a red marble?

A. $\frac{5}{6}$

B. $\frac{1}{6}$

C. $\frac{2}{11}$

D. $\frac{1}{12}$

2. A fair spinner has 8 equal sections. 6 sections are orange. The spinner is spun once.

What is the probability that the spinner lands on orange? Write your answer as a decimal.

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.2.3-X

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

1. A bag holds 12 marbles, and 2 of them are red. One marble is drawn at random.

What is the probability of drawing a red marble?

A. $\frac{5}{6}$

B. $\frac{1}{6}$

C. $\frac{2}{11}$

D. $\frac{1}{12}$

2. A fair spinner has 8 equal sections. 6 sections are orange. The spinner is spun once.

What is the probability that the spinner lands on orange? Write your answer as a decimal.

Score: ____ **/ 2** → **copy to your progress strip**

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

Learning goal: I can use a simulation to find experimental probability. (circle your level after scoring)

4	I can design a simulation for a situation and predict results for a larger number of trials.
3	I can use results from a simulation to find an experimental probability and compare it to theoretical.
2	I can record the results of trials in a table.
1	I need help to start this benchmark.

1. A fair 12-sided die is rolled 300 times. It lands on a 9 exactly 24 times.

Part A. What is the experimental probability of rolling a 9? Write it as a fraction in simplest form.
Part B. What is the theoretical probability of rolling a 9? Write it as a fraction in simplest form.
Part C. Is the experimental probability greater than or less than the theoretical probability?

2. A fair coin is tossed 400 times and lands on heads 204 times.

What is the experimental probability of landing on heads? Write your answer as a decimal.

3. Two students each simulate rolling a fair die. One runs 5,900 trials and the other runs 370 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. The 5,900-trial student, because more trials usually bring the result closer to the theory.

B. Both students equally, because a fair experiment gives the same result for any number of trials.

C. The 370-trial student, because fewer trials leave fewer chances for a result to go wrong.

D. Neither student, because an experimental result never comes close to the theoretical one.

Turn over →

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____

4. A spinner has 10 equal sections, and 8 of them are red. In a simulation the spinner is spun 400 times and lands on red 311 times.

Which statement correctly compares the experimental and theoretical probabilities of landing on red?

A. The experimental probability, $\frac{311}{400}$, is a little less than the theoretical probability, $\frac{4}{5}$.

B. The theoretical probability is $\frac{311}{400}$, because that is what actually happened.

C. The experimental probability, $\frac{311}{400}$, is a little greater than the theoretical probability, $\frac{4}{5}$.

D. The two probabilities must be equal because the spinner is fair.

5. A bag holds 10 marbles. Each marble is green, purple or blue, but the number of each color is not known. A student draws a marble at random, records its color and puts it back. The table shows the results of 200 draws.

Green	Purple	Blue
121	43	36

Part A. What is the experimental probability of drawing a purple marble? Write it as a decimal.
Part B. Use your answer to Part A. If the student makes 2,000 more draws, about how many times should a purple marble be drawn?
Part C. About how many of the 10 marbles in the bag are purple?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.2.4 · MA.7.DP.2.4 · Experimental vs theoretical probability

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **Experimental: $\frac{2}{25}$; theoretical: $\frac{1}{12}$. The experimental probability is slightly less than the theoretical probability. Score: 1 point for all three parts; the comparison alone earns 0.**

Part A. Experimental probability = times it happened ÷ trials = $\frac{24}{300} = \frac{2}{25} = 0.08$.

Part B. Theoretical probability = $1 \text{ favorable face} \div 12 \text{ faces} = \frac{1}{12} \approx 0.0833$.

Part C. The experimental probability is slightly less than the theoretical probability. They are close but not equal; with more trials the experimental probability usually gets closer to $\frac{1}{12}$.

2. Regular **0.51**

Experimental probability = successes ÷ trials.

$204 \div 400 = 0.51$.

3. Regular **A. The 5,900-trial student, because more trials usually bring the result closer to the theory.**

Experimental probabilities vary from trial to trial, but as the number of trials increases they typically get closer to the theoretical probability.

The 5,900-trial simulation is expected to be closer.

4. Regular **A. The experimental probability, $\frac{311}{400}$, is a little less than the theoretical probability, $\frac{4}{5}$.**

Theoretical: $\frac{8}{10} = \frac{4}{5} = 0.8$.

Experimental: $\frac{311}{400} = 0.7775$.

The experimental probability is a little less than the theoretical probability because of random variation.

5. Rigor **Part A: 0.215. Part B: about 430 times. Part C: about 2 purple marbles. Score: 1 point for all three parts; Part A alone earns 0.**

Part A. Experimental probability = purple draws ÷ all $\text{draws} = \frac{43}{200} = 0.215$.

Part B. The bag has not changed, so about the same part of the new draws should be purple: $0.215 \times 2{,}000 = 430$ times. (The actual count will vary a little.)

Part C. The experimental probability estimates the theoretical probability, which is the number of purple marbles out of 10: $0.215 \times 10 = 2.15$, so about 2 of the marbles are purple ($\frac{2}{10} = 0.2$, close to 0.215).

Re-check strip (3 items, C-DP.2.4-R)

1. Regular **B. The 8,000-trial student, because more trials usually bring the result closer to the theory.**

Experimental probabilities vary from trial to trial, but as the number of trials increases they typically get closer to the theoretical probability.

The 8,000-trial simulation is expected to be closer.

2. Regular **0.47**

Experimental probability = successes ÷ trials.

$376 \div 800 = 0.47$.

3. Regular **about 100 times**

The theoretical probability of drawing purple is 0.25, so about 25% of the draws should be purple.

$0.25 \times 400 = 100$.

About 100 of the 400 draws would be expected to be purple. The actual number will usually differ a little, because of random variation.

Comparison strip (2 items, C-DP.2.4-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **C. The 4,700-trial student, because more trials usually bring the result closer to the theory.**

Experimental probabilities vary from trial to trial, but as the number of trials increases they typically get closer to the theoretical probability.

The 4,700-trial simulation is expected to be closer.

2. Regular **Experimental: $\frac{23}{200}$; theoretical: $\frac{1}{10}$. The experimental probability is slightly greater than the theoretical probability. Score: 1 point for all three parts; the comparison alone earns 0.**

Part A. Experimental probability = times it happened ÷ trials = $\frac{23}{200} = 0.115$.

Part B. Theoretical probability = $1 \text{ favorable face} \div 10 \text{ faces} = \frac{1}{10} = 0.1$.

Part C. The experimental probability is slightly greater than the theoretical probability. They are close but not equal; with more trials the experimental probability usually gets closer to $\frac{1}{10}$.

Reteach plan — C-DP.2.4

MA.7.DP.2.4 · Experimental vs theoretical probability

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.2.4-R

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

1. Two students each simulate spinning a fair spinner. One runs 210 trials and the other runs 8,000 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. The 210-trial student, because fewer trials leave fewer chances for a result to go wrong.

B. The 8,000-trial student, because more trials usually bring the result closer to the theory.

C. Both students equally, because a fair experiment gives the same result for any number of trials.

D. Neither student, because an experimental result never comes close to the theoretical one.

2. A fair coin is tossed 800 times and lands on heads 376 times.

What is the experimental probability of landing on heads? Write your answer as a decimal.

3. A bag contains only purple marbles and blue marbles. When one marble is drawn at random, the probability that it is purple is 0.25. A student draws a marble, records its color and returns it to the bag. The student repeats this 400 times.

About how many times would you expect the student to draw a purple marble?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-DP.2.4-R

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

1. Two students each simulate spinning a fair spinner. One runs 210 trials and the other runs 8,000 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. The 210-trial student, because fewer trials leave fewer chances for a result to go wrong.

B. The 8,000-trial student, because more trials usually bring the result closer to the theory.

C. Both students equally, because a fair experiment gives the same result for any number of trials.

D. Neither student, because an experimental result never comes close to the theoretical one.

2. A fair coin is tossed 800 times and lands on heads 376 times.

What is the experimental probability of landing on heads? Write your answer as a decimal.

3. A bag contains only purple marbles and blue marbles. When one marble is drawn at random, the probability that it is purple is 0.25. A student draws a marble, records its color and returns it to the bag. The student repeats this 400 times.

About how many times would you expect the student to draw a purple marble?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-DP.2.4-X

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

1. Two students each simulate rolling a fair die. One runs 170 trials and the other runs 4,700 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. Neither student, because an experimental result never comes close to the theoretical one.

B. The 170-trial student, because fewer trials leave fewer chances for a result to go wrong.

C. The 4,700-trial student, because more trials usually bring the result closer to the theory.

D. Both students equally, because a fair experiment gives the same result for any number of trials.

2. A fair 10-sided die is rolled 200 times. It lands on a 3 exactly 23 times.

Part A. What is the experimental probability of rolling a 3? Write it as a fraction in simplest form.
Part B. What is the theoretical probability of rolling a 3? Write it as a fraction in simplest form.
Part C. Is the experimental probability greater than or less than the theoretical probability?

Score: ____ / 2 → copy to your progress strip

7th Grade Math · Comparison items (2)

C-DP.2.4-X

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

1. Two students each simulate rolling a fair die. One runs 170 trials and the other runs 4,700 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. Neither student, because an experimental result never comes close to the theoretical one.

B. The 170-trial student, because fewer trials leave fewer chances for a result to go wrong.

C. The 4,700-trial student, because more trials usually bring the result closer to the theory.

D. Both students equally, because a fair experiment gives the same result for any number of trials.

2. A fair 10-sided die is rolled 200 times. It lands on a 3 exactly 23 times.

Part A. What is the experimental probability of rolling a 3? Write it as a fraction in simplest form.
Part B. What is the theoretical probability of rolling a 3? Write it as a fraction in simplest form.
Part C. Is the experimental probability greater than or less than the theoretical probability?

Score: ____ / 2 → copy to your progress strip

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers Student code: _____ Calculator: no

Learning goal: I can add, subtract, multiply, and divide rational numbers. (circle your level after scoring)

4	I can solve a multi-step computation with rational numbers and check that the sign and size make sense.
3	I can add, subtract, multiply, and divide fractions, decimals, and negatives accurately.
2	I can add and subtract integers and multiply and divide with the correct sign.
1	I need help to start this benchmark.

1. What is the value of the expression?

$4.13 - 2.91 - 17.5$

2. What is the value of the expression? Write your answer in simplest form.

$-0.64 \div \left(-\frac{4}{5}\right)$

Turn over →

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers Student code: _____

3. What number makes the equation true?

$-2\frac{2}{5} + __ = -3.61$

4. Evaluate the expression. Write your answer in simplest form.

$2\frac{1}{2} \times (-0.1)$

5. A student evaluated the expression $-\frac{1}{6} \div 1\frac{4}{5} - \frac{1}{2}$ as shown.

Step 1: $-\frac{1}{6} \times \frac{5}{9} = \frac{5}{54}$
Step 2: $\frac{5}{54} - \frac{1}{2} = -\frac{11}{27}$

In which step did the student make the first error? What is the correct value of the expression?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.2.2 · MA.7.NSO.2.2 · Four operations with rational numbers

Unit 3: Operations with Rational Numbers. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **−16.28**

Subtract from left to right.

$4.13 - 2.91 = 1.22$.

$1.22 - 17.5 = -16.28$.

2. Regular $\frac{4}{5}$ **(or 0.8)**

Write the decimal as a fraction: $-0.64 = -\frac{16}{25}$.

Dividing by a fraction means multiplying by its reciprocal: $-\frac{16}{25} \times \left(-\frac{5}{4}\right) = \frac{4}{5}$.

Two negatives give a positive.

3. Regular $-1\frac{21}{100} = -\frac{121}{100} (\text{or } -1.21)$

The missing term is the difference of the two known numbers.

Write the fraction as a decimal: $-2\frac{2}{5} = -2.4$.

$-3.61 - (-2.4) = -1.21$, which is $-1\frac{21}{100}$.

Check: $-2\frac{2}{5} + \left(-1\frac{21}{100}\right) = -3.61$.

4. Regular $-\frac{1}{4} (\text{or } -0.25)$

Write the decimal as a fraction: $-0.1 = -\frac{1}{10}$.

Write the mixed number as an improper fraction: $2\frac{1}{2} = \frac{5}{2}$.

$\frac{5}{2} \times \left(-\frac{1}{10}\right) = -\frac{1}{4}$. A negative times a positive is negative.

5. Rigor **Step 1; the correct value is $-\frac{16}{27}$. Score: 1 point for the step and the correct value; the step alone earns 0.**

Step 1 has the wrong sign: a negative divided by a positive (or a positive by a negative) is negative, and two negatives give a positive.

Correct work: $-\frac{1}{6} \times \frac{5}{9} = -\frac{5}{54}$; $-\frac{5}{54} - \frac{1}{2} = -\frac{16}{27}$.

Re-check strip (3 items, C-NSO.2.2-R)

1. Regular **−45.75**

Subtract from left to right.

$6.49 - 2.34 = 4.15$.

$4.15 - 49.9 = -45.75$.

2. Regular **−6**

Write the decimal as a fraction: $-1.5 = -\frac{3}{2}$.

Dividing by a fraction means multiplying by its reciprocal: $-\frac{3}{2} \times \frac{4}{1} = -\frac{12}{2} = -6$.

A negative divided by a positive (or a positive by a negative) is negative.

3. Regular $\frac{5}{6}$

LCD of 3 and 6 is 6.

$\frac{4}{6} - \left(-\frac{1}{6}\right) = \frac{5}{6}$.

Comparison strip (2 items, C-NSO.2.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $-\frac{3}{4}$

LCD of 2 and 4 is 4.

$-\frac{2}{4} - \frac{1}{4} = -\frac{3}{4}$.

2. Regular $-\frac{8}{15} (\text{or } -0.5\overline{3})$

Write the decimal as a fraction: $-1.6 = -\frac{8}{5}$.

$\frac{1}{3} \times \left(-\frac{8}{5}\right) = -\frac{8}{15}$. A negative times a positive is negative.

Reteach plan — C-NSO.2.2

MA.7.NSO.2.2 · Four operations with rational numbers

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.2.2-R

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers Student code: _____ Calculator: no

1. What is the value of the expression?

$6.49 - 2.34 - 49.9$

2. What is the value of the expression? Write your answer in simplest form.

$-1.5 \div \frac{1}{4}$

3. Subtract. Write your answer in simplest form.

$\frac{2}{3} - \left(-\frac{1}{6}\right)$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-NSO.2.2-R

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers Student code: _____ Calculator: no

1. What is the value of the expression?

$6.49 - 2.34 - 49.9$

2. What is the value of the expression? Write your answer in simplest form.

$-1.5 \div \frac{1}{4}$

3. Subtract. Write your answer in simplest form.

$\frac{2}{3} - \left(-\frac{1}{6}\right)$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-NSO.2.2-X

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers

Student code: _____ Calculator: no

1. Evaluate. Write your answer in simplest form.

$-\frac{1}{2} - \frac{1}{4}$

2. Evaluate the expression. Write your answer in simplest form.

$\frac{1}{3} \times (-1.6)$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-NSO.2.2-X

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers

Student code: _____ Calculator: no

1. Evaluate. Write your answer in simplest form.

$-\frac{1}{2} - \frac{1}{4}$

2. Evaluate the expression. Write your answer in simplest form.

$\frac{1}{3} \times (-1.6)$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Weekly Check · Unit 3 — C-NSO.2.3

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

Learning goal: I can solve real-world problems with rational numbers. (circle your level after scoring)

4	I can solve a multi-step real-world problem with rational numbers and explain whether the answer is reasonable.
3	I can solve a real-world problem involving rational numbers and state the answer with units.
2	I can pick the operation a word problem needs.
1	I need help to start this benchmark.

1. A tank contains $76\frac{3}{4}$ gallons of water. Water leaks out at 0.25 gallons per hour for 5 hours, and then $14\frac{1}{2}$ gallons are added.

How much water is in the tank now?

2. A group of 5 friends owes $83.60 for a canoe rental. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Jamal pays $\frac{3}{8}$ of his share.

What is Jamal's balance now?

Turn over →

3. At 6 a.m. the temperature at a weather station was -5.2°C. The temperature then fell 3.5°C each hour for 6 hours. Later in the day it rose 4.1°C.

What was the temperature after all of these changes?

4. The price of one share of a company's stock changed by these amounts over 4 days.

+$1.85, −$1.30, −$0.65, −$2.30

What was the mean (average) daily change in the price?

5. The temperature outside a cabin in January is 9.3°F. The temperature drops $3\frac{1}{2}^\circ\text{F}$ each hour.

After how many whole hours will the temperature first be below -3°F?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.2.3 · MA.7.NSO.2.3 · Real-world problems with rational numbers

Unit 3: Operations with Rational Numbers. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **90 gallons**

Water lost: $5 \times 0.25 = 1.25$ gallons.

After the loss: $76\frac{3}{4} - 1.25 = 75\frac{1}{2}$ gallons.

After adding water: $75\frac{1}{2} + 14\frac{1}{2} = 90$ gallons.

2. Regular **−$10.45**

Each share: $-\$83.60 \div 5 = -\$16.72$.

Paying $\frac{3}{8}$ of the share leaves $1 - \frac{3}{8} = \frac{5}{8}$ of it: $\frac{5}{8} \times (-\$16.72) = -\$10.45$.

Jamal's balance is −$10.45.

3. Regular -22.1°C

Total fall: $3.5 \times 6 = 21^\circ\text{C}$.

$-5.2 - 21 = -26.2$.

$-26.2 + 4.1 = -22.1^\circ\text{C}$.

4. Regular **−$0.60**

Add the 4 changes: the total change is −$2.40.

Divide by the number of days: $-\$2.40 \div 4 = -\$0.60$.

5. Rigor **4 hours**

The temperature after h hours is $9.3 - h \times 3\frac{1}{2}$.

After 3 hours: $9.3 - 3 \times 3.5 = -1.2^\circ\text{F}$, which is not below -3°F.

After 4 hours: $9.3 - 4 \times 3.5 = -4.7^\circ\text{F}$, which is below -3°F.

So the temperature is first below -3°F after 4 hours.

Re-check strip (3 items, C-NSO.2.3-R)

1. Regular -20.8°C

Total fall: $3.5 \times 5 = 17.5^\circ\text{C}$.

$-8.7 - 17.5 = -26.2$.

$-26.2 + 5.4 = -20.8^\circ\text{C}$.

2. Regular **−$0.33**

Add the 5 changes: the total change is −$1.65.

Divide by the number of days: $-\$1.65 \div 5 = -\$0.33$.

3. Regular **−$4.70**

Each share: $-\$141.00 \div 6 = -\$23.50$.

Paying $\frac{4}{5}$ of the share leaves $1 - \frac{4}{5} = \frac{1}{5}$ of it: $\frac{1}{5} \times (-\$23.50) = -\$4.70$.

Maya's balance is −$4.70.

Comparison strip (2 items, C-NSO.2.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **−$1.01**

Add the 5 changes: the total change is −$5.05.

Divide by the number of days: $-\$5.05 \div 5 = -\$1.01$.

2. Regular **−$5.70**

Each share: $-\$136.80 \div 4 = -\$34.20$.

Paying $\frac{5}{6}$ of the share leaves $1 - \frac{5}{6} = \frac{1}{6}$ of it: $\frac{1}{6} \times (-\$34.20) = -\$5.70$.

Ava's balance is −$5.70.

Reteach plan — C-NSO.2.3

MA.7.NSO.2.3 · Real-world problems with rational numbers

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.2.3-R

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

1. At 6 a.m. the temperature at a weather station was -8.7°C. The temperature then fell 3.5°C each hour for 5 hours. Later in the day it rose 5.4°C.

What was the temperature after all of these changes?

2. The price of one share of a company's stock changed by these amounts over 5 days.

−$0.25, +$1.55, −$0.70, −$2.20, −$0.05

What was the mean (average) daily change in the price?

3. A group of 6 friends owes $141.00 for a rented party tent. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Maya pays $\frac{4}{5}$ of her share.

What is Maya's balance now?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-NSO.2.3-R

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

1. At 6 a.m. the temperature at a weather station was -8.7°C. The temperature then fell 3.5°C each hour for 5 hours. Later in the day it rose 5.4°C.

What was the temperature after all of these changes?

2. The price of one share of a company's stock changed by these amounts over 5 days.

−$0.25, +$1.55, −$0.70, −$2.20, −$0.05

What was the mean (average) daily change in the price?

3. A group of 6 friends owes $141.00 for a rented party tent. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Maya pays $\frac{4}{5}$ of her share.

What is Maya's balance now?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.2.3-X

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

1. The price of one share of a company's stock changed by these amounts over 5 days.

+$0.85, −$2.55, −$1.55, −$0.70, −$1.10

What was the mean (average) daily change in the price?

2. A group of 4 friends owes $136.80 for a rented party tent. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Ava pays $\frac{5}{6}$ of her share.

What is Ava's balance now?

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.2.3-X

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

1. The price of one share of a company's stock changed by these amounts over 5 days.

+$0.85, −$2.55, −$1.55, −$0.70, −$1.10

What was the mean (average) daily change in the price?

2. A group of 4 friends owes $136.80 for a rented party tent. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Ava pays $\frac{5}{6}$ of her share.

What is Ava's balance now?

Score: ____ / 2 → **copy to your progress strip**

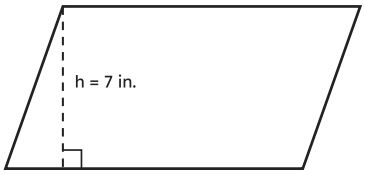
Benchmark: **MA.7.GR.1.1** · Area of trapezoids, parallelograms, and rhombi Student code: _____ Calculator: yes

Learning goal: I can find the area of trapezoids, parallelograms, and rhombi. (circle your level after scoring)

4	I can find a missing dimension when the area is given, or explain where a formula comes from.
3	I can use the formulas to find the area of a trapezoid, parallelogram, or rhombus.
2	I can find the area of a rectangle and a triangle.
1	I need help to start this benchmark.

1. The parallelogram shown has a base of 18 inches and a vertical height of 7 inches.

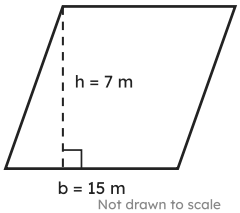
What is its area, in square inches?



Not drawn to scale

2. The rhombus shown has a base of 15 meters and a vertical height of 7 meters.

What is its area, in square meters?



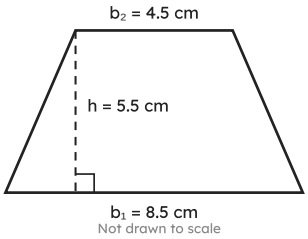
Not drawn to scale

Turn over →

Benchmark: **MA.7.GR.1.1** · Area of trapezoids, parallelograms, and rhombi Student code: _____

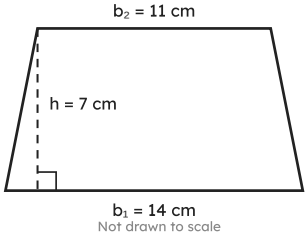
3. A trapezoid has bases of 8.5 centimeters and 4.5 centimeters and a vertical height of 5.5 centimeters.

What is its area, in square centimeters?



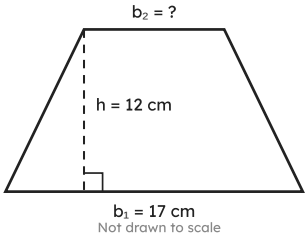
4. A trapezoid has bases of 14 cm and 11 cm and a vertical height of 7 cm.

What is its area, in square centimeters?



5. The trapezoid shown has a base of 17 centimeters and a vertical height of 12 centimeters. Its area is equal to the area of a parallelogram with a base of 15 centimeters and a vertical height of 10 centimeters.

What is the length of the trapezoid's other base, in centimeters?



Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.1.1 · MA.7.GR.1.1 · Area of trapezoids, parallelograms, and rhombi

Unit 4: Area. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $126\,\text{in.}^2$

$A = bh = 18 \times 7 = 126\,\text{in.}^2$

2. Regular $105\,\text{m}^2$

A rhombus is a parallelogram, so $A = bh$.

$A = 15 \times 7 = 105\,\text{m}^2$.

3. Regular $35.75\,\text{cm}^2$

$A = \frac{1}{2}(b_1 + b_2)h$.

$\frac{1}{2}(8.5 + 4.5)(5.5) = \frac{1}{2}(13)(5.5) = 35.75\,\text{cm}^2$.

4. Regular $87.5\,\text{cm}^2$

$A = \frac{1}{2}(b_1 + b_2)h$.

$\frac{1}{2}(14 + 11)(7) = \frac{1}{2}(25)(7) = 87.5\,\text{cm}^2$.

5. Rigor **8 cm**

Area of the parallelogram: $A = bh = 15 \times 10 = 150\,\text{cm}^2$.

The trapezoid has the same area: $A = \frac{1}{2}(b_1 + b_2)h$, so $150 = \frac{1}{2}(17 + b)(12) = 6(17 + b)$.

$17 + b = 150 \div 6 = 25$.

$b = 25 - 17 = 8\,\text{cm}$.

Re-check strip (3 items, C-GR.1.1-R)

1. Regular $153\,\text{in.}^2$

$A = bh = 17 \times 9 = 153\,\text{in.}^2$

2. Regular **C.** $180\,\text{in.}^2$

A rhombus is a parallelogram, so $A = bh$.

$18 \times 10 = 180\,\text{in.}^2$

3. Regular $99\,\text{cm}^2$

$A = \frac{1}{2}(b_1 + b_2)h$.

$\frac{1}{2}(14 + 8)(9) = \frac{1}{2}(22)(9) = 99\,\text{cm}^2$.

Comparison strip (2 items, C-GR.1.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $35\,\text{m}^2$

A rhombus is a parallelogram, so $A = bh$.

$A = 7 \times 5 = 35\,\text{m}^2$.

2. Regular $66\,\text{in.}^2$

$A = bh = 6 \times 11 = 66\,\text{in.}^2$

Reteach plan — C-GR.1.1

MA.7.GR.1.1 · Area of trapezoids, parallelograms, and rhombi

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

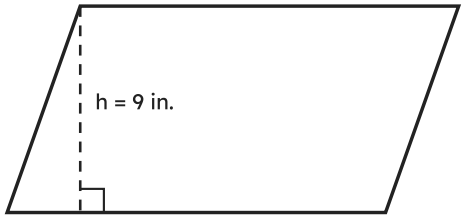
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.1.1-R

Benchmark: **MA.7.GR.1.1** · Area of trapezoids, parallelograms, and rhombi Student code: _____ Calculator: yes

1. The parallelogram shown has a base of 17 inches and a vertical height of 9 inches.

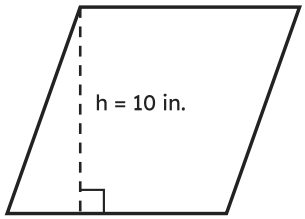
What is its area, in square inches?



b = 17 in.
Not drawn to scale

2. A rhombus has a base of 18 inches and a vertical height of 10 inches.

What is its area, in square inches?



b = 18 in.
Not drawn to scale

A. $324\,\text{in.}^2$

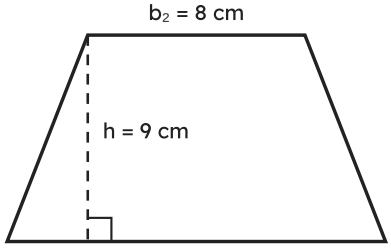
B. $90\,\text{in.}^2$

C. $180\,\text{in.}^2$

D. $72\,\text{in.}^2$

3. A trapezoid has bases of 14 cm and 8 cm and a vertical height of 9 cm.

What is its area, in square centimeters?



Not drawn to scale

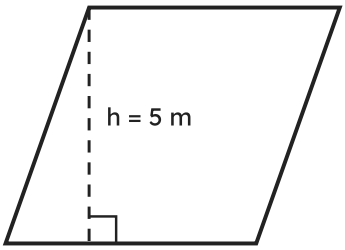
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-GR.1.1-X

Benchmark: **MA.7.GR.1.1** · Area of trapezoids, parallelograms, and rhombi Student code: _____ Calculator: yes

1. The rhombus shown has a base of 7 meters and a vertical height of 5 meters.

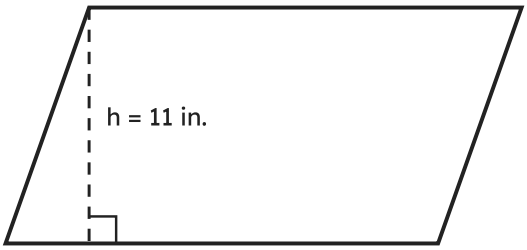
What is its area, in square meters?



Not drawn to scale

2. The parallelogram shown has a base of 6 inches and a vertical height of 11 inches.

What is its area, in square inches?



Not drawn to scale

Score: ____ / 2 → copy to your progress strip

7th Grade Math · Weekly Check · Unit 4 C-GR.1.2

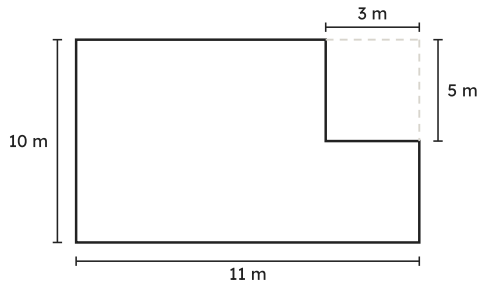
Benchmark: **MA.7.GR.1.2** · Area of polygons and composite figures Student code: _____ Calculator: yes

Learning goal: I can find the area of polygons and composite figures. (circle your level after scoring)

4	I can solve a real-world problem with a composite figure, including one with a region removed.
3	I can find the area of a polygon or composite figure by adding or subtracting areas.
2	I can split a figure into shapes I know.
1	I need help to start this benchmark.

1. A garden is shaped like the figure shown: an 11 m by 10 m rectangle with a 3 m by 5 m rectangular corner removed. The dashed lines show the part that was removed.

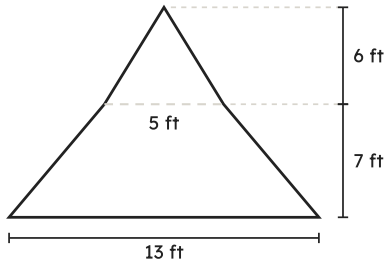
What is the area of the garden, in square meters?



Not drawn to scale

2. The front of a tent is shaped like the figure shown: a trapezoid with bases of 13 ft and 5 ft and a vertical height of 7 ft, with a triangle on top. The triangle's base is the 5 ft top base of the trapezoid, and its vertical height is 6 ft.

What is the area of the front of the tent, in square feet?



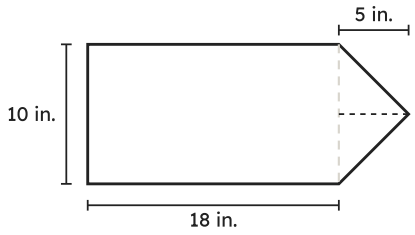
Not drawn to scale

Turn over →

Benchmark: **MA.7.GR.1.2** · Area of polygons and composite figures Student code: _____

3. A tabletop is shaped like the figure shown: an $18\,\text{in.} \times 10\,\text{in.}$ rectangle with a triangle attached to one end. The triangle's base is the 10 in. side and its height is 5 in.

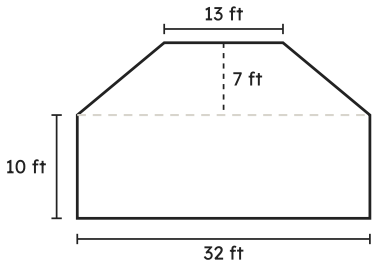
What is the area of the tabletop, in square inches?



Not drawn to scale

4. The end wall of a barn is shaped like the figure shown: a rectangle 32 ft wide and 10 ft tall, with a trapezoid on top. The trapezoid's bases are the 32 ft top edge of the rectangle and a 13 ft top side, and its vertical height is 7 ft.

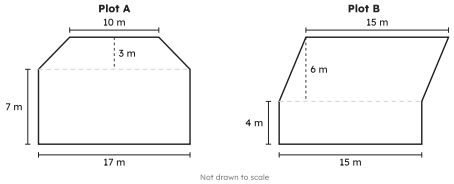
What is the area of the end wall, in square feet?



Not drawn to scale

5. Two garden plots are shown. Plot A is a trapezoid with bases of 17 m and 10 m and a vertical height of 3 m, joined along its longer base to a rectangle 17 m by 7 m. Plot B is a parallelogram with a base of 15 m and a vertical height of 6 m, joined along its base to a rectangle 15 m by 4 m.

Which plot has the greater area, and how much greater is it?



Not drawn to scale

Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — C-GR.1.2 · MA.7.GR.1.2 · Area of polygons and composite figures

Unit 4: Area. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $95\,\text{m}^2$

Whole rectangle: $11 \times 10 = 110\,\text{m}^2$.
Removed corner: $3 \times 5 = 15\,\text{m}^2$.
Area: $110 - 15 = 95\,\text{m}^2$.

2. Regular $78\,\text{ft}^2$

Trapezoid: $\frac{1}{2}(13 + 5)(7) = 63\,\text{ft}^2$.
Triangle: $5 \times 6 \div 2 = 15\,\text{ft}^2$.
Total: $63 + 15 = 78\,\text{ft}^2$.

3. Regular $205\,\text{in.}^2$

Rectangle: $18 \times 10 = 180$.
Triangle: $\frac{1}{2} \times 10 \times 5 = 25$.
Total: $180 + 25 = 205\,\text{in.}^2$.

4. Regular $477.5\,\text{ft}^2$

Rectangle: $32 \times 10 = 320\,\text{ft}^2$.
Trapezoid: $\frac{1}{2}(32 + 13)(7) = 157.5\,\text{ft}^2$.
Total: $320 + 157.5 = 477.5\,\text{ft}^2$.

5. Rigor **Plot A, by $9.5\,\text{m}^2$. Score: 1 point — the plot and the difference; the plot alone earns 0.**

Plot A: trapezoid $\frac{1}{2}(17 + 10)(3) = 40.5\,\text{m}^2$, rectangle $17 \times 7 = 119\,\text{m}^2$, so $159.5\,\text{m}^2$.
Plot B: parallelogram $15 \times 6 = 90\,\text{m}^2$, rectangle $15 \times 4 = 60\,\text{m}^2$, so $150\,\text{m}^2$.
$159.5 - 150 = 9.5$, so Plot A is greater by $9.5\,\text{m}^2$.

Re-check strip (3 items, C-GR.1.2-R)

1. Regular $108\,\text{ft}^2$

Whole rectangle: $12 \times 11 = 132\,\text{ft}^2$.
Removed notch: $6 \times 4 = 24\,\text{ft}^2$.
Area: $132 - 24 = 108\,\text{ft}^2$.

2. Regular $70\,\text{in.}^2$

Trapezoid: $\frac{1}{2}(10 + 8)(6) = 54\,\text{in.}^2$.
Triangle: $8 \times 4 \div 2 = 16\,\text{in.}^2$.
Total: $54 + 16 = 70\,\text{in.}^2$.

3. Regular $117\,\text{ft}^2$

Rectangle: $18 \times 6 = 108$.
Triangle: $\frac{1}{2} \times 6 \times 3 = 9$.
Total: $108 + 9 = 117\,\text{ft}^2$.

Comparison strip (2 items, C-GR.1.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $165\,\text{ft}^2$

Whole rectangle: $15 \times 13 = 195\,\text{ft}^2$.
Removed corner: $5 \times 6 = 30\,\text{ft}^2$.
Area: $195 - 30 = 165\,\text{ft}^2$.

2. Regular $850\,\text{in.}^2$

One trapezoid: $\frac{1}{2}(32 + 18)(17) = 425\,\text{in.}^2$.
The two trapezoids are congruent, so the hexagon is $2 \times 425 = 850\,\text{in.}^2$.

Reteach plan — C-GR.1.2

MA.7.GR.1.2 · Area of polygons and composite figures

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

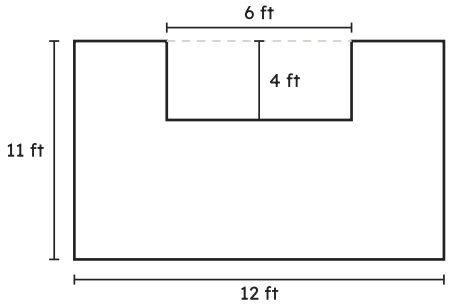
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.1.2-R

Benchmark: **MA.7.GR.1.2** · Area of polygons and composite figures Student code: _____ Calculator: yes

1. A kitchen floor is shaped like the figure shown: a 12 ft by 11 ft rectangle with a 6 ft by 4 ft rectangular notch removed from the middle of one edge. The dashed lines show the part that was removed.

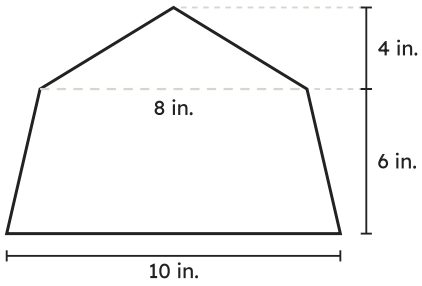
What is the area of the floor, in square feet?



Not drawn to scale

2. A stained-glass panel is shaped like the figure shown: a trapezoid with bases of 10 in. and 8 in. and a vertical height of 6 in., with a triangle on top. The triangle's base is the 8 in. top base of the trapezoid, and its vertical height is 4 in.

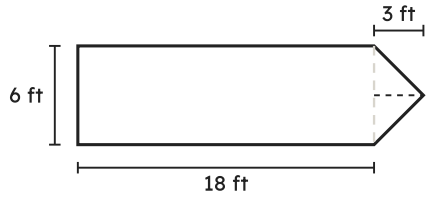
What is the area of the panel, in square inches?



Not drawn to scale

3. A garden bed is shaped like the figure shown: an $18\,\text{ft} \times 6\,\text{ft}$ rectangle with a triangle attached to one end. The triangle's base is the 6 ft side and its height is 3 ft.

What is the area of the garden bed, in square feet?



Not drawn to scale

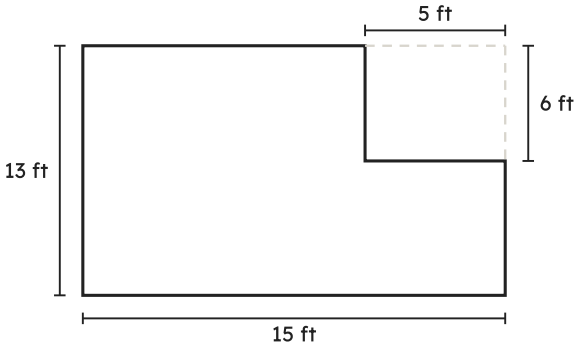
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-GR.1.2-X

Benchmark: **MA.7.GR.1.2** · Area of polygons and composite figures Student code: _____ Calculator: yes

1. A kitchen floor is shaped like the figure shown: a 15 ft by 13 ft rectangle with a 5 ft by 6 ft rectangular corner removed. The dashed lines show the part that was removed.

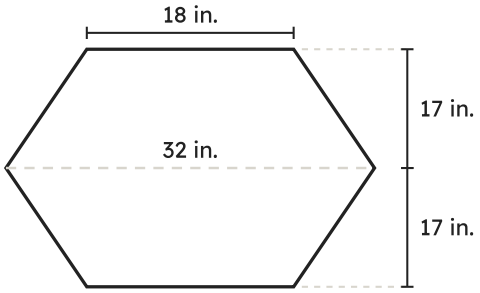
What is the area of the floor, in square feet?



Not drawn to scale

2. A hexagonal tabletop is shaped like the hexagon shown. It is made of two congruent trapezoids that share the 32 in. base drawn as a dashed line. Each trapezoid has a vertical height of 17 in., and its other base is 18 in.

What is the area of the tabletop, in square inches?



Not drawn to scale

Score: ____ / 2 → copy to your progress strip

Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____ Calculator: yes

Learning goal: I can use the relationship between circumference and diameter. (circle your level after scoring)

4	I can explain pi as the ratio of circumference to diameter and use it in a real-world problem.
3	I can find circumference from radius or diameter, and radius or diameter from circumference.
2	I can identify the radius, diameter, and circumference of a circle.
1	I need help to start this benchmark.

1. A circle has a radius of 11 centimeters.

What is the circumference of the circle? Write your answer in terms of π.

2. The circumference of a round pool is 106.81 feet.

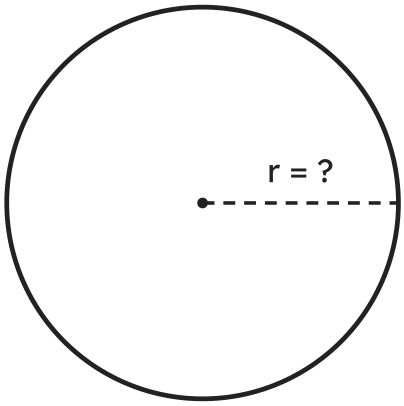
What is the diameter of the round pool? Round your answer to the nearest hundredth. Use the π key on your calculator.

Turn over →

Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____

3. A circular clock face has a circumference of 62.80 centimeters, as shown.

What is the radius of the circular clock face? Round your answer to the nearest hundredth. Use 3.14 for π.



C = 62.80 cm

Not drawn to scale

4. A circular tabletop has a radius of 2 feet.

What is the circumference of the tabletop? Round your answer to the nearest hundredth. Use the π key on your calculator.

5. A scooter wheel has a diameter of 12 inches. The wheel rolls in a straight line for 200 feet.

How many complete turns does the wheel make? ($1 \text{ foot} = 12$ inches) Use 3.14 for π.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.1.3 · MA.7.GR.1.3 · Circumference of a circle

Unit 4: Area. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **22π centimeters**

$C = 2\pi r = 2 \times \pi \times 11 = 22\pi$ centimeters.

2. Regular **34 feet (34.00 also accepted)**

$C = \pi d$, so $d = C \div \pi$.

With the π key: $106.81 \div \pi \approx 34.00$ feet.

3. Regular **10 centimeters (10.00 also accepted)**

$C = 2\pi r$, so $r = C \div (2\pi)$.

$2 \times 3.14 = 6.28$, and $62.80 \div 6.28 = 10.00$ centimeters.

4. Regular **12.57 feet**

$C = 2\pi r = 2\pi(2) = 4\pi$.

With the π key: $4 \times \pi \approx 12.57$ feet.

5. Rigor **63 complete turns**

In one turn, the wheel rolls a distance equal to its circumference: $C = \pi d = 3.14 \times 12 = 37.68$ inches.

$200 \text{ feet} = 200 \times 12 = 2400 \text{ inches}$.

$2400 \div 37.68 \approx 63.69$ turns.

The wheel makes 63 complete turns; the last turn is not finished.

Re-check strip (3 items, C-GR.1.3-R)

1. Regular **13 centimeters (13.00 also accepted)**

$C = 2\pi r$, so $r = C \div (2\pi)$.

With the π key: $81.68 \div (2\pi) \approx 13.00$ centimeters.

2. Regular **3.14, which is an approximation of π**

The constant of proportionality is circumference ÷ diameter.

$25.13 \div 8 \approx 3.14$

$43.98 \div 14 \approx 3.14$

$62.83 \div 20 \approx 3.14$

The ratio is the same for every circle. That constant is π, and 3.14 is the common approximation of it.

3. Regular **30 feet**

$C = 2\pi r$, so $60\pi = 2\pi r$.

Divide both sides by 2π: $r = 60 \div 2 = 30$ feet.

Comparison strip (2 items, C-GR.1.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **36π inches**

$C = 2\pi r = 2 \times \pi \times 18 = 36\pi$ inches.

2. Regular **8 centimeters (8.00 also accepted)**

$C = 2\pi r$, so $r = C \div (2\pi)$.

$2 \times 3.14 = 6.28$, and $50.24 \div 6.28 = 8.00$ centimeters.

Reteach plan — C-GR.1.3

MA.7.GR.1.3 · Circumference of a circle

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

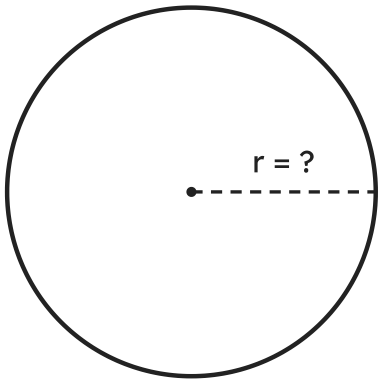
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.1.3-R

Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____ Calculator: yes

1. A round plate has a circumference of 81.68 centimeters, as shown.

What is the radius of the round plate? Round your answer to the nearest hundredth. Use the π key on your calculator.



C = 81.68 cm

Not drawn to scale

2. A student measured the diameter and the circumference of three paper cups, in centimeters. The results are shown in the table.

Diameter (cm)	Circumference (cm)
8	25.13
14	43.98
20	62.83

The circumference of a circle is proportional to its diameter. Based on the table, what is the constant of proportionality, rounded to the nearest hundredth?

3. The circumference of a circle is 60π feet.

What is the radius of the circle?

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-GR.1.3-X

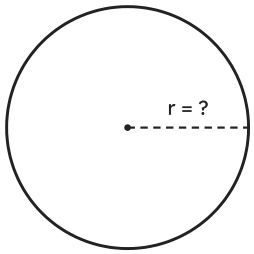
Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____ Calculator: yes

1. A circle has a radius of 18 inches.

What is the circumference of the circle? Write your answer in terms of π.

2. A circular clock face has a circumference of 50.24 centimeters, as shown.

What is the radius of the circular clock face? Round your answer to the nearest hundredth. Use 3.14 for π.



C = 50.24 cm

Not drawn to scale

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-GR.1.3-X

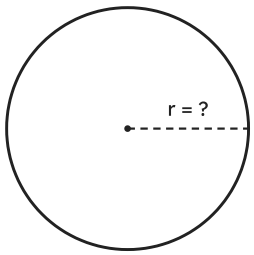
Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____ Calculator: yes

1. A circle has a radius of 18 inches.

What is the circumference of the circle? Write your answer in terms of π.

2. A circular clock face has a circumference of 50.24 centimeters, as shown.

What is the radius of the circular clock face? Round your answer to the nearest hundredth. Use 3.14 for π.



C = 50.24 cm

Not drawn to scale

Score: ____ **/ 2** → **copy to your progress strip**

Benchmark: **MA.7.GR.1.4** · Area of a circle Student code: _____ Calculator: yes

Learning goal: I can find the area of a circle. (circle your level after scoring)

4	I can solve a real-world problem using circumference and area together, including semicircles.
3	I can find the area of a circle from its radius or diameter.
2	I can find the radius from the diameter and square a number.
1	I need help to start this benchmark.

1. A circle has a diameter of 12 meters.

What is the area of the circle? Write your answer in terms of π.

2. A circular garden has a diameter of 14 feet.

What is the area of the circular garden? Round your answer to the nearest hundredth. Use 3.14 for π.

3. A 24-inch pizza is cut into 9 equal slices. (The size of a pizza is its diameter.) A student eats 5 slices.

How many square inches of pizza did the student eat? Write your answer in terms of π.

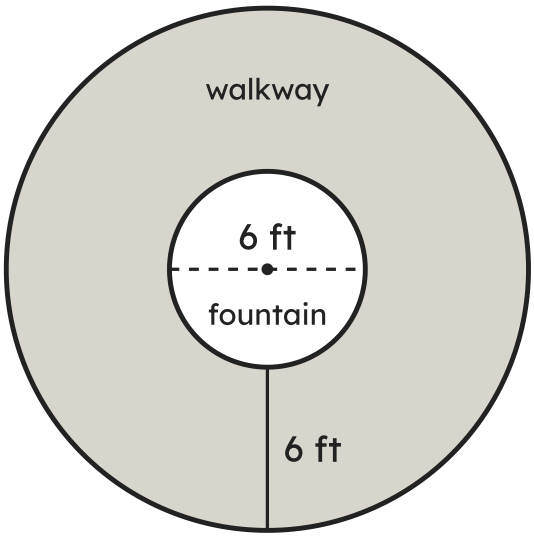
Turn over →

4. A circular skating rink has a radius of 21 meters.

What is the area of the circular skating rink? Round your answer to the nearest hundredth. Use 3.14 for π.

5. A circular fountain has a diameter of 6 feet. A walkway 6 feet wide surrounds the fountain, as shown.

What is the area of the walkway, in square feet?



Not drawn to scale

A. 72π

B. 36π

C. 108π

D. 81π

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.1.4 · MA.7.GR.1.4 · Area of a circle

Unit 4: Area. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **36π square meters**

The radius is $12 \div 2 = 6$ meters.

$A = \pi r^2 = \pi \times 6^2 = 36\pi$ square meters.

2. Regular **153.86 square feet**

The radius is $14 \div 2 = 7$ feet.

$A = \pi r^2 = \pi \times 7^2 = 49\pi$.

With 3.14 for π: $49 \times 3.14 = 153.86$ square feet.

3. Regular **80π square inches**

The diameter is 24 inches, so the radius is $24 \div 2 = 12$ inches and the whole pizza has area $\pi(12)^2 = 144\pi$ square inches.

The student ate $\frac{5}{9}$ of the pizza.

$\frac{5}{9} \times 144\pi = 80\pi$ square inches.

4. Regular **1384.74 square meters**

$A = \pi r^2 = \pi \times 21^2 = 441\pi$.

With 3.14 for π: $441 \times 3.14 = 1384.74$ square meters.

5. Rigor **A.** 72π

The fountain's radius is $6 \div 2 = 3$ feet, so the outer edge of the walkway is $3 + 6 = 9$ feet from the center.

Fountain and walkway together: $\pi(9)^2 = 81\pi$.

Fountain alone: $\pi(3)^2 = 9\pi$.

Walkway: $81\pi - 9\pi = 72\pi$ square feet.

The walkway is a ring around the fountain, not a circle with a radius of 6 feet.

Re-check strip (3 items, C-GR.1.4-R)

1. Regular **196π square inches**

The radius is $28 \div 2 = 14$ inches.

$A = \pi r^2 = \pi \times 14^2 = 196\pi$ square inches.

2. Regular **803.84 square inches**

$A = \pi r^2 = \pi \times 16^2 = 256\pi$.

With 3.14 for π: $256 \times 3.14 = 803.84$ square inches.

3. Regular **75π square inches**

The diameter is 30 inches, so the radius is $30 \div 2 = 15$ inches and the whole pizza has area $\pi(15)^2 = 225\pi$ square inches.

The student ate $\frac{1}{3}$ of the pizza.

$\frac{1}{3} \times 225\pi = 75\pi$ square inches.

Comparison strip (2 items, C-GR.1.4-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B.** 484π

$A = \pi r^2 = \pi \times 22 \times 22 = 484\pi$.

The value 44π would be the circumference instead.

2. Regular **2289.06 square meters**

$A = \pi r^2 = \pi \times 27^2 = 729\pi$.

With 3.14 for π: $729 \times 3.14 = 2289.06$ square meters.

Reteach plan — C-GR.1.4

MA.7.GR.1.4 · Area of a circle

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items)

C-GR.1.4-R

Benchmark: **MA.7.GR.1.4** · Area of a circle

Student code: _____ Calculator: yes

1. A circle has a diameter of 28 inches.

What is the area of the circle? Write your answer in terms of π.

2. A round tabletop has a radius of 16 inches.

What is the area of the round tabletop? Round your answer to the nearest hundredth. Use 3.14 for π.

3. A 30-inch pizza is cut into 3 equal slices. (The size of a pizza is its diameter.) A student eats 1 slice.

How many square inches of pizza did the student eat? Write your answer in terms of π.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items)

C-GR.1.4-R

Benchmark: **MA.7.GR.1.4** · Area of a circle

Student code: _____ Calculator: yes

1. A circle has a diameter of 28 inches.

What is the area of the circle? Write your answer in terms of π.

2. A round tabletop has a radius of 16 inches.

What is the area of the round tabletop? Round your answer to the nearest hundredth. Use 3.14 for π.

3. A 30-inch pizza is cut into 3 equal slices. (The size of a pizza is its diameter.) A student eats 1 slice.

How many square inches of pizza did the student eat? Write your answer in terms of π.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-GR.1.4-X

Benchmark: **MA.7.GR.1.4** · Area of a circle Student code: _____ Calculator: yes

1. A circle has a radius of 22 feet.

What is the area of the circle, in square feet?

A. 44π

B. 484π

C. 1936π

D. 22π

2. A circular skating rink has a radius of 27 meters.

What is the area of the circular skating rink? Round your answer to the nearest hundredth. Use 3.14 for π.

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-GR.1.4-X

Benchmark: **MA.7.GR.1.4** · Area of a circle Student code: _____ Calculator: yes

1. A circle has a radius of 22 feet.

What is the area of the circle, in square feet?

A. 44π

B. 484π

C. 1936π

D. 22π

2. A circular skating rink has a radius of 27 meters.

What is the area of the circular skating rink? Round your answer to the nearest hundredth. Use 3.14 for π.

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____ Calculator: yes

Learning goal: I can use a random sample to make a prediction about a population. (circle your level after scoring)

4	I can judge whether a prediction is reasonable and explain how sample size or bias affects it.
3	I can use a proportion from a sample to predict a value for the whole population.
2	I can tell whether a sample is random and representative.
1	I need help to start this benchmark.

1. In a random sample of 100 passengers, 35 check a bag.

Based on this sample, about how many of the 800 passengers in the population check a bag?

A. 520

B. 431

C. 35

D. 280

2. To predict how many of the 4000 residents in the town support building a new park, Priya asked the 24 people at the skate park on Saturday, and 22 said yes. She predicts that about 3670 of the 4000 residents support building a new park.

Why is this prediction not valid?

A. A prediction about the town can only be made by asking every one of the 4000 residents

B. The prediction should be 22 residents, the number who said yes, not 3670 residents

C. The sample is too small; asking a larger group of people at the skate park on Saturday would make the prediction valid

D. The people at the skate park on Saturday are not a random sample of all the residents, so they may not represent the whole town

Turn over →

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____

3. A library has 1200 readers. Based on a random sample of 240 readers, it is predicted that about 480 of the 1200 readers borrow e-books.

How many readers in the sample said they borrow e-books?

4. A factory checks a random sample of 400 marbles and finds that 100 of them are red.

Based on this sample, about how many of 1000 marbles would be expected to be red?

5. The principal wants to predict how many of the 2100 students in the school would buy a school T-shirt. Two random samples of students were surveyed.
Sample A: 600 students, and 210 said yes.
Sample B: 120 students, and 54 said yes.

The two samples give different predictions. Which sample's prediction should the principal trust more, and about how many of the 2100 students does it predict? Explain your choice.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.3 · MA.7.DP.1.3 · Predictions from a random sample

Unit 5: Scaling With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **D. 280**

Sample proportion: $35 \div 100 = 0.35$.
Prediction: $0.35 \times 800 = 280$.

2. Regular **D. The people at the skate park on Saturday are not a random sample of all the residents, so they may not represent the whole town**

Priya's group is the people at the skate park on Saturday: a particular group that may feel differently from the rest of the town, so its proportion, $\frac{22}{24}$, cannot be applied to all 4000 residents.
A valid prediction needs a random sample of the whole town, where every resident has the same chance of being chosen. A larger group of people at the skate park on Saturday would still not be random, and a random sample does not have to include everyone.

3. Regular **96 readers**

A prediction from a random sample uses the sample's proportion, so the sample had the same proportion as the prediction: $480 \div 1200 = 0.4 = 40\%$.
40% of the sample of 240: $0.4 \times 240 = 96$ readers said they borrow e-books.

4. Regular **about 250**

The sample rate is $\frac{100}{400} = \frac{1}{4}$, or 1 out of every 4.
$1000 \div 4 = 250$.

5. Rigor **Sample A, about 735 students, because it is the larger random sample, so its proportion is usually closer to the proportion in the whole school. Score: 1 point — the larger sample chosen, its prediction and the reason (sample size); the prediction alone earns 0.**

Sample A: $\frac{210}{600} = 35\%$, and $35\% \text{ of } 2100 = 735 \text{ students}$.
Sample B: $\frac{54}{120} = 45\%$, and $45\% \text{ of } 2100 = 945 \text{ students}$.
Both samples are random, so both predictions are reasonable, but a larger random sample varies less from the whole school, so its prediction, about 735, is the one to trust more.

Re-check strip (3 items, C-DP.1.3-R)

1. Regular **60%**

A prediction from a random sample uses the sample's proportion, so the sample had the same proportion as the prediction: $1140 \div 1900 = 0.6 = 60\%$.
So 60% of the 400 students in the sample said they ride the bus to school.

2. Regular **about 800 students**

In the sample, 100 out of 250 students prefer soy milk.
$\frac{100}{250} = \frac{x}{2000}$, so $x = 100 \div 250 \times 2000 = 800$.
About 800 students at the school prefer soy milk.

3. Regular **about 500**

The sample rate is $\frac{80}{400} = \frac{1}{5}$, or 1 out of every 5.
$2500 \div 5 = 500$.

Comparison strip (2 items, C-DP.1.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **D. The members of the drama club are not a random sample of all the students, so they may not represent the whole school**

Lena's group is the members of the drama club: a particular group that may feel differently from the rest of the school, so its proportion, $\frac{22}{29}$, cannot be applied to all 1200 students.
A valid prediction needs a random sample of the whole school, where every student has the same chance of being chosen. A larger group of members of the drama club would still not be random, and a random sample does not have to include everyone.

2. Regular **75%**

A prediction from a random sample uses the sample's proportion, so the sample had the same proportion as the prediction: $2550 \div 3400 = 0.75 = 75\%$.
So 75% of the 400 voters in the sample said they support the new park.

Reteach plan — C-DP.1.3

MA.7.DP.1.3 · Predictions from a random sample

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.1.3-R

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____ Calculator: yes

1. A school has 1900 students. Based on a random sample of 400 students, it is predicted that about 1140 of the 1900 students ride the bus to school.

What percent of the students in the sample said they ride the bus to school?

2. A random sample of 250 students at a school were asked which kind of milk they prefer. The results are shown in the table.

Cow's milk	Soy milk	Almond milk
100	100	50

There are 2000 students at the school. Based on this sample, about how many students at the school prefer soy milk?

3. A factory checks a random sample of 400 seeds and finds that 80 of them sprouted.

Based on this sample, about how many of 2500 seeds would be expected to sprout?

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.3-X

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____ Calculator: yes

1. To predict how many of the 1200 students in the school would buy a yearbook, Lena asked the 29 members of the drama club, and 22 said yes. She predicts that about 910 of the 1200 students would buy a yearbook.

Why is this prediction not valid?

A. The sample is too small; asking a larger group of members of the drama club would make the prediction valid

B. The prediction should be 22 students, the number who said yes, not 910 students

C. A prediction about the school can only be made by asking every one of the 1200 students

D. The members of the drama club are not a random sample of all the students, so they may not represent the whole school

2. A town has 3400 voters. Based on a random sample of 400 voters, it is predicted that about 2550 of the 3400 voters support the new park.

What percent of the voters in the sample said they support the new park?

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.1.3-X

Benchmark: **MA.7.DP.1.3** · Predictions from a random sample Student code: _____ Calculator: yes

1. To predict how many of the 1200 students in the school would buy a yearbook, Lena asked the 29 members of the drama club, and 22 said yes. She predicts that about 910 of the 1200 students would buy a yearbook.

Why is this prediction not valid?

A. The sample is too small; asking a larger group of members of the drama club would make the prediction valid

B. The prediction should be 22 students, the number who said yes, not 910 students

C. A prediction about the school can only be made by asking every one of the 1200 students

D. The members of the drama club are not a random sample of all the students, so they may not represent the whole school

2. A town has 3400 voters. Based on a random sample of 400 voters, it is predicted that about 2550 of the 3400 voters support the new park.

What percent of the voters in the sample said they support the new park?

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor Student code: _____ Calculator: yes

Learning goal: I can solve problems with scale drawings and scale factors. (circle your level after scoring)

4	I can explain how the scale factor changes perimeter versus area and use that in a real-world problem.
3	I can find actual lengths, perimeters, and areas from a scale drawing or scale factor.
2	I can find a scale factor between two figures.
1	I need help to start this benchmark.

1. A rectangle measures 7 units by 10 units. It is enlarged by a scale factor of 2.

What is the area of the enlarged rectangle, in square units?

2. A photo is reduced by a scale factor of $\frac{1}{8}$. A side that measures 72 inches in the original is scaled down.

What is the length of the corresponding side in the reduction, in inches?

Turn over →

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor Student code: _____

3. Figure 2 is a scale drawing of Figure 1. A side that measures 2 centimeters in Figure 1 measures 12 centimeters in the corresponding position in Figure 2.

What scale factor was used to produce Figure 2 from Figure 1? Write the scale factor as a whole number or a fraction.

4. A model of a building uses the scale 1 centimeter : 5 meters. On the model, a doorway is 1.5 centimeters long.

How long is the actual doorway, in meters?

5. On a blueprint, the rectangular front wall of a gymnasium measures 4.5 inches long and 5 inches tall. The actual wall is 22.5 feet long.

What is the area of the actual wall, in square feet?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.1.5 · MA.7.GR.1.5 · Scale drawings and scale factor

Unit 5: Scaling With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **280 square units**

The enlarged rectangle measures 14 by 20, so its area is 280 square units.

That is $2^2 = 4$ times the original area of 70.

2. Regular **9 inches**

Multiply by the scale factor: $72 \times \frac{1}{8} = 9$ inches.

3. Regular **6**

The scale factor is the constant of proportionality between corresponding lengths: new length ÷ original length.

$12 \div 2 = 6$.

The factor is greater than 1, so Figure 2 is an enlargement.

4. Regular **7.5 meters**

Every centimeter on the model stands for 5 meters, so multiply: $1.5 \times 5 = 7.5$ meters.

5. Rigor **562.5 square feet**

Find the scale factor between the blueprint and the wall: $22.5 \div 4.5 = 5$ feet for each inch.

Actual height: $5 \times 5 = 25$ feet.

Area: $22.5 \times 25 = 562.5$ square feet.

Re-check strip (3 items, C-GR.1.5-R)

1. Regular **14 centimeters**

Multiply the length by the scale factor: $4 \times 3.5 = 14$ centimeters.

2. Regular **B. 64**

Every side is multiplied by 4, and a perimeter is a sum of side lengths, so the perimeter is also multiplied by 4: $16 \times 4 = 64$ centimeters.

Only an area is multiplied by $4^2 = 16$.

3. Regular **4**

The scale factor is the constant of proportionality between corresponding lengths: new length ÷ original length.

$12 \div 3 = 4$.

The factor is greater than 1, so Figure 2 is an enlargement.

Comparison strip (2 items, C-GR.1.5-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **C. 252**

Every side is multiplied by 6, and a perimeter is a sum of side lengths, so the perimeter is also multiplied by 6: $42 \times 6 = 252$ centimeters.

Only an area is multiplied by $6^2 = 36$.

2. Regular **32 centimeters**

Multiply the length by the scale factor: $8 \times 4 = 32$ centimeters.

Reteach plan — C-GR.1.5

MA.7.GR.1.5 · Scale drawings and scale factor

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.1.5-R

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor Student code: _____ Calculator: yes

1. A drawing is enlarged by a scale factor of 3.5. A side that measures 4 centimeters in the drawing is scaled up.

What is the length of the corresponding side in the enlargement, in centimeters?

2. A rhombus has a perimeter of 16 centimeters. It is enlarged by a scale factor of 4.

What is the perimeter of the enlarged rhombus, in centimeters?

A. 20
B. 64
C. 256
D. 4

3. Figure 2 is a scale drawing of Figure 1. A side that measures 3 centimeters in Figure 1 measures 12 centimeters in the corresponding position in Figure 2.

What scale factor was used to produce Figure 2 from Figure 1? Write the scale factor as a whole number or a fraction.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-GR.1.5-R

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor Student code: _____ Calculator: yes

1. A drawing is enlarged by a scale factor of 3.5. A side that measures 4 centimeters in the drawing is scaled up.

What is the length of the corresponding side in the enlargement, in centimeters?

2. A rhombus has a perimeter of 16 centimeters. It is enlarged by a scale factor of 4.

What is the perimeter of the enlarged rhombus, in centimeters?

A. 20
B. 64
C. 256
D. 4

3. Figure 2 is a scale drawing of Figure 1. A side that measures 3 centimeters in Figure 1 measures 12 centimeters in the corresponding position in Figure 2.

What scale factor was used to produce Figure 2 from Figure 1? Write the scale factor as a whole number or a fraction.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-GR.1.5-X

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor

Student code: _____ Calculator: yes

1. A trapezoid has a perimeter of 42 centimeters. It is enlarged by a scale factor of 6.

What is the perimeter of the enlarged trapezoid, in centimeters?

A. 1512

B. 48

C. 252

D. 7

2. A drawing is enlarged by a scale factor of 4. A side that measures 8 centimeters in the drawing is scaled up.

What is the length of the corresponding side in the enlargement, in centimeters?

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-GR.1.5-X

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor

Student code: _____ Calculator: yes

1. A trapezoid has a perimeter of 42 centimeters. It is enlarged by a scale factor of 6.

What is the perimeter of the enlarged trapezoid, in centimeters?

A. 1512

B. 48

C. 252

D. 7

2. A drawing is enlarged by a scale factor of 4. A side that measures 8 centimeters in the drawing is scaled up.

What is the length of the corresponding side in the enlargement, in centimeters?

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

Learning goal: I can convert units using proportional reasoning. (circle your level after scoring)

4	I can solve a multi-step conversion problem across measurement systems and check the result.
3	I can convert between measurement systems for length, area, weight or mass, volume, or money, using a conversion factor.
2	I can set up a ratio between two units.
1	I need help to start this benchmark.

1.

$1 \text{ kilogram} \approx 2.2 \text{ pounds}.$

A suitcase weighs 40 pounds. What is its mass in kilograms, to the nearest tenth?

A. 1.8 kilograms

B. 37.8 kilograms

C. 18.2 kilograms

D. 88.0 kilograms

2. On one day, 1 U.S. dollar could be exchanged for 0.9 Swiss francs.

Maya exchanged $100.00 for Swiss francs and then spent 31.25 Swiss francs on a meal. How many Swiss francs did Maya have left?

Turn over →

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____

3.

$1 \text{ mile} \approx 1.609 \text{ kilometers}$.

A race course is 36 kilometers long. How many miles is that? Round to the nearest tenth.

4. $1 \text{ hour} = 60 \text{ minutes}$, and $1 \text{ gallon} \approx 3.785 \text{ liters}$.

A pump moves 20 liters of water each minute. It runs for 3 hours 15 minutes. How many gallons does it move? Round to the nearest tenth.

5. Store A sells cheddar cheese for $7.29 per pound. Store B sells cheddar cheese for $17.05 per kilogram.

$1 \text{ kilogram} \approx 2.2 \text{ pounds}$.

Which store charges less per pound, and how much less per pound?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.3.3 · MA.7.AR.3.3 · Unit conversions across measurement systems

Unit 6: Multi-Step Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **C. 18.2 kilograms**

Each kilogram is about 2.2 pounds, so there are fewer kilograms than pounds: divide by 2.2.

$40 \div 2.2 \approx 18.182$, which rounds to 18.2 kilograms.

2. Regular **58.75 Swiss francs**

Each dollar becomes 0.9 Swiss francs: $100 \times 0.9 = 90.00$ Swiss francs.

Subtract the meal: $90.00 - 31.25 = 58.75$ Swiss francs.

3. Regular **22.4 miles**

Each mile is about 1.609 kilometers, so divide the kilometers by 1.609: $36 \div 1.609 \approx 22.374$.

Rounded to the nearest tenth: 22.4 miles.

There are fewer miles than kilometers, so the answer must be smaller than 36.

4. Regular **1030.4 gallons**

Convert the time to minutes: $3 \times 60 + 15 = 195$ minutes.

Liters moved: $20 \times 195 = 3900$ liters.

Liters to gallons: $3900 \div 3.785 \approx 1030.383$, which rounds to 1030.4 gallons.

5. Rigor **Store A, by $0.46 per pound. Score: 1 point for the store and the amount per pound; the store alone earns 0.**

Store B's price is for about 2.2 pounds, so divide by 2.2 to get the price per pound: $\$17.05 \div 2.2 = \$7.75$ per pound.

Store A charges $7.29 per pound.

$\$7.75 - \$7.29 = \$0.46$, so Store A charges $0.46 less per pound.

Re-check strip (3 items, C-AR.3.3-R)

1. Regular **D. 26.4 kilograms**

Each kilogram is about 2.2 pounds, so there are fewer kilograms than pounds: divide by 2.2.

$58 \div 2.2 \approx 26.364$, which rounds to 26.4 kilograms.

2. Regular **69.75 euros**

Each dollar becomes 0.88 euros: $100 \times 0.88 = 88.00$ euros.

Subtract the meal: $88.00 - 18.25 = 69.75$ euros.

3. Regular **30.6 gallons**

Each gallon is about 3.785 liters, so divide the liters by 3.785: $116 \div 3.785 \approx 30.647$.

Rounded to the nearest tenth: 30.6 gallons.

Comparison strip (2 items, C-AR.3.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B. 12.7 kilograms**

Each kilogram is about 2.2 pounds, so there are fewer kilograms than pounds: divide by 2.2.

$28 \div 2.2 \approx 12.727$, which rounds to 12.7 kilograms.

2. Regular **96.5 kilometers (96.6 also accepted if the factor was rounded to 1.61)**

Convert minutes to hours: $75 \div 60 = 1.25$ hours.

Distance: $48 \times 1.25 = 60$ miles.

Miles to kilometers: $60 \times 1.609 \approx 96.540$, which rounds to 96.5 kilometers.

Reteach plan — C-AR.3.3

MA.7.AR.3.3 · Unit conversions across measurement systems

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items)

C-AR.3.3-R

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

1.

1 kilogram $\approx$ 2.2 pounds.

A suitcase weighs 58 pounds. What is its mass in kilograms, to the nearest tenth?

A. 55.8 kilograms

B. 127.6 kilograms

C. 2.6 kilograms

D. 26.4 kilograms

2. On one day, 1 U.S. dollar could be exchanged for 0.88 euros.

Maya exchanged $100.00 for euros and then spent 18.25 euros on a meal. How many euros did Maya have left?

3.

1 gallon $\approx$ 3.785 liters.

A water heater holds 116 liters. How many gallons is that? Round to the nearest tenth.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items)

C-AR.3.3-R

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

1.

1 kilogram $\approx$ 2.2 pounds.

A suitcase weighs 58 pounds. What is its mass in kilograms, to the nearest tenth?

A. 55.8 kilograms

B. 127.6 kilograms

C. 2.6 kilograms

D. 26.4 kilograms

2. On one day, 1 U.S. dollar could be exchanged for 0.88 euros.

Maya exchanged $100.00 for euros and then spent 18.25 euros on a meal. How many euros did Maya have left?

3.

1 gallon $\approx$ 3.785 liters.

A water heater holds 116 liters. How many gallons is that? Round to the nearest tenth.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.3-X

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

1.

$1 \text{ kilogram} \approx 2.2 \text{ pounds}.$

A suitcase weighs 28 pounds. What is its mass in kilograms, to the nearest tenth?

A. 25.8 kilograms

B. 12.7 kilograms

C. 61.6 kilograms

D. 1.3 kilograms

2. $1 \text{ hour} = 60 \text{ minutes}$, and $1 \text{ mile} \approx 1.609 \text{ kilometers}$.

A car travels at 48 miles per hour for 75 minutes. How many kilometers does it travel? Round to the nearest tenth.

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.3-X

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

1.

$1 \text{ kilogram} \approx 2.2 \text{ pounds}.$

A suitcase weighs 28 pounds. What is its mass in kilograms, to the nearest tenth?

A. 25.8 kilograms

B. 12.7 kilograms

C. 61.6 kilograms

D. 1.3 kilograms

2. $1 \text{ hour} = 60 \text{ minutes}$, and $1 \text{ mile} \approx 1.609 \text{ kilometers}$.

A car travels at 48 miles per hour for 75 minutes. How many kilometers does it travel? Round to the nearest tenth.

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____ Calculator: yes

Learning goal: I can set up and solve proportions for real-world problems. (circle your level after scoring)

4	I can solve a multi-step problem with proportions and explain why the setup makes sense.
3	I can write a proportion from a situation and solve it.
2	I can tell whether two ratios are equivalent.
1	I need help to start this benchmark.

1. In a trail mix, the ratio of peanuts to raisins is $28{:}16$. A hiker uses 20 ounces of raisins.

Write a proportion that can be used to find p, the number of ounces of peanuts that keeps the same ratio. Then solve the proportion.

2. A drip line uses 3.6 gallons of water every 4 hours.

How many gallons of water does the drip line use in 9 hours?

Turn over →

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____

3. The table shows what different amounts of ground beef cost at one store. The cost is proportional to the number of pounds.

Pounds of ground beef	Cost
2	$8.88
2.5	$11.10
3	$13.32
4.5	$19.98

At this store, what will 6 pounds of ground beef cost?

4. A recipe uses $\frac{1}{2}$ cup of oil for every 2 dozen muffins. The amounts are proportional.

How much oil is needed for 5 dozen muffins? Write your answer as a fraction or a mixed number in simplest form.

5. Tomás usually runs 10-kilometer races, and 10 kilometers is about 6.2 miles. Tomás is training for a trail race, which is 15 miles. Tomás runs at a steady pace of 6 minutes per kilometer.

About how many kilometers is the trail race, to the nearest tenth? About how long will Tomás take to run it at that pace, to the nearest minute?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.3.2 · MA.7.AR.3.2 · Solve problems involving proportions

Unit 6: Multi-Step Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\frac{28}{16} = 1\frac{12}{16} = \frac{p}{20}$ **(or an equivalent proportion); $p = 35$. Score: 1 point for the proportion (or an equivalent proportion) and its solution; the solution alone earns 0.**

Keep peanuts over raisins on both sides: $\frac{28}{16} = \frac{p}{20}$.

$28{:}16$ in simplest form is $7{:}4$, and $20 \div 4 = 5$, so both parts are multiplied by 5: $p = 7 \times 5 = 35$.

Check by cross-multiplying: $28 \times 20 = 560$ and $16 \times 35 = 560$.

2. Regular **8.1 gallons**

Unit rate: $3.6 \div 4 = 0.9$ gallons per hour.

$9 \times 0.9 = 8.1$.

3. Regular **$26.64**

The cost is proportional to the weight, so find the cost of one pound.

$\$8.88 \div 2 = \$4.44$ for one pound.

$\$4.44 \times 6 = \$26.64$.

4. Regular $1\frac{1}{4} = \frac{5}{4}$ **cups**

Oil for 1 dozen: $\frac{1}{2} \div 2 = \frac{1}{4}$ cup.

Oil for 5 dozen: $5 \times \frac{1}{4} = 1\frac{1}{4}$ cups.

5. Rigor **about 24.2 kilometers; about 145 minutes (2 hours 25 minutes). Score: 1 point for the distance and the time; one alone earns 0.**

Kilometers: set up a proportion with kilometers over miles, $10/6.2 = \frac{k}{15}$.

$k = 15 \times 10 \div 6.2 \approx 24.194$, which rounds to 24.2 kilometers.

Time: the pace is a second rate, 6 minutes for each kilometer, so the time is about $24.194 \times 6 \approx 145.16$ minutes, which rounds to 145 minutes (2 hours 25 minutes).

Using the rounded 24.2 kilometers gives 145.20, which rounds the same way.

Re-check strip (3 items, C-AR.3.2-R)

1. Regular **B. 11.25 ft**

The tree's shadow is $20 \div 8 = 2.5$ times as long as the person's.

The tree is 2.5 times as tall: $4.5 \times 2.5 = 11.25\,\text{ft}$.

2. Regular **B. 37.50 km**

Scale: $15 \div 4 = 3.75\,\text{km}$ for every 1 cm.

$10 \times 3.75 = 37.50\,\text{km}$.

Adding 6 cm to 15 km treats the scale as additive; $4 \times 10 \div 15$ sets up the ratio upside down.

3. Regular **17.5 gallons**

Unit rate: $5.25 \div 3 = 1.75$ gallons per hour.

$10 \times 1.75 = 17.5$.

Comparison strip (2 items, C-AR.3.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $\frac{3}{4}$ **cup**

Oil for 1 dozen: $\frac{1}{3} \div 4 = \frac{1}{12}$ cup.

Oil for 9 dozen: $9 \times \frac{1}{12} = \frac{3}{4}$ cup.

2. Regular **C. 9.60 km**

Scale: $8 \div 5 = 1.6\,\text{km}$ for every 1 cm.

$6 \times 1.6 = 9.60\,\text{km}$.

Adding 1 cm to 8 km treats the scale as additive; $5 \times 6 \div 8$ sets up the ratio upside down.

Reteach plan — C-AR.3.2

MA.7.AR.3.2 · Solve problems involving proportions

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

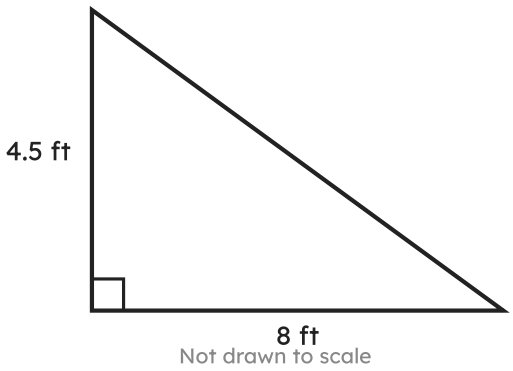
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.3.2-R

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____ Calculator: yes

1. A 4.5-foot person casts an 8-foot shadow. At the same time a tree casts a 20-foot shadow.

How tall is the tree, in feet?



Not drawn to scale

A. 35.56 ft

B. 11.25 ft

C. 4.44 ft

D. 16.50 ft

2. On a map, 4 cm represents 15 km.

How many kilometers does 10 cm represent?

A. 21.00 km

B. 37.50 km

C. 2.67 km

D. 150.00 km

3. A drip line uses 5.25 gallons of water every 3 hours.

How many gallons of water does the drip line use in 10 hours?

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-AR.3.2-X

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____ Calculator: yes

1. A recipe uses $\frac{1}{3}$ cup of oil for every 4 dozen muffins. The amounts are proportional.

How much oil is needed for 9 dozen muffins? Write your answer as a fraction or a mixed number in simplest form.

2. On a map, 5 cm represents 8 km.

How many kilometers does 6 cm represent?

A. 3.75 km

B. 48.00 km

C. 9.60 km

D. 9.00 km

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.2-X

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____ Calculator: yes

1. A recipe uses $\frac{1}{3}$ cup of oil for every 4 dozen muffins. The amounts are proportional.

How much oil is needed for 9 dozen muffins? Write your answer as a fraction or a mixed number in simplest form.

2. On a map, 5 cm represents 8 km.

How many kilometers does 6 cm represent?

A. 3.75 km

B. 48.00 km

C. 9.60 km

D. 9.00 km

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems Student code: _____ Calculator: yes

Learning goal: I can solve multi-step problems with proportional relationships. (circle your level after scoring)

4	I can compare two proportional situations and justify which is the better choice.
3	I can solve a real-world problem with a proportional relationship in two or more steps.
2	I can find a unit rate.
1	I need help to start this benchmark.

1. A horse-drawn carriage travels at a constant rate. It travels $\frac{1}{2}$ mile in 10 minutes.

At this rate, how many minutes will it take the carriage to travel $2\frac{3}{4}$ miles?

2. A 3-D printer uses $\frac{3}{5}$ ounce of plastic every $\frac{1}{3}$ hour, at a constant rate.

How many ounces of plastic does the printer use in 12 hours? Write your answer as a fraction or a mixed number in simplest form.

3. A store sells ribbon at a constant price. $4\frac{1}{4}$ meters of ribbon cost $15.30.

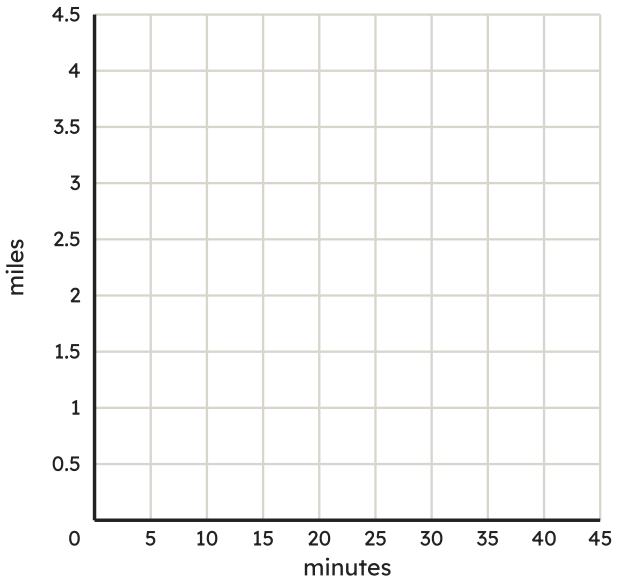
At that rate, what is the cost of $7\frac{1}{2}$ meters of ribbon?

Turn over →

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems Student code: _____

4. A carriage travels at a constant rate. After 5 minutes, it has traveled $\frac{1}{2}$ mile. The relationship between minutes and miles is proportional.

Graph the relationship on the grid. Then use the graph to find how many minutes it takes the carriage to complete a 4-mile ride.



5. Store A sells rope for \$6.30 for $2\frac{1}{4}$ meters. Store B sells the same rope for \$13.75 for 5 meters. At each store, the cost is proportional to the amount bought.

Which store is the better buy? Explain, and tell how much lower its price is per meter.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.4.5 · MA.7.AR.4.5 · Real-world proportional problems

Unit 6: Multi-Step Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **55 minutes**

Minutes for 1 mile: $10 \div \frac{1}{2} = 10 \times 2 = 20$ minutes.

Minutes for $2\frac{3}{4}$ miles: $2\frac{3}{4} \times 20 = 55$ minutes.

2. Regular $21\frac{3}{5} = \frac{108}{5}$ **ounces**

Unit rate: $\frac{3}{5} \div \frac{1}{3} = \frac{3}{5} \times 3 = 1\frac{4}{5}$ ounces per hour.

In 12 hours: $12 \times 1\frac{4}{5} = 21\frac{3}{5}$ ounces.

3. Regular **$27.00**

Write the lengths as decimals: $4\frac{1}{4}$ meters is 4.25 meters and $7\frac{1}{2}$ meters is 7.5 meters.

Unit price: $\$15.30 \div 4.25 = \$3.60$ per meter.

Cost: $7.5 \times \$3.60 = \$27.00$.

4. Regular **A straight line through $(0,\ 0)$ and $(5,\ 0.5)$; the 4-mile ride takes 40 minutes. Score: 1 point for the graph and the minutes; the graph alone earns 0.**

Plot $(0,\ 0)$ and $\left(5,\ \frac{1}{2}\right)$ and draw the straight line through them.

The carriage takes $5 \div \frac{1}{2} = 5 \times 2 = 10$ minutes per mile, so the line passes through $(40,\ 4)$: $4 \times 10 = 40$.

Read across from 4 on the miles axis to the line, then down to 40 minutes.

5. Rigor **Store B; its price per meter is lower, $2.75 compared with $2.80, so each meter costs less there; lower by $0.05 per meter. Score: 1 point for the store with the unit-price explanation and the amount per meter; the store alone earns 0.**

Store A: $\$6.30 \div 2.25 = \$2.80$ per meter.

Store B: $\$13.75 \div 5 = \$2.75$ per meter.

$\$2.80 - \$2.75 = \$0.05$, so Store B has the lower price, by $0.05 per meter.

Re-check strip (3 items, C-AR.4.5-R)

1. Regular **A straight line through $(0,\ 0)$ and $(12,\ 0.75)$; the 2-mile ride takes 32 minutes. Score: 1 point for the graph and the minutes; the graph alone earns 0.**

Plot $(0,\ 0)$ and $\left(12,\ \frac{3}{4}\right)$ and draw the straight line through them.

The carriage takes $12 \div \frac{3}{4} = 12 \times \frac{4}{3} = 16$ minutes per mile, so the line passes through $(32,\ 2)$: $2 \times 16 = 32$.

Read across from 2 on the miles axis to the line, then down to 32 minutes.

2. Regular **24 minutes**

Minutes for 1 mile: $8 \div \frac{2}{3} = 8 \times \frac{3}{2} = 12$ minutes.

Minutes for 2 miles: $2 \times 12 = 24$ minutes.

3. Regular **9 miles**

Unit rate: $\frac{1}{2} \div \frac{1}{6} = \frac{1}{2} \times 6 = 3$ miles per hour.

In 3 hours: $3 \times 3 = 9$ miles.

Comparison strip (2 items, C-AR.4.5-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **D. 10.5 gallons**

Unit rate: $85 \div 2.5 = 34$ miles per gallon.

$357 \div 34 = 10.5$ gallons.

$2.5 \times 85 \div 357$ turns the ratio upside down, $357 \div 2.5$ treats gallons as miles per gallon, and $2.5 + 357 \div 85$ adds the scale factor instead of multiplying by it.

2. Regular **$16.20**

Write the lengths as decimals: $3\frac{3}{4}$ meters is 3.75 meters and $6\frac{3}{4}$ meters is 6.75 meters.

Unit price: $\$9.00 \div 3.75 = \$2.40$ per meter.

Cost: $6.75 \times \$2.40 = \$16.20$.

Reteach plan — C-AR.4.5

MA.7.AR.4.5 · Real-world proportional problems

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

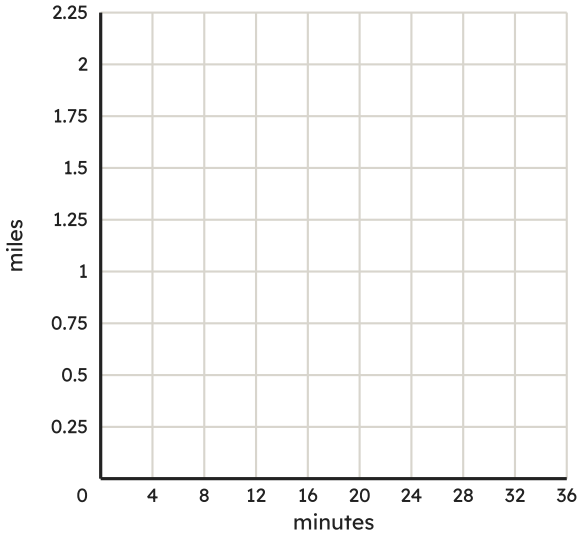
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.4.5-R

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems Student code: _____ Calculator: yes

1. A carriage travels at a constant rate. After 12 minutes, it has traveled $\frac{3}{4}$ mile. The relationship between minutes and miles is proportional.

Graph the relationship on the grid. Then use the graph to find how many minutes it takes the carriage to complete a 2-mile ride.



2. A horse-drawn carriage travels at a constant rate. It travels $\frac{2}{3}$ mile in 8 minutes.

At this rate, how many minutes will it take the carriage to travel 2 miles?

3. A hiker walks $\frac{1}{2}$ mile every $\frac{1}{6}$ hour, at a constant rate.

How many miles does the hiker walk in 3 hours?

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2)

C-AR.4.5-X

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems

Student code: _____ Calculator: yes

1. A car uses gas at a constant rate. It travels 85 miles on 2.5 gallons of gas.

How many gallons of gas are needed to travel 357 miles?

A. 142.8 gallons

B. 6.7 gallons

C. 0.6 gallons

D. 10.5 gallons

2. A store sells ribbon at a constant price. $3\frac{3}{4}$ meters of ribbon cost $9.00.

At that rate, what is the cost of $6\frac{3}{4}$ meters of ribbon?

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-AR.4.5-X

Benchmark: **MA.7.AR.4.5** · Real-world proportional problems

Student code: _____ Calculator: yes

1. A car uses gas at a constant rate. It travels 85 miles on 2.5 gallons of gas.

How many gallons of gas are needed to travel 357 miles?

A. 142.8 gallons

B. 6.7 gallons

C. 0.6 gallons

D. 10.5 gallons

2. A store sells ribbon at a constant price. $3\frac{3}{4}$ meters of ribbon cost $9.00.

At that rate, what is the cost of $6\frac{3}{4}$ meters of ribbon?

Score: ____ **/ 2** → **copy to your progress strip**

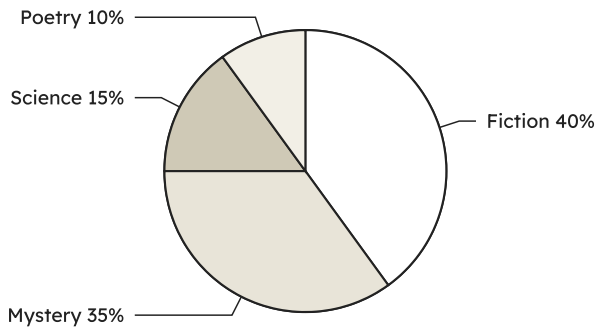
Benchmark: **MA.7.DP.1.4** · Circle graphs Student code: _____ Calculator: yes

Learning goal: I can create and interpret circle graphs. (circle your level after scoring)

4	I can construct a circle graph from data and use it to answer a comparison question.
3	I can find the sector angle or percent for each category and interpret a circle graph.
2	I can read a percent or category from a circle graph.
1	I need help to start this benchmark.

1. The circle graph shows how 540 people answered a survey.

How many more people chose Fiction than Science?



2. A survey asked 320 people which pet they would most like to have. The results are shown in the table.

Dogs	Cats	Fish	Birds
128	112	32	48

In a circle graph of the data, what percent of the circle is the Fish section?

3. 420 people were asked which color they like best. In a circle graph of the results, the Purple section has a central angle of 54°.

How many people chose Purple?

A. 15

B. 357

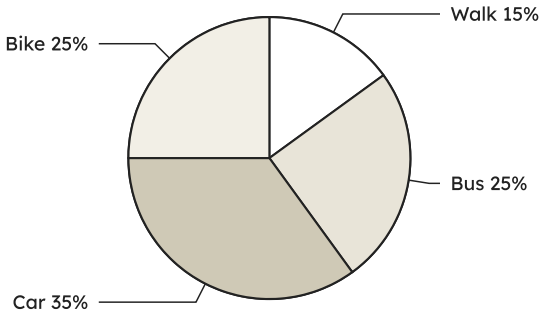
C. 63

D. 54

Turn over →

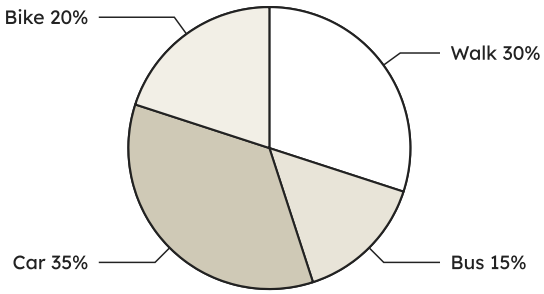
4. The circle graph shows how 740 students answered a survey.

How many students chose Bus?



5. 360 students were asked how they get to school. The circle graph shows the results.

Select **all** the statements that the circle graph supports.



Select **all** the correct answers.

A. ☐ Walk was chosen by twice as many students as Car.

B. ☐ Bus and Bike together were chosen by more than half of the students.

C. ☐ 18 more students chose Car than Walk.

D. ☐ The Bike section has a central angle of 20°.

E. ☐ 54 of the 360 students chose Bus.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.4 · MA.7.DP.1.4 · Circle graphs

Unit 7: Percentage Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **135 more people**

Fiction: $0.4 \times 540 = 216$ people.

Science: $0.15 \times 540 = 81$ people.

$216 - 81 = 135$ more people chose Fiction.

2. Regular **10%**

Fish was chosen by 32 of the 320 people: $\frac{32}{320} = \frac{1}{10} = 10\%$.

The Fish section is 10% of the circle.

3. Regular **C. 63**

The section is 54° of the whole circle, 360°: $\frac{54}{360} = \frac{3}{20} = 15\%$.

So 15% of the 420 people chose Purple: $0.15 \times 420 = 63$ people.

4. Regular **185 students**

Bus is 25% of the circle.

$0.25 \times 740 = 185$ students.

5. Rigor **C. 18 more students chose Car than Walk; E. 54 of the 360 students chose Bus.**

Counts: Walk $0.3 \times 360 = 108$; Bus $0.15 \times 360 = 54$; Car $0.35 \times 360 = 126$; Bike $0.2 \times 360 = 72$.

Central angles: Walk $30\% \text{ of } 360^\circ = 108^\circ$; Bus $15\% \text{ of } 360^\circ = 54^\circ$; Car $35\% \text{ of } 360^\circ = 126^\circ$; Bike $20\% \text{ of } 360^\circ = 72^\circ$.

True: 54 of the 360 students chose Bus. 18 more students chose Car than Walk.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Re-check strip (3 items, C-DP.1.4-R)

1. Regular **36°**

Bus was chosen by 22 of the 220 students: $\frac{22}{220} = \frac{1}{10} = 10\%$.

A whole circle is 360°, so the central angle is $10\% \text{ of } 360^\circ = 0.1 \times 360 = 36^\circ$.

2. Regular **D. 60**

The section is 90° of the whole circle, 360°: $\frac{90}{360} = \frac{1}{4} = 25\%$.

So 25% of the 240 students chose Pizza: $0.25 \times 240 = 60$ students.

3. Regular **23 more people**

Summer: $0.2 \times 460 = 92$ people.

Spring: $0.15 \times 460 = 69$ people.

$92 - 69 = 23$ more people chose Summer.

Comparison strip (2 items, C-DP.1.4-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **15%**

$\frac{3}{20} = 15\%$ and $\frac{2}{5} = 40\%$.

A circle graph always totals 100%: $15 + 30 + 40 = 85$.

$100 - 85 = 15\%$.

2. Regular **C. 16**

The section is 36° of the whole circle, 360°: $\frac{36}{360} = \frac{1}{10} = 10\%$.

So 10% of the 160 students chose Music: $0.1 \times 160 = 16$ students.

Reteach plan — C-DP.1.4

MA.7.DP.1.4 · Circle graphs

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

1. A survey asked 220 students how they get to school. The results are shown in the table.

Walk	Bicycle	Car	Bus
44	110	44	22

In a circle graph of the data, what is the measure of the central angle of the Bus section?

2. 240 students were asked which lunch they like best. In a circle graph of the results, the Pizza section has a central angle of 90°.

How many students chose Pizza?

A. 25

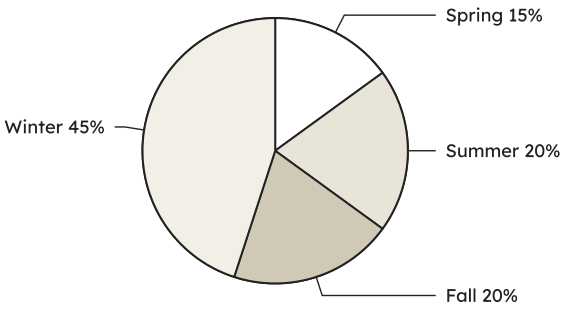
B. 90

C. 180

D. 60

3. The circle graph shows how 460 people answered a survey.

How many more people chose Summer than Spring?



Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.4-X

Benchmark: **MA.7.DP.1.4** · Circle graphs Student code: _____ Calculator: yes

1. A circle graph of survey results has four sections: Cats, Dogs, Birds and Fish. The Cats section is $\frac{3}{20}$ of the graph, the Dogs section is 30%, and the Birds section is $\frac{2}{5}$ of the graph. The percent for Fish is not given.

What percent of the circle graph is Fish?

2. 160 students were asked what they like to do after school. In a circle graph of the results, the Music section has a central angle of 36°.

How many students chose Music?

A. 36 **B.** 144

C. 16 **D.** 10

Score: ____ / 2 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.4-X

Benchmark: **MA.7.DP.1.4** · Circle graphs Student code: _____ Calculator: yes

1. A circle graph of survey results has four sections: Cats, Dogs, Birds and Fish. The Cats section is $\frac{3}{20}$ of the graph, the Dogs section is 30%, and the Birds section is $\frac{2}{5}$ of the graph. The percent for Fish is not given.

What percent of the circle graph is Fish?

2. 160 students were asked what they like to do after school. In a circle graph of the results, the Music section has a central angle of 36°.

How many students chose Music?

A. 36 **B.** 144

C. 16 **D.** 10

Score: ____ / 2 → copy to your progress strip

7th Grade Math · Weekly Check · Unit 7 **C-AR.3.1**

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems Student code: _____ Calculator: yes

Learning goal: I can solve percent problems: discount, markup, tax, tip, interest, percent change, percent error. (circle your level after scoring)

4	I can solve a multi-step percent problem, including percent change or percent error, and explain the result.
3	I can solve a one-step percent problem such as a discount, tax, tip, or simple interest.
2	I can find a percent of a number.
1	I need help to start this benchmark.

1. A video game costs $76.98 and is on sale for 12% off.

What is the sale price, to the nearest cent?

2. Devon deposits $3150.00 in an account that pays 3.5% simple interest each year. Devon leaves the money in the account for 3 years, then withdraws everything.

How much does Devon withdraw?

Turn over →

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems Student code: _____

3. The ratio of dogs to cats among the pets at a shelter is $11:9$. Every pet at the shelter is a dog or a cat.

What percent of the pets at the shelter are dogs? If there are 180 pets at the shelter, how many are dogs?

4. The number of visitors to a museum each month changed from 320 to 344.

What is the percent increase?

5. A store takes 28% off the original price of $162.50, then takes another 32% off the reduced price.

What is the final price?

A. $117.00

B. $110.50

C. $65.00

D. $79.56

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.3.1 · MA.7.AR.3.1 · Multi-step real-world percent problems

Unit 7: Percentage Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **$67.74**

Discount: $\$76.98 \times 0.12 = \$9.2376 \approx \$9.24$.

Sale price: $\$76.98 - \$9.24 = \$67.74$.

12% must be written as 0.12 before multiplying.

2. Regular **$3480.75**

Interest: $I = Prt = 3150 \times 0.035 \times 3 = \$330.75$.

Withdrawal is the principal plus the interest: $\$3150.00 + \$330.75 = \$3480.75$.

3. Regular **55%; 99 dogs. Score: 1 point for the percent and the count; one alone earns 0.**

For every 11 dogs there are 9 cats, so 11 of every 20 are dogs.

$11 \div 20 = 0.55$, which is 55%.

55% of $180 = 0.55 \times 180 = 99$ dogs.

4. Regular **7.5%**

Change: $344 - 320 = 24$, an increase of 24.

Compare the change with the ORIGINAL amount: $24 \div 320 = 0.075$.

$0.075 \times 100 = 7.5\%$.

5. Rigor **D. $79.56**

First discount: $\$162.50 \times 72\% = \$162.50 \times 0.72 = \$117.00$.

The second discount applies to $117.00, not to the original price: $\$117.00 \times 0.68 = \$79.56$.

Taking 28% and then 32% is not the same as taking $60\%(\$65.00)$, and the second discount is not taken from the original price $(\$110.50)$.

Re-check strip (3 items, C-AR.3.1-R)

1. Regular **32.5%**

Change: $162 - 240 = -78$, a decrease of 78.

Compare the change with the ORIGINAL amount: $78 \div 240 = 0.325$.

$0.325 \times 100 = 32.5\%$.

2. Regular **$92.51**

Tax: $\$86.46 \times 0.07 = \$6.0522 \approx \$6.05$.

Total: $\$86.46 + \$6.05 = \$92.51$.

3. Regular **6.25%**

Percent error compares the size of the error with the ACTUAL amount.

Error: $|285 - 304| = 19$.

$19 \div 304 = 0.0625$.

$0.0625 \times 100 = 6.25\%$.

Comparison strip (2 items, C-AR.3.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **$2706.00**

Interest: $I = Prt = 2400 \times 0.0425 \times 3 = \$306.00$.

Withdrawal is the principal plus the interest: $\$2400.00 + \$306.00 = \$2706.00$.

2. Regular **$58.89**

Tax: $\$54.78 \times 0.075 = \$4.1085 \approx \$4.11$.

Total: $\$54.78 + \$4.11 = \$58.89$.

Reteach plan — C-AR.3.1

MA.7.AR.3.1 · Multi-step real-world percent problems

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items)

C-AR.3.1-R

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems

Student code: _____ Calculator: yes

1. The number of visitors to a museum each month changed from 240 to 162.

What is the percent decrease?

2. A bike helmet costs $86.46. The sales tax rate is 7%.

What is the total cost, including tax, to the nearest cent?

3. A student estimated the number of steps on a trail to be 285. The actual number was 304 steps.

What is the percent error of the estimate?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items)

C-AR.3.1-R

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems

Student code: _____ Calculator: yes

1. The number of visitors to a museum each month changed from 240 to 162.

What is the percent decrease?

2. A bike helmet costs $86.46. The sales tax rate is 7%.

What is the total cost, including tax, to the nearest cent?

3. A student estimated the number of steps on a trail to be 285. The actual number was 304 steps.

What is the percent error of the estimate?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.1-X

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems Student code: _____ Calculator: yes

1. Andre deposits $2400.00 in an account that pays 4.25% simple interest each year. Andre leaves the money in the account for 3 years, then withdraws everything.

How much does Andre withdraw?

2. A tent costs $54.78. The sales tax rate is 7.5%.

What is the total cost, including tax, to the nearest cent?

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.1-X

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems Student code: _____ Calculator: yes

1. Andre deposits $2400.00 in an account that pays 4.25% simple interest each year. Andre leaves the money in the account for 3 years, then withdraws everything.

How much does Andre withdraw?

2. A tent costs $54.78. The sales tax rate is 7.5%.

What is the total cost, including tax, to the nearest cent?

Score: ____ **/ 2** → **copy to your progress strip**

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

Learning goal: I can apply the laws of exponents with whole-number exponents. (circle your level after scoring)

4	I can combine several exponent rules in one expression and explain each step.
3	I can use the product, quotient, and power rules to write an equivalent expression.
2	I can evaluate a power with a whole-number exponent.
1	I need help to start this benchmark.

1. Evaluate the expression. Write your answer as a fraction in simplest form.

$$\left(-\frac{2}{3}\right)^3 \times \left(-\frac{2}{3}\right)^3$$

2. Which expression is equivalent to $4^5 \times 4^2$?

A. 4^7

B. 16^7

C. 4^3

D. 4^{10}

3. Evaluate the expression.

$$(-10)^6 \div (-10)^4$$

Turn over →

4. The equation shown is true.

$$(6^4)^n = 6^{16}$$

What is the value of n?

5. A student wrote $\left(\left(\frac{1}{2}\right)^5\right)^3 \div \left(\frac{1}{2}\right)^4$ as a single power.

Step 1: $\left(\frac{1}{2}\right)^8 \div \left(\frac{1}{2}\right)^4$

Step 2: $\left(\frac{1}{2}\right)^4$

In which step did the student make the first error? Write the expression correctly as a single power.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.1.1 · MA.7.NSO.1.1 · Laws of exponents (whole-number exponents)

Unit 8: Evaluating Numeric Expressions. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\frac{64}{729}$

Multiplying powers with the same base adds the exponents: $3 + 3 = 6$, so the expression equals $\left(-\frac{2}{3}\right)^6$.

$\left(-\frac{2}{3}\right)^6 = \frac{64}{729}$. A negative base to an even power is positive.

2. Regular **A.** 4^7

When multiplying powers with the same base, add the exponents.

$4^5 \times 4^2 = 4^7$.

3. Regular **100**

The bases are the same, so subtract the exponents: $6 - 4 = 2$.

$(-10)^2 = 100$.

A negative base raised to an even power is positive. Only the exponents change — the base stays −10.

4. Regular **4**

A power raised to a power multiplies the exponents, so $4 \times n = 16$.

$n = 16 \div 4 = 4$.

5. Rigor **Step 1; $\left(\frac{1}{2}\right)^{11}$. Score: 1 point for the step and the correct single power; the step alone earns 0.**

Step 1 adds 5 and 3; a power raised to a power multiplies them: $5 \times 3 = 15$.

Correct work: $\left(\left(\frac{1}{2}\right)^5\right)^3 = \left(\frac{1}{2}\right)^{15}$, and $\left(\frac{1}{2}\right)^{15} \div \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^{11}$.

Re-check strip (3 items, C-NSO.1.1-R)

1. Regular **243**

Multiplying powers with the same base adds the exponents: $3 + 2 = 5$, so the expression equals 3^5.

$3^5 = 243$.

2. Regular 9^8

A power raised to a power multiplies the exponents: $4 \times 2 = 8$.

So $(9^4)^2 = 9^8$.

Adding the exponents here would give 9^6; exponents are added only when powers are multiplied.

3. Regular **1000**

Dividing powers with the same base subtracts the exponents: $7 - 4 = 3$.

$10^3 = 1000$.

Only the exponents change — the base stays 10, it is not divided.

Comparison strip (2 items, C-NSO.1.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **5**

Write 8 as a power of 2: $8 = 2^3$.

Dividing powers with the same base subtracts the exponents, so $n - 2 = 3$.

$n = 3 + 2 = 5$.

2. Regular **625**

Dividing powers with the same base subtracts the exponents: $7 - 3 = 4$.

$5^4 = 625$.

Only the exponents change — the base stays 5, it is not divided.

Reteach plan — C-NSO.1.1

MA.7.NSO.1.1 · Laws of exponents (whole-number exponents)

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.1.1-R

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

1. Evaluate the expression.

$3^3 \times 3^2$

2. Write the expression as a single power.

$(9^4)^2$

3. Evaluate the expression.

$10^7 \div 10^4$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-NSO.1.1-R

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

1. Evaluate the expression.

$3^3 \times 3^2$

2. Write the expression as a single power.

$(9^4)^2$

3. Evaluate the expression.

$10^7 \div 10^4$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) — C-NSO.1.1-X

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

1. The equation shown is true.

$2^n \div 2^2 = 8$

What is the value of n?

2. Evaluate the expression.

$5^7 \div 5^3$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) — C-NSO.1.1-X

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

1. The equation shown is true.

$2^n \div 2^2 = 8$

What is the value of n?

2. Evaluate the expression.

$5^7 \div 5^3$

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

Learning goal: I can use order of operations with rational numbers. (circle your level after scoring)

4	I can find and explain an error in someone else's work with order of operations.
3	I can evaluate an expression with rational numbers, exponents, and grouping symbols.
2	I can use order of operations with whole numbers.
1	I need help to start this benchmark.

1. Evaluate the expression.

$4 \times (|4.57 - 8.3| - 0.8^2)$

2. What is the value of the expression?

$7.3 - [0.75 + (-0.5 - 2)^2] \div 0.5$

A. 0.60

B. −6.70

C. 19.05

D. 18.30

Turn over →

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____

3. Evaluate the expression. Write your answer as a fraction in simplest form.

$$9 - 2[\left(1\frac{1}{2} - 2\right)^2 + \frac{1}{2} \div 6]$$

4. What is the value of the expression? Write your answer as a fraction in simplest form.

$$[\left(2\frac{1}{2} - 3\right)^2 + |\frac{3}{8} - 4|] \times 3$$

5. Where should one pair of parentheses be placed so that the expression has a value of 147.75?

$$-8.5 + 3 \times 5 - 2.5^2$$

A. $-8.5 + 3 \times (5 - 2.5^2)$

B. $-8.5 + 3 \times (5 - 2.5)^2$

C. $-8.5 + (3 \times 5 - 2.5)^2$

D. $(-8.5 + 3) \times 5 - 2.5^2$

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.2.1 · MA.7.NSO.2.1 · Order of operations with rational numbers

Unit 8: Evaluating Numeric Expressions. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **12.36**

   Inside the absolute value: $4.57 - 8.3 = -3.73$, so $|4.57 - 8.3| = 3.73$.
   The exponent: $0.8^2 = 0.64$.
   Inside the parentheses: $3.73 - 0.64 = 3.09$.
   Multiply: $4 \times 3.09 = 12.36$.

2. Regular **B. −6.70**

   Innermost grouping first: $-0.5 - 2 = -2.5$.
   The exponent: $(-2.5)^2 = 6.25$.
   Inside the brackets: $0.75 + 6.25 = 7$.
   Divide: $7 \div 0.5 = 14$.
   Subtract: $7.3 - 14 = -6.7$.
   Subtracting before dividing gives 0.60; dropping the brackets gives 19.05.

3. Regular $8\frac{1}{3} = \frac{25}{3}$

   Inside the parentheses: $1\frac{1}{2} - 2 = -\frac{1}{2}$.
   The exponent: $\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$.
   Divide: $\frac{1}{2} \div 6 = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.
   Inside the brackets: $\frac{1}{4} \div \frac{1}{12} = \frac{1}{3}$.
   Multiply: $2 \times \frac{1}{3} = \frac{2}{3}$.
   Subtract: $9 - \frac{2}{3} = 8\frac{1}{3}$.

4. Regular $11\frac{5}{8} = \frac{93}{8}$

   Inside the parentheses: $2\frac{1}{2} - 3 = -\frac{1}{2}$.
   The exponent: $\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$.
   Inside the absolute value: $\frac{3}{8} - 4 = -3\frac{5}{8}$, so $\left|\frac{3}{8} - 4\right| = 3\frac{5}{8}$.
   Inside the brackets: $\frac{1}{4} + 3\frac{5}{8} = 3\frac{7}{8}$.
   Multiply: $3\frac{7}{8} \times 3 = 11\frac{5}{8}$.

5. Rigor **C.** $-8.5 + (3 \times 5 - 2.5)^2$

   Evaluate each choice.
   $(-8.5 + 3) \times 5 - 2.5^2 = -33.75$
   $-8.5 + 3 \times (5 - 2.5)^2 = 10.25$
   $-8.5 + (3 \times 5 - 2.5)^2 = 147.75$
   $-8.5 + 3 \times (5 - 2.5^2) = -12.25$
   Without parentheses the expression is 0.25. Only $-8.5 + (3 \times 5 - 2.5)^2$ has the value 147.75.

Re-check strip (3 items, C-NSO.2.1-R)

1. Regular $4\frac{5}{8} = \frac{37}{8}$

   Inside the parentheses: $1\frac{1}{4} - 2 = -\frac{3}{4}$.
   The exponent: $\left(-\frac{3}{4}\right)^2 = \frac{9}{16}$.
   Divide: $\frac{1}{4} \div 2 = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$.
   Inside the brackets: $\frac{9}{16} + \frac{1}{8} = \frac{11}{16}$.
   Multiply: $2 \times \frac{11}{16} = 1\frac{3}{8}$.
   Subtract: $6 - 1\frac{3}{8} = 4\frac{5}{8}$.

2. Regular **B. −16.30**

   Innermost grouping first: $1.5 - 3 = -1.5$.
   The exponent: $(-1.5)^2 = 2.25$.
   Inside the brackets: $3.75 + 2.25 = 6$.
   Divide: $6 \div 0.5 = 12$.
   Subtract: $-4.3 - 12 = -16.3$.
   Subtracting before dividing gives −20.60; dropping the brackets gives −3.55.

3. Regular $2\frac{5}{8} = \frac{21}{8}$ **(or 2.625)**

   Write the decimal as a fraction: $1.75 = 1\frac{3}{4}$.
   Inside the parentheses: $-2\frac{1}{2} - 1 = -3\frac{1}{2}$.
   The exponent: $\left(-3\frac{1}{2}\right)^2 = 12\frac{1}{4}$. A negative number squared is positive.
   Inside the brackets: $12\frac{1}{4} - 1\frac{3}{4} = 10\frac{1}{2}$.
   Divide: $10\frac{1}{2} \div 4 = 2\frac{5}{8}$.

Comparison strip (2 items, C-NSO.2.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **31.25**

   Inside the absolute value: $4.6 - 7.8 = -3.2$, so $|4.6 - 7.8| = 3.2$.
   Inside the parentheses: $5.7 - 3.2 = 2.5$.
   The exponent: $(2.5)^3 = 15.625$.
   Divide: $15.625 \div 0.5 = 31.25$.

2. Regular $2\frac{1}{4} = \frac{9}{4}$ **(or 2.25)**

   Write the decimal as a fraction: $1.75 = 1\frac{3}{4}$.
   Inside the parentheses: $-1\frac{1}{2} - 1 = -2\frac{1}{2}$.
   The exponent: $\left(-2\frac{1}{2}\right)^2 = 6\frac{1}{4}$. A negative number squared is positive.
   Inside the brackets: $6\frac{1}{4} - 1\frac{3}{4} = 4\frac{1}{2}$.
   Divide: $4\frac{1}{2} \div 2 = 2\frac{1}{4}$.

Reteach plan — C-NSO.2.1

MA.7.NSO.2.1 · Order of operations with rational numbers

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.2.1-R

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

1. Evaluate the expression. Write your answer as a fraction in simplest form.

$$6 - 2[\left(1\frac{1}{4} - 2\right)^2 + \frac{1}{4} \div 2]$$

2. What is the value of the expression?

$$-4.3 - [3.75 + (1.5 - 3)^2] \div 0.5$$

A. −7.30 **B.** −16.30

C. −3.55 **D.** −20.60

3. Evaluate the expression.

$$[\left(-2\frac{1}{2} - 1\right)^2 - 1.75] \div 4$$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-NSO.2.1-R

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

1. Evaluate the expression. Write your answer as a fraction in simplest form.

$$6 - 2[\left(1\frac{1}{4} - 2\right)^2 + \frac{1}{4} \div 2]$$

2. What is the value of the expression?

$$-4.3 - [3.75 + (1.5 - 3)^2] \div 0.5$$

A. −7.30 **B.** −16.30

C. −3.55 **D.** −20.60

3. Evaluate the expression.

$$[\left(-2\frac{1}{2} - 1\right)^2 - 1.75] \div 4$$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.2.1-X

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

1. Evaluate the expression. Write your answer as a decimal.

$(5.7 - |4.6 - 7.8|)^3 \div 0.5$

2. Evaluate the expression.

$[\left(-1\frac{1}{2} - 1\right)^2 - 1.75] \div 2$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.2.1-X

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

1. Evaluate the expression. Write your answer as a decimal.

$(5.7 - |4.6 - 7.8|)^3 \div 0.5$

2. Evaluate the expression.

$[\left(-1\frac{1}{2} - 1\right)^2 - 1.75] \div 2$

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

Learning goal: I can add and subtract linear expressions. (circle your level after scoring)

4	I can write and simplify an expression from a real-world situation.
3	I can add and subtract linear expressions with rational coefficients.
2	I can combine like terms with integer coefficients.
1	I need help to start this benchmark.

1. Write the expression using the fewest possible terms.

$$\left(3+\left(\frac{5}{6}\right)p\right)+\left(\left(\frac{5}{6}\right)p-1\right)$$

2. Which expression is equivalent to the expression shown?

$(0.4x+2)+(1.6x-2.5)$

A. $2x-0.5$

B. $1.5x$

C. $0.64x-0.5$

D. $2x+0.5$

Turn over →

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____

3. What expression makes the equation true?

$$\left(\left(\frac{2}{3}\right)x + 6\right) - (__) = \left(-\frac{3}{8}\right)x - 2$$

4. Write the expression using the fewest possible terms.

$$\left(\left(\frac{4}{3}\right)x - 8\right) - \left(\left(\frac{7}{3}\right)x + 4\right)$$

5. A rectangle has a length of $\left(\left(\frac{5}{2}\right)x + 0.5\right)$ meters and a width of $(x - 2)$ meters.

Write an expression for the perimeter of the rectangle, in meters. Then write it using the fewest possible terms.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.1.1 · MA.7.AR.1.1 · Simplify linear expressions

Unit 9: Algebraic Expressions. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\left(\frac{5}{3} = 1\frac{2}{3}\right)p + 2$

Group the like terms: $\left(\left(\frac{5}{6}\right)p + \left(\frac{5}{6}\right)p\right) + (3 - 1)$.
p-terms: $\frac{5}{6} + \frac{5}{6} = \frac{5}{3}$, so the p-term is $\left(\frac{5}{3}\right)p$.
Constants: $3 - 1 = 2$.
$= \left(\frac{5}{3}\right)p + 2$

2. Regular **A.** $2x - 0.5$

Add the x-terms: $0.4x + 1.6x = 2x$. Add the constants: $2 - 2.5 = -0.5$.
An x-term and a constant cannot be combined into one term.

3. Regular $\left(\frac{25}{24} = 1\frac{1}{24}\right)x + 8$

The missing expression is the first expression minus the result.
$\left(\left(\frac{2}{3}\right)x + 6\right) - \left(\left(-\frac{3}{8}\right)x - 2\right)$: x-terms $\left(\frac{2}{3}\right)x - \left(-\frac{3}{8}\right)x = \left(\frac{25}{24}\right)x$; constants $6 - (-2) = 8$.
The missing expression is $\left(\frac{25}{24}\right)x + 8$.
Check: $\left(\left(\frac{2}{3}\right)x + 6\right) - \left(\left(\frac{25}{24}\right)x + 8\right) = \left(-\frac{3}{8}\right)x - 2$.

4. Regular $-x - 12$

Subtracting the whole expression changes the sign of each of its terms: $\left(\frac{4}{3}\right)x - 8 - \left(\frac{7}{3}\right)x - 4$.
x-terms: $\left(\frac{4}{3}\right)x - \left(\frac{7}{3}\right)x = -x$. Constants: $-8 - 4 = -12$.
$= -x - 12$

5. Rigor $2\left(\left(\frac{5}{2} = 2\frac{1}{2}\right)x + 0.5\right) + 2(x - 2)$ $\left(\text{or } \left(\left(\frac{5}{2} = 2\frac{1}{2}\right)x + 0.5\right) + (x - 2) + \left(\left(\frac{5}{2} = 2\frac{1}{2}\right)x + 0.5\right) + (x - 2)\right)$**; simplified:** $7x - 3$**. Score: 1 point for a correct perimeter expression and the simplified form** $7x - 3$**; either alone earns 0.**

The perimeter is twice the length plus twice the width: $2\left(\left(\frac{5}{2}\right)x + 0.5\right) + 2(x - 2)$.
Distribute: $2\left(\left(\frac{5}{2}\right)x + 0.5\right) = 5x + 1$ and $2(x - 2) = 2x - 4$.
Combine like terms: $5x + 2x = 7x$, and $1 - 4 = -3$.
The perimeter, in meters, is $7x - 3$.

Re-check strip (3 items, C-AR.1.1-R)

1. Regular $4.93x - 1$

Combine the two x-terms inside the second expression: $0.49x + 0.58x = 1.07x$.
Subtracting the whole expression changes both signs: $6x - 6 - 1.07x + 5$.
x-terms: $6x - 1.07x = 4.93x$. Constants: $-6 + 5 = -1$.
$= 4.93x - 1$

2. Regular **D.** $7.7x + 1.5$

Add the x-terms: $3.2x + 4.5x = 7.7x$. Add the constants: $2 - 0.5 = 1.5$.
An x-term and a constant cannot be combined into one term.

3. Regular $\left(\frac{19}{6} = 3\frac{1}{6}\right)x + 9$

The missing expression is the first expression minus the result.
$\left(\left(\frac{3}{2}\right)x + 8\right) - \left(\left(-\frac{5}{3}\right)x - 1\right)$: x-terms $\left(\frac{3}{2}\right)x - \left(-\frac{5}{3}\right)x = \left(\frac{19}{6}\right)x$; constants $8 - (-1) = 9$.
The missing expression is $\left(\frac{19}{6}\right)x + 9$.
Check: $\left(\left(\frac{3}{2}\right)x + 8\right) - \left(\left(\frac{19}{6}\right)x + 9\right) = \left(-\frac{5}{3}\right)x - 1$.

Comparison strip (2 items, C-AR.1.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $0.97x - 3$

Combine the two x-terms inside the second expression: $0.59x + 0.44x = 1.03x$.
Subtracting the whole expression changes both signs: $2x - 4 - 1.03x + 1$.
x-terms: $2x - 1.03x = 0.97x$. Constants: $-4 + 1 = -3$.
$= 0.97x - 3$

2. Regular $\left(\frac{1}{12}\right)x + 5$

The missing expression is the result minus the first expression.
$\left(\left(\frac{7}{4}\right)x - 2\right) - \left(\left(\frac{5}{3}\right)x - 7\right)$: x-terms $\left(\frac{7}{4}\right)x - \left(\frac{5}{3}\right)x = \left(\frac{1}{12}\right)x$; constants $-2 - (-7) = 5$.
The missing expression is $\left(\frac{1}{12}\right)x + 5$.
Check: $\left(\left(\frac{5}{3}\right)x - 7\right) + \left(\left(\frac{1}{12}\right)x + 5\right) = \left(\frac{7}{4}\right)x - 2$.

Reteach plan — C-AR.1.1

MA.7.AR.1.1 · Simplify linear expressions

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.1.1-R

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

1. Write the expression using the fewest possible terms.

$(6x - 6) - (0.49x - 5 + 0.58x)$

2. Which expression is equivalent to the expression shown?

$(3.2x + 2) + (4.5x - 0.5)$

A. $14.4x + 1.5$

B. $7.7x - 1.5$

C. $9.2x$

D. $7.7x + 1.5$

3. What expression makes the equation true?

$\left(\left(\frac{3}{2}\right)x + 8\right) - (__) = \left(-\frac{5}{3}\right)x - 1$

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Re-check (3 items) C-AR.1.1-R

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

1. Write the expression using the fewest possible terms.

$(6x - 6) - (0.49x - 5 + 0.58x)$

2. Which expression is equivalent to the expression shown?

$(3.2x + 2) + (4.5x - 0.5)$

A. $14.4x + 1.5$

B. $7.7x - 1.5$

C. $9.2x$

D. $7.7x + 1.5$

3. What expression makes the equation true?

$\left(\left(\frac{3}{2}\right)x + 8\right) - (__) = \left(-\frac{5}{3}\right)x - 1$

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) — C-AR.1.1-X

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

1. Write the expression using the fewest possible terms.

$(2x - 4) - (0.59x - 1 + 0.44x)$

2. What expression makes the equation true?

$\left(\left(\frac{5}{3}\right)x - 7\right) + (__) = \left(\frac{7}{4}\right)x - 2$

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) — C-AR.1.1-X

Benchmark: **MA.7.AR.1.1** · Simplify linear expressions Student code: _____ Calculator: no

1. Write the expression using the fewest possible terms.

$(2x - 4) - (0.59x - 1 + 0.44x)$

2. What expression makes the equation true?

$\left(\left(\frac{5}{3}\right)x - 7\right) + (__) = \left(\frac{7}{4}\right)x - 2$

Score: ____ **/ 2** → **copy to your progress strip**

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent Student code: _____ Calculator: no

Learning goal: I can tell whether two expressions are equivalent. (circle your level after scoring)

4	I can justify equivalence using properties of operations, including factoring.
3	I can decide whether two linear expressions are equivalent and rewrite one as the other.
2	I can use the distributive property on a simple expression.
1	I need help to start this benchmark.

1. Select **all** the expressions that are equivalent to $\left(\frac{5}{4}\right)(12x + 20)$.

Select **all** the correct answers.

A. ☐ $\left(\frac{5}{4}\right)(12x) + \left(\frac{5}{4}\right)(20)$

B. ☐ $\left(\frac{53}{4}\right)x + \frac{85}{4}$

C. ☐ $15x + 25$

D. ☐ $15x + 20$

E. ☐ $40x$

2. Match each expression to an equivalent expression.

Check one box in each row.

	$5x - 4$	$7x + 4$	$20x + 15$
$2.5(8x + 6)$	☐	☐	☐
$\left(\frac{1}{2}\right)(10x - 8)$	☐	☐	☐
$(2.5x + 6) + (4.5x - 2)$	☐	☐	☐

Turn over →

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent Student code: _____

3. Which expression is equivalent to $1.5(6x - 2)$?

A. $9x - 2$

B. $9x - 3$

C. $6x$

D. $7.5x - 0.5$

4. Are the two expressions equivalent?

$2.5(8x - 2) + 4x + 8$ and $24x + 3.5$

A. Yes, because both of them simplify to $24x + 3.5$.

B. No, because $2.5(8x - 2)$ is $20x - 2$, giving $24x + 6$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $24x + 3$, not $24x + 3.5$.

5. Two expressions are shown.

$-\left(\frac{3}{4}\right)(4x - 4) + \left(\frac{3}{2}\right)x$ and $\left(\frac{11}{2}\right)x - \left(\frac{3}{4}\right)(8x - 4)$

Are the two expressions equivalent? Write each expression using the fewest possible terms, and use those forms to explain your answer.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.1.2 · MA.7.AR.1.2 · Determine if two expressions are equivalent

Unit 9: Algebraic Expressions. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A.** $\left(\frac{5}{4} = 1\frac{1}{4}\right)(12x) + \left(\frac{5}{4} = 1\frac{1}{4}\right)(20)$**; C.** $15x + 25$

Distribute $\frac{5}{4}$ to both terms: $\frac{5}{4} \times 12x = 15x$ and $\frac{5}{4} \times 20 = 25$.
So $\left(\frac{5}{4}\right)(12x + 20) = 15x + 25$, which is also written $\left(\frac{5}{4}\right)(12x) + \left(\frac{5}{4}\right)(20)$.
Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

2. Regular $2.5(8x + 6) \rightarrow 20x + 15$**;** $\left(\frac{1}{2}\right)(10x - 8) \rightarrow 5x - 4$**;** $(2.5x + 6) + (4.5x - 2) \rightarrow 7x + 4$

Use the properties of operations to write each expression with the fewest possible terms.
$\left(\frac{1}{2}\right)(10x - 8) = 5x - 4$
$(2.5x + 6) + (4.5x - 2) = 7x + 4$
$2.5(8x + 6) = 20x + 15$
Scoring: 1 point only if every match is correct; anything else earns 0.

3. Regular **B.** $9x - 3$

Distribute 1.5 to both terms inside the parentheses.
$1.5 \times 6x = 9x$ and $1.5 \times (-2) = -3$, so the result is $9x - 3$.

4. Regular **D. No, because the first simplifies to $24x + 3$, not $24x + 3.5$.**

Distribute and combine: $2.5(8x - 2) = 20x - 5$, then + $4x + 8$ gives $24x + 3$.
The second expression is $24x + 3.5$. The constants differ (3 and 3.5), so they are not equivalent.
The 2.5 multiplies both terms in the parentheses.

5. Rigor **No; the first expression simplifies to $-\left(\frac{3}{2} = 1\frac{1}{2}\right)x + 3$ and the second to $-\left(\frac{1}{2}\right)x + 3$. Score: 1 point for the correct yes or no with both simplified forms; yes or no alone earns 0.**

First expression: distribute $-\frac{3}{4}$: $-\frac{3}{4} \times 4x = -3x$ and $-\frac{3}{4} \times (-4) = 3$. Combine the x-terms: $-3x + \left(\frac{3}{2}\right)x = -\left(\frac{3}{2}\right)x$. So the first expression is $-\left(\frac{3}{2}\right)x + 3$.
Second expression: distribute $-\frac{3}{4}$: $-\frac{3}{4} \times 8x = -6x$ and $-\frac{3}{4} \times (-4) = 3$. Combine the x-terms: $\left(\frac{11}{2}\right)x - 6x = -\left(\frac{1}{2}\right)x$. So the second expression is $-\left(\frac{1}{2}\right)x + 3$.
The constants match but the x-terms differ, so the expressions are not equivalent.

Re-check strip (3 items, C-AR.1.2-R)

1. Regular **D. No, because the first simplifies to $7x - 1$, not $7x - 2$.**

Distribute and combine: $1.5(4x - 6) = 6x - 9$, then + $x + 8$ gives $7x - 1$.
The second expression is $7x - 2$. The constants differ $(-1 \text{ and } -2)$, so they are not equivalent.
The 1.5 multiplies both terms in the parentheses.

2. Regular **A.** $1.5(4x + 5)$

Both terms of $6x + 7.5$ are 1.5 times a whole number: $6 = 1.5 \times 4$ and $7.5 = 1.5 \times 5$.
So $6x + 7.5 = 1.5(4x + 5)$. Check by distributing: $1.5 \times 4x = 6x$ and $1.5 \times 5 = 7.5$.

3. Regular **A.** $\left(\frac{3}{2} = 1\frac{1}{2}\right)(6x) + \left(\frac{3}{2} = 1\frac{1}{2}\right)(2)$**; E.** $9x + 3$

Distribute $\frac{3}{2}$ to both terms: $\frac{3}{2} \times 6x = 9x$ and $\frac{3}{2} \times 2 = 3$.
So $\left(\frac{3}{2}\right)(6x + 2) = 9x + 3$, which is also written $\left(\frac{3}{2}\right)(6x) + \left(\frac{3}{2}\right)(2)$.
Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Comparison strip (2 items, C-AR.1.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B.** $\left(\frac{2}{5}\right)(20x + 15)$

Both terms of $8x + 6$ are $\frac{2}{5}$ times a whole number: $8 = \frac{2}{5} \times 20$ and $6 = \frac{2}{5} \times 15$.
So $8x + 6 = \left(\frac{2}{5}\right)(20x + 15)$. Check by distributing: $\frac{2}{5} \times 20x = 8x$ and $\frac{2}{5} \times 15 = 6$.
$\left(\frac{5}{2}\right)(20x + 15)$ uses the reciprocal factor and does not give $8x + 6$.

2. Regular $1.5(2x + 2) \rightarrow 3x + 3$**;** $\left(\frac{1}{4}\right)(16x - 8) \rightarrow 4x - 2$**;** $(2.5x + 7) + (1.5x - 5) \rightarrow 4x + 2$

Use the properties of operations to write each expression with the fewest possible terms.
$\left(\frac{1}{4}\right)(16x - 8) = 4x - 2$
$(2.5x + 7) + (1.5x - 5) = 4x + 2$
$1.5(2x + 2) = 3x + 3$
Scoring: 1 point only if every match is correct; anything else earns 0.

Reteach plan — C-AR.1.2

MA.7.AR.1.2 · Determine if two expressions are equivalent

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.1.2-R

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent Student code: _____ Calculator: no

1. Are the two expressions equivalent?

$1.5(4x - 6) + x + 8$ and $7x - 2$

A. No, because $1.5(4x - 6)$ is $6x - 6$, giving $7x + 2$.

B. Yes, because both of them simplify to $7x - 2$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $7x - 1$, not $7x - 2$.

2. Which expression is equivalent to $6x + 7.5$?

A. $1.5(4x + 5)$

B. $6(x + 5)$

C. $1.5(6x + 5)$

D. $1.5(4x + 7.5)$

3. Select **all** the expressions that are equivalent to $\left(\frac{3}{2}\right)(6x + 2)$.

Select **all** the correct answers.

A. ☐ $\left(\frac{3}{2}\right)(6x) + \left(\frac{3}{2}\right)(2)$

B. ☐ $\left(\frac{15}{2}\right)x + \frac{7}{2}$

C. ☐ $12x$

D. ☐ $9x + 2$

E. ☐ $9x + 3$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-AR.1.2-R

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent Student code: _____ Calculator: no

1. Are the two expressions equivalent?

$1.5(4x - 6) + x + 8$ and $7x - 2$

A. No, because $1.5(4x - 6)$ is $6x - 6$, giving $7x + 2$.

B. Yes, because both of them simplify to $7x - 2$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $7x - 1$, not $7x - 2$.

2. Which expression is equivalent to $6x + 7.5$?

A. $1.5(4x + 5)$

B. $6(x + 5)$

C. $1.5(6x + 5)$

D. $1.5(4x + 7.5)$

3. Select **all** the expressions that are equivalent to $\left(\frac{3}{2}\right)(6x + 2)$.

Select **all** the correct answers.

A. ☐ $\left(\frac{3}{2}\right)(6x) + \left(\frac{3}{2}\right)(2)$

B. ☐ $\left(\frac{15}{2}\right)x + \frac{7}{2}$

C. ☐ $12x$

D. ☐ $9x + 2$

E. ☐ $9x + 3$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-AR.1.2-X

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent

Student code: _____ Calculator: no

1. Which expression is equivalent to $8x + 6$?

A. $\left(\frac{2}{5}\right)(20x + 6)$

B. $\left(\frac{2}{5}\right)(20x + 15)$

C. $\left(\frac{5}{2}\right)(20x + 15)$

D. $\left(\frac{2}{5}\right)(8x + 15)$

2. Match each expression to an equivalent expression.

Check one box in each row.

	$4x - 2$	$4x + 2$	$3x + 3$
$1.5(2x + 2)$	☐	☐	☐
$\left(\frac{1}{4}\right)(16x - 8)$	☐	☐	☐
$(2.5x + 7) + (1.5x - 5)$	☐	☐	☐

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-AR.1.2-X

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent

Student code: _____ Calculator: no

1. Which expression is equivalent to $8x + 6$?

A. $\left(\frac{2}{5}\right)(20x + 6)$

B. $\left(\frac{2}{5}\right)(20x + 15)$

C. $\left(\frac{5}{2}\right)(20x + 15)$

D. $\left(\frac{2}{5}\right)(8x + 15)$

2. Match each expression to an equivalent expression.

Check one box in each row.

	$4x - 2$	$4x + 2$	$3x + 3$
$1.5(2x + 2)$	☐	☐	☐
$\left(\frac{1}{4}\right)(16x - 8)$	☐	☐	☐
$(2.5x + 7) + (1.5x - 5)$	☐	☐	☐

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Weekly Check · Unit 10

C-AR.2.1

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities

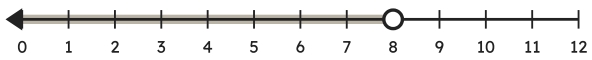
Student code: _____ Calculator: yes

Learning goal: I can write and solve one-step inequalities and graph the solutions. (circle your level after scoring)

4	I can write an inequality from a real-world situation, solve it, and interpret the solution.
3	I can solve a one-step inequality, including with negative numbers, and graph the solution.
2	I can graph an inequality on a number line.
1	I need help to start this benchmark.

1. The graph shows the solution of an inequality.

Which inequality has this solution?



A. $2.5x \leq 20$

B. $x + 2.5 < 20$

C. $2.5x > 20$

D. $2.5x < 20$

2. Solve the inequality.

$-3 + t \geq -6.9$

Turn over →

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities Student code: _____

3. Solve the inequality.

$-0.5n > 0.5$

Which number line shows the solution?

A. Open circle at –1, shaded left

B. Closed circle at –1, shaded left

C. Open circle at 1, shaded left

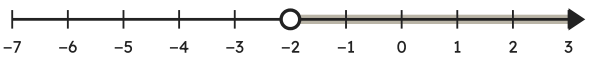
D. Open circle at –1, shaded right

4. The product of a number x and –0.5 is less than –5.

What are the possible values of x?

5. The number line shows the solution of one of the inequalities below.

Which inequality has this solution?



A. $x + 3.5 \geq 1.5$

B. $\frac{x}{4} < -0.5$

C. $x - 4.5 > -8.5$

D. $-1.5x < 3$

Score: ____ **/ 5** **My level:** **1** **2** **3** **4** **→ copy both to your progress strip**

Answer key — C-AR.2.1 · MA.7.AR.2.1 · Solve one-step inequalities

Unit 10: Equations and Inequalities. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **D.** $2.5x < 20$

The open circle at 8 shows that 8 is not included, and the shading to the left shows $x < 8$.

$2.5x < 20$ has this solution: dividing both sides by 2.5 gives $x < 8$.

$2.5x \leq 20$ would need a closed circle; $2.5x > 20$ shades the other direction; $x + 2.5 < 20$ has the boundary 17.5.

2. Regular $t \geq -3.9$

Add 3 to both sides.

$t \geq -6.9 + 3 = -3.9$.

3. Regular **A. Open circle at –1, shaded left**

Divide both sides by –0.5. Dividing by a negative number reverses the inequality symbol, so $n < -1$.

An open circle at –1 shows that –1 is not included, and the shading goes to the left.

4. Regular $x > 10$

"The product of a number x and –0.5" is $-0.5x$, and "less than –5" gives $-0.5x < -5$.

Divide both sides by –0.5. Because it is negative, the inequality symbol reverses: $x > 10$.

5. Rigor **D.** $-1.5x < 3$

The graph shows $x > -2$: an open circle at –2, shaded to the right.

Solve each choice. Only $-1.5x < 3$ gives $x > -2$ (dividing by the negative coefficient reverses the symbol).

The other three solve to a different direction, the other kind of circle, or a different boundary.

Re-check strip (3 items, C-AR.2.1-R)

1. Regular $x > 8$

"The product of a number x and –1.5" is $-1.5x$, and "less than –12" gives $-1.5x < -12$.

Divide both sides by –1.5. Because it is negative, the inequality symbol reverses: $x > 8$.

2. Regular **A. Closed circle at –3, shaded right**

Divide both sides by –1.5. Dividing by a negative number reverses the inequality symbol, so $c \geq -3$.

A closed circle at –3 shows that –3 is included, and the shading goes to the right.

3. Regular **$n \leq -5$; graph: closed circle at –5, shaded to the left. Score: 1 point for the solution (or an equivalent inequality) and a matching graph; either alone earns 0.**

Multiply both sides by –4. Multiplying by a negative number reverses the inequality symbol.

$n \leq 1.25 \times (-4) = -5$

Graph: –5 is included, so draw a closed circle at –5, and shade to the left because the solutions are less than or equal to –5.

Comparison strip (2 items, C-AR.2.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $x \leq 8$

"The product of a number x and –2.5" is $-2.5x$, and "at least –20" gives $-2.5x \geq -20$.

Divide both sides by –2.5. Because it is negative, the inequality symbol reverses: $x \leq 8$.

2. Regular **$w \leq 5$; graph: closed circle at 5, shaded to the left. Score: 1 point for the solution (or an equivalent inequality) and a matching graph; either alone earns 0.**

Divide both sides by 0.5.

$w \leq 2.5 \div 0.5 = 5$

Graph: 5 is included, so draw a closed circle at 5, and shade to the left because the solutions are less than or equal to 5.

Reteach plan — C-AR.2.1

MA.7.AR.2.1 · Solve one-step inequalities

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities Student code: _____ Calculator: yes

1. The product of a number x and −1.5 is less than −12.

What are the possible values of x?

2. Solve the inequality.

$-1.5c \leq 4.5$

Which number line shows the solution?

A. Closed circle at −3, shaded right

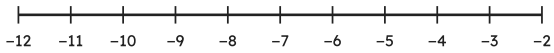
B. Closed circle at 3, shaded right

C. Open circle at −3, shaded right

D. Closed circle at −3, shaded left

3. Solve the inequality. Then graph the solution on the number line.

$\frac{n}{-4} \geq 1.25$



Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2)

C-AR.2.1-X

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities

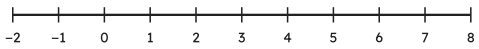
Student code: _____ Calculator: yes

1. The product of a number x and −2.5 is at least −20.

What are the possible values of x?

2. Solve the inequality. Then graph the solution on the number line.

$0.5w \leq 2.5$



Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-AR.2.1-X

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities

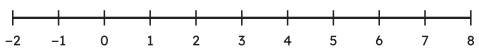
Student code: _____ Calculator: yes

1. The product of a number x and −2.5 is at least −20.

What are the possible values of x?

2. Solve the inequality. Then graph the solution on the number line.

$0.5w \leq 2.5$



Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Weekly Check · Unit 10 C-AR.2.2

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____ Calculator: yes

Learning goal: I can write and solve two-step equations. (circle your level after scoring)

4	I can write a two-step equation from a real-world context, solve it, and check the answer.
3	I can solve a two-step equation, including with rational numbers or the distributive property.
2	I can solve a one-step equation.
1	I need help to start this benchmark.

1. A park charges a $25.00 entry fee plus $12.50 for each hour of skate rental. A visitor paid $62.50 in all.

Write an equation that can be used to find h, the number of hours of skate rental the visitor paid for. Then solve the equation.

2. Solve the equation for h.

$1.5(h + 1.6) = -0.6$

Turn over →

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____

3. 3 students each saved the same amount, s dollars, and then each earned \$5.00 more. Together they now have \$105.00.

Write an equation that can be used to find s. Then solve the equation.

4. What is the exact value of g in the equation shown?

$$-1\frac{7}{9} = \left(\frac{2}{3}\right) g - 4$$

5. Gym A charges \$24.50 each month. Jordan used a coupon worth \$35.00 off the total and paid Gym A \$185.50 in all.

Write and solve an equation to find n, the number of months Jordan paid for. Gym B charges \$26.50 each month and takes no coupons. For the same number of months, which gym would have cost less, and by how much?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.2.2 · MA.7.AR.2.2 · Two-step equations with rational numbers

Unit 10: Equations and Inequalities. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $12.5h + 25 = 62.5$; $h = 3$. **Score: 1 point for the equation (or an equivalent equation) and its solution; the solution alone earns 0.**

The cost of the hours of skate rental plus the entry fee equals the total paid: $12.5h + 25 = 62.5$.
Subtract 25 from both sides: $12.5h = 37.5$.
Divide both sides by 12.5: $h = 3$.

2. Regular $h = -2$

Divide both sides by 1.5: $h + 1.6 = -0.6 \div 1.5 = -0.4$.
Subtract 1.6 from both sides: $h = -0.4 - 1.6 = -2$.

3. Regular $3(s + 5) = 105$ **(or an equivalent equation); $s = 30$. Score: 1 point for the equation (or an equivalent equation) and its solution; the solution alone earns 0.**

Each student now has $s + 5$, and there are 3 students: $3(s + 5) = 105$.
Divide both sides by 3: $s + 5 = 35$.
Subtract 5 from both sides: $s = 30$.

4. Regular $g = 3\frac{1}{3} = \frac{10}{3}$

Add 4 to both sides: $-1\frac{7}{9} + 4 = 2\frac{2}{9}$, so $2\frac{2}{9} = \left(\frac{2}{3}\right) g$.
Divide both sides by $\frac{2}{3}$, which means multiplying by its reciprocal $\frac{3}{2}$: $g = \frac{20}{9} \times \frac{3}{2} = 3\frac{1}{3}$.

5. Rigor $24.5n - 35 = 185.5$; $n = 9$; **Gym A, by $53.00. Score: 1 point for the equation (or an equivalent equation), $n = 9$ and the correct gym with the amount; n alone earns 0.**

Monthly charges minus the coupon equal the total: $24.5n - 35 = 185.5$.
Add 35 to both sides: $24.5n = 220.5$. Divide both sides by 24.5: $n = 9$.
Gym B for 9 months: $9 \times \$26.50 = \$238.50$.
$\$238.50 - \$185.50 = \$53.00$, so Gym A would have cost $53.00 less.

Re-check strip (3 items, C-AR.2.2-R)

1. Regular $x = -9$

Subtract 4 from both sides: $36 = -4x$.
Divide both sides by −4: $x = -9$.

2. Regular $6(b - 1.25) = 40.5$ **(or an equivalent equation); $b = 8$. Score: 1 point for the equation (or an equivalent equation) and its solution; the solution alone earns 0.**

Each box without packaging weighs $b - 1.25$, and there are 6 boxes: $6(b - 1.25) = 40.5$.
Divide both sides by 6: $b - 1.25 = 6.75$.
Add 1.25 to both sides: $b = 8$.

3. Regular **A.** $3(x - 8) = 3$

"Decreased by 8" is $(x - 8)$.
Multiplying that result by 3 gives $3(x - 8)$, and that equals 3.

Comparison strip (2 items, C-AR.2.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $3.75g + 18 = 44.25$; $g = 7$. **Score: 1 point for the equation (or an equivalent equation) and its solution; the solution alone earns 0.**

The cost of the games plus the entry fee equals the total paid: $3.75g + 18 = 44.25$.
Subtract 18 from both sides: $3.75g = 26.25$.
Divide both sides by 3.75: $g = 7$.

2. Regular $t = 2\frac{1}{3} = \frac{7}{3}$

Add 9 to both sides: $-5\frac{8}{9} + 9 = 3\frac{1}{9}$, so $3\frac{1}{9} = \left(\frac{4}{3}\right) t$.
Divide both sides by $\frac{4}{3}$, which means multiplying by its reciprocal $\frac{3}{4}$: $t = \frac{28}{9} \times \frac{3}{4} = 2\frac{1}{3}$.

Reteach plan — C-AR.2.2

MA.7.AR.2.2 · Two-step equations with rational numbers

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.2.2-R

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____ Calculator: yes

1. Solve the equation for x.

$40 = -4x + 4$

2. 6 identical boxes each weigh b pounds with the packaging. After 1.25 pounds of packaging is removed from each box, the 6 boxes weigh 40.5 pounds in all.

Write an equation that can be used to find b. Then solve the equation.

3. A number, x, is decreased by 8. That result is then multiplied by 3, giving 3.

Which equation represents this situation?

A. $3(x - 8) = 3$

B. $3x - 24 = 9$

C. $3(x + 8) = 3$

D. $3x - 8 = 3$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-AR.2.2-R

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____ Calculator: yes

1. Solve the equation for x.

$40 = -4x + 4$

2. 6 identical boxes each weigh b pounds with the packaging. After 1.25 pounds of packaging is removed from each box, the 6 boxes weigh 40.5 pounds in all.

Write an equation that can be used to find b. Then solve the equation.

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A. $3(x - 8) = 3$

B. $3x - 24 = 9$

C. $3(x + 8) = 3$

D. $3x - 8 = 3$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.2.2-X

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____ Calculator: yes

1. A park charges a $18.00 entry fee plus $3.75 for each game. A visitor paid $44.25 in all.

Write an equation that can be used to find g, the number of games the visitor paid for. Then solve the equation.

2. What is the exact value of t in the equation shown?

$$-5\frac{8}{9} = \left(\frac{4}{3}\right)t - 9$$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.2.2-X

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____ Calculator: yes

1. A park charges a $18.00 entry fee plus $3.75 for each game. A visitor paid $44.25 in all.

Write an equation that can be used to find g, the number of games the visitor paid for. Then solve the equation.

2. What is the exact value of t in the equation shown?

$$-5\frac{8}{9} = \left(\frac{4}{3}\right)t - 9$$

Score: ____ / 2 → **copy to your progress strip**

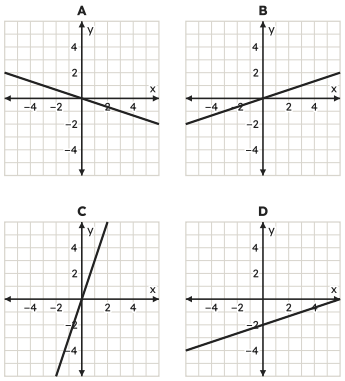
Benchmark: **MA.7.AR.4.3** · Graph proportional relationships Student code: _____ Calculator: yes

Learning goal: I can graph proportional relationships. (circle your level after scoring)

4	I can use a graph to find the constant of proportionality and interpret it in context.
3	I can graph a proportional relationship and identify the points (0, 0) and (1, r).
2	I can plot points from a table.
1	I need help to start this benchmark.

1. A proportional relationship between x and y has a constant of proportionality of $\frac{1}{3}$.

Which graph represents this relationship?



Choose the graph that matches.

A. Graph A

B. Graph B

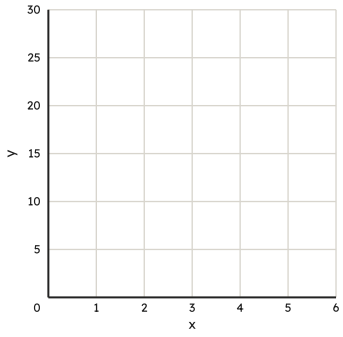
C. Graph C

D. Graph D

2. The table shows a proportional relationship between x and y.

x	y
4	20
5	25
6	30

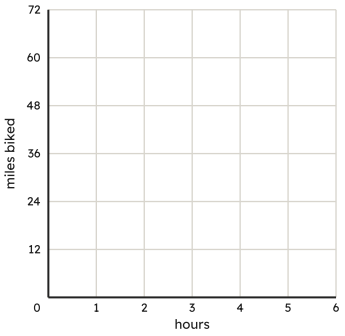
Graph the relationship on the grid.



Turn over →

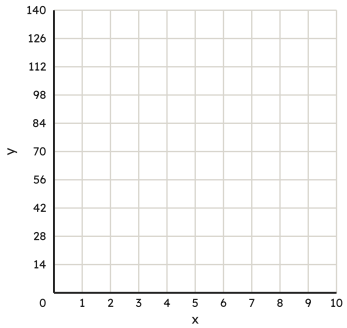
3. The relationship between hours (x) and miles biked (y) is proportional, with a constant of proportionality of 12.

Graph the relationship on the grid for 0 to 6 hours. What does the constant of proportionality mean in this situation?



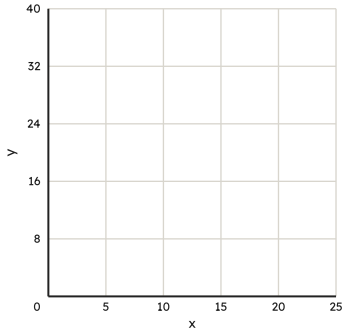
4. A proportional relationship is given by the equation $y = 14x$.

Graph the relationship on the grid for x-values from 0 to 10.



5. The constant of proportionality for a proportional relationship between x and y is $\frac{8}{5}$.

Graph the relationship on the grid for x-values from 0 to 25, and mark and label the point on the line whose x-coordinate is 5. Is the point $(40, 64)$ on the line? Explain.



Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — C-AR.4.3 · MA.7.AR.4.3 · Graph proportional relationships

Unit 11: Representing Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **Graph B**

A proportional relationship is a straight line through the origin, so the graph passes through $(0, 0)$.
The constant of proportionality $\frac{1}{3}$ is the slope: from $(0, 0)$ the line rises 1 for every 3 to the right, so it passes through $(3, 1)$, inside the labeled part of each graph.
The reciprocal 3 would pass through $(1, 3)$ instead, which switches x and y; a line that falls to the right has a negative constant.

2. Regular **The points $(4, 20)$, $(5, 25)$, $(6, 30)$ and a straight line through them and the origin.**

Plot each row of the table as a point: $(4, 20)$, $(5, 25)$, $(6, 30)$.
Each y-value is 5 times its x-value, so the line is proportional and passes through $(0, 0)$; draw the straight line through the origin and the points.

3. Regular **A straight line from the origin to $(6, 72)$, through $(1, 12)$, $(2, 24)$ and $(3, 36)$; the constant means each hour, 12 more miles are biked. Score: 1 point for the graph and the meaning of the constant; the graph alone earns 0.**

The constant of proportionality 12 means $y = 12x$: each hour, 12 more miles are biked.
Plot $(0, 0)$, $(1, 12)$, $(2, 24)$ and draw the line through them to $x = 6$.

4. Regular **A straight line from $(0, 0)$ to $(10, 140)$, through points such as $(2, 28)$, $(3, 42)$, $(7, 98)$.**

In $y = 14x$ the constant of proportionality is 14, so every point has $y = 14 \times x$.
Plot $(0, 0)$, $(2, 28)$, $(3, 42)$, $(7, 98)$ and draw the straight line through them from $x = 0$ to $x = 10$.

5. Rigor **A straight line through the origin and $(5, 8)$, $(10, 16)$, $(15, 24)$, continued to $(25, 40)$; the point $(5, 8)$ marked and labeled; yes — $\frac{8}{5} = 1\frac{3}{5} \times 40$ is 64, so $(40, 64)$ is on the line. Score: 1 point for the graph with $(5, 8)$ labeled and the correct yes or no with its reason; the graph alone earns 0.**

The constant of proportionality $\frac{8}{5}$ means $y = \left(\frac{8}{5}\right) x$. Choose x-values that are multiples of 5 so y is a whole number: $x = 5$ gives $y = \frac{8}{5} \times 5 = 8$; $x = 10$ gives $y = \frac{8}{5} \times 10 = 16$; $x = 15$ gives $y = \frac{8}{5} \times 15 = 24$.
Plot those points and $(0, 0)$, then draw the line through them to $x = 25$.
At $x = 5$, $y = \frac{8}{5} \times 5 = 8$, so the point $(5, 8)$ is on the line, where the gridlines $x = 5$ and $y = 8$ cross; mark and label it.
A point is on the line when $y = \frac{8}{5} \times x$: $\frac{8}{5} \times 40 = 64$, which is 64, so $(40, 64)$ is on the line.

Re-check strip (3 items, C-AR.4.3-R)

1. Regular **The points $(2, 32)$, $(4, 64)$, $(5, 80)$ and a straight line through them and the origin.**

Plot each row of the table as a point: $(2, 32)$, $(4, 64)$, $(5, 80)$.
Each y-value is 16 times its x-value, so the line is proportional and passes through $(0, 0)$; draw the straight line through the origin and the points.

2. Regular **A straight line from the origin to $(6, 90)$, through $(1, 15)$, $(2, 30)$ and $(3, 45)$; the constant means each box holds 15 books. Score: 1 point for the graph and the meaning of the constant; the graph alone earns 0.**

The constant of proportionality 15 means $y = 15x$: each box holds 15 books.
Plot $(0, 0)$, $(1, 15)$, $(2, 30)$ and draw the line through them to $x = 6$.

3. Regular **A straight line from $(0, 0)$ to $(6, 138)$, through points such as $(1, 23)$, $(2, 46)$, $(6, 138)$.**

In $y = 23x$ the constant of proportionality is 23, so every point has $y = 23 \times x$.
Plot $(0, 0)$, $(1, 23)$, $(2, 46)$, $(6, 138)$ and draw the straight line through them from $x = 0$ to $x = 6$.

Comparison strip (2 items, C-AR.4.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **Graph C**

Each y-value is $\frac{1}{3}$ times its x-value, so the constant of proportionality is $\frac{1}{3}$.
The graph is a straight line through the origin that rises 1 for every 3 to the right, so it passes through $(3, 1)$, inside the labeled part of each graph.
A line through $(1, 3)$ would switch x and y; a line that falls to the right has a negative constant.

2. Regular **Graph A**

A proportional relationship is a straight line through the origin, so the graph passes through $(0, 0)$.
The constant of proportionality 2 is the slope: from $(0, 0)$ the line rises 2 for every 1 to the right, so it passes through $(1, 2)$, inside the labeled part of each graph.
The reciprocal $\frac{1}{2}$ would pass through $(2, 1)$ instead, which switches x and y; a line that falls to the right has a negative constant.

Reteach plan — C-AR.4.3

MA.7.AR.4.3 · Graph proportional relationships

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

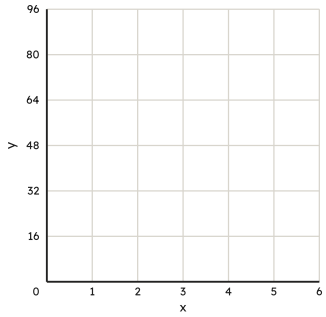
Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

1. The table shows a proportional relationship between x and y.

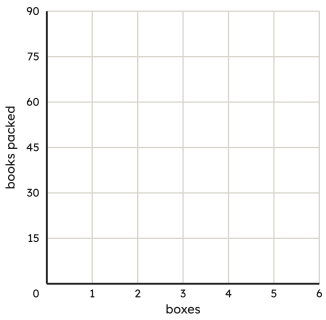
x	y
2	32
4	64
5	80

Graph the relationship on the grid.



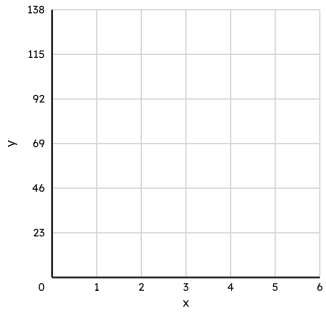
2. The relationship between boxes (x) and books packed (y) is proportional, with a constant of proportionality of 15.

Graph the relationship on the grid for 0 to 6 boxes. What does the constant of proportionality mean in this situation?



3. A proportional relationship is given by the equation $y = 23x$.

Graph the relationship on the grid for x-values from 0 to 6.

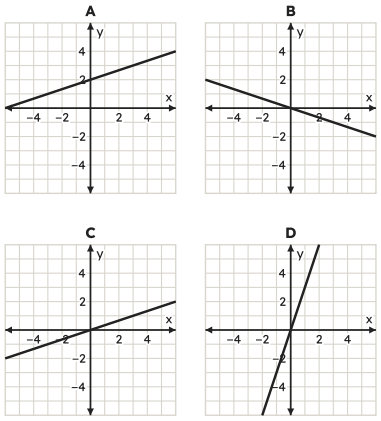


Score: ____ / 3 **→ copy to your progress strip**

1. The table shows a proportional relationship between x and y.

x	y
3	1
9	3
12	4

Which graph represents this relationship?



Choose the graph that matches.

A. Graph A

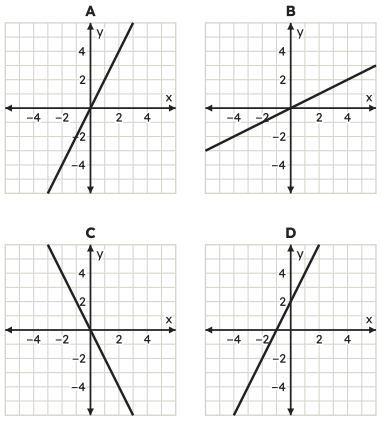
B. Graph B

C. Graph C

D. Graph D

2. A proportional relationship between x and y has a constant of proportionality of 2.

Which graph represents this relationship?



Choose the graph that matches.

A. Graph A

B. Graph B

C. Graph C

D. Graph D

Score: ____ / 2 → **copy to your progress strip**

Benchmark: **MA.7.AR.4.2** · Constant of proportionality Student code: _____ Calculator: yes

Learning goal: I can find the constant of proportionality. (circle your level after scoring)

4	I can explain what the constant of proportionality means in the situation.
3	I can find the constant of proportionality from a table, graph, or written description.
2	I can find a unit rate from a ratio.
1	I need help to start this benchmark.

1. In a proportional relationship, 64.05 cups of water are used in 7 minutes.

What is the constant of proportionality, in cups of water per minute?

A. 9.15

B. 64.05

C. 71.05

D. 0.11

2. The daily fee for docking a boat at a marina is proportional to the length of the boat. The table shows the fee for four different boat lengths.

Boat length (feet)	Daily fee
20	$62.00
22	$68.20
24	$74.40
28	$86.80

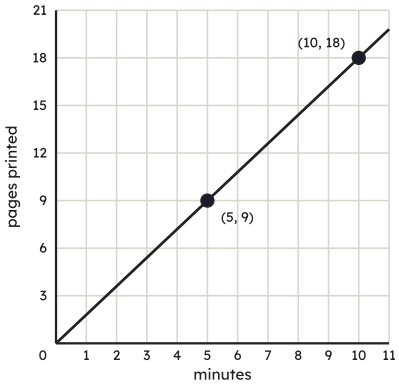
What is the constant of proportionality, in dollars per foot?

Turn over →

Benchmark: **MA.7.AR.4.2** · Constant of proportionality Student code: _____

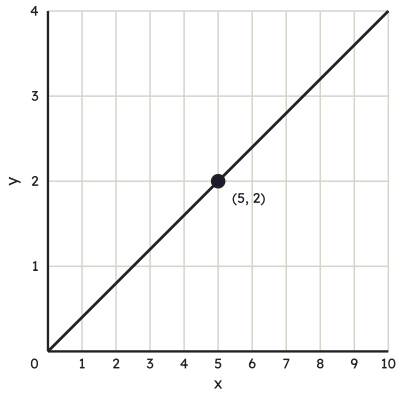
3. The graph shows the proportional relationship between minutes and pages printed.

What is the constant of proportionality, in pages per minute?



4. The graph shows a proportional relationship between x and y.

What is the constant of proportionality? Write your answer as a fraction in simplest form.



5. Each batch of paint is made with 2 cups of blue paint and 12 cups of white paint. The relationship between cups of blue paint and cups of white paint is proportional. Let x be the number of cups of white paint and let y be the number of cups of blue paint.

What is the constant of proportionality? What does it mean in this situation?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.4.2 · MA.7.AR.4.2 · Constant of proportionality

Unit 11: Representing Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. 9.15**

The constant of proportionality is cups of water per minute: $64.05 \div 7 = 9.15$.

0.11 divides the other way, 64.05 is the total rather than the rate, and adding or subtracting the two amounts does not give a rate.

2. Regular **3.1**

The constant of proportionality is the fee divided by the length.

$62 \div 20 = 3.1$, and $68.2 \div 22 = 3.1$.

The constant of proportionality is 3.1 dollars per foot.

3. Regular **1.8 pages per minute**

Use a labeled point: $(5,\ 9)$.

The constant of proportionality is $y \div x = 9 \div 5 = 1.8$ pages per minute.

It is not the y-value of a labeled point, because no labeled point has $x = 1$.

4. Regular $\dfrac{2}{5}$

Use the labeled point: $y \div x = 2 \div 5 = \dfrac{2}{5}$.

5. Rigor **$k = \dfrac{1}{6}$; each cup of white paint is used with $\dfrac{1}{6}$ cup of blue paint. Score: 1 point for the constant (or an equivalent form of it) and its meaning; the constant alone earns 0.**

The constant of proportionality is $y \div x$: $2 \div 12 = \dfrac{1}{6}$.

So $y = \left(\dfrac{1}{6}\right) x$: each cup of white paint is used with $\dfrac{1}{6}$ cup of blue paint.

Dividing the other way, $12 \div 2 = 6$, would be the constant if x were the cups of blue paint; here x is the cups of white paint.

Re-check strip (3 items, C-AR.4.2-R)

1. Regular **0.9 cups of flour per batch**

Use a labeled point: $(5,\ 4.5)$.

The constant of proportionality is $y \div x = 4.5 \div 5 = 0.9$ cups of flour per batch.

It is not the y-value of a labeled point, because no labeled point has $x = 1$.

2. Regular **3.95**

The constant of proportionality is the fee divided by the length.

$63.2 \div 16 = 3.95$, and $79 \div 20 = 3.95$.

The constant of proportionality is 3.95 dollars per foot.

3. Regular **D. 3.15**

The constant of proportionality is liters of paint per wall: $12.6 \div 4 = 3.15$.

0.32 divides the other way, 12.60 is the total rather than the rate, and adding or subtracting the two amounts does not give a rate.

Comparison strip (2 items, C-AR.4.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $\dfrac{9}{7} = 1\dfrac{2}{7}$

Use the labeled point: $y \div x = 9 \div 7 = \dfrac{9}{7}$.

2. Regular $\dfrac{8}{11}$

The constant of proportionality is $y \div x$.

$8 \div 11 = \dfrac{8}{11}$, and every pair gives the same ratio.

Reteach plan — C-AR.4.2

MA.7.AR.4.2 · Constant of proportionality

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

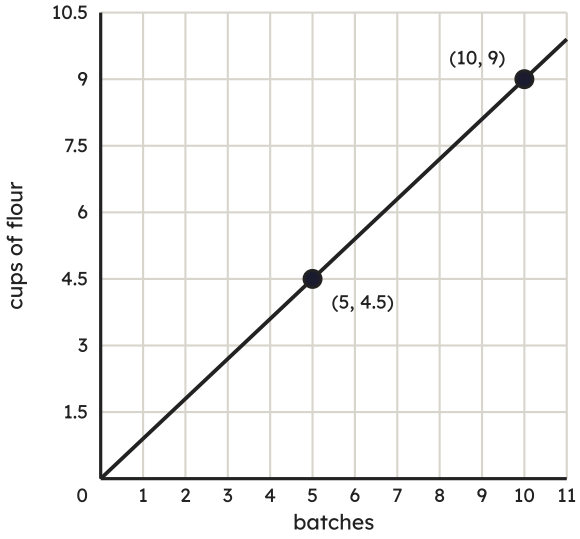
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.4.2-R

Benchmark: **MA.7.AR.4.2** · Constant of proportionality Student code: _____ Calculator: yes

1. The graph shows the proportional relationship between batches and cups of flour.

What is the constant of proportionality, in cups of flour per batch?



2. The daily fee for docking a boat at a marina is proportional to the length of the boat. The table shows the fee for four different boat lengths.

Boat length (feet)	Daily fee
16	$63.20
20	$79.00
22	$86.90
26	$102.70

What is the constant of proportionality, in dollars per foot?

3. In a proportional relationship, 12.6 liters of paint are needed to cover 4 walls.

What is the constant of proportionality, in liters of paint per wall?

A. 0.32

B. 16.60

C. 12.60

D. 3.15

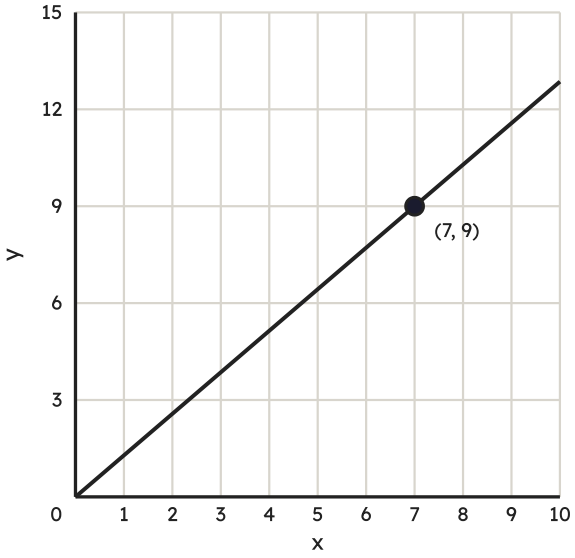
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-AR.4.2-X

Benchmark: **MA.7.AR.4.2** · Constant of proportionality Student code: _____ Calculator: yes

1. The graph shows a proportional relationship between x and y.

What is the constant of proportionality? Write your answer as a fraction in simplest form.



2. The table shows a proportional relationship between x and y.

x	y
11	8
22	16
55	40
66	48

What is the constant of proportionality? Write your answer as a fraction in simplest form.

Score: ____ / 2 → copy to your progress strip

Benchmark: **MA.7.AR.4.1** · Is it proportional? (table, graph, description) Student code: _____ Calculator: yes

Learning goal: I can decide whether a relationship is proportional. (circle your level after scoring)

4	I can explain why a relationship is not proportional using more than one representation.
3	I can decide whether a table, graph, or written description shows a proportional relationship and say why.
2	I can check whether two ratios are equal.
1	I need help to start this benchmark.

1. Select **all** the situations that describe a proportional relationship.

Select **all** the correct answers.

A. ☐ A printer prints 30 pages every minute.

B. ☐ A phone plan costs $36 plus $0.10 for each text message.

C. ☐ A candle is 12 inches tall and burns down 1 inch each hour (its height compared with the hours burned).

D. ☐ A taxi charges $3.00 plus $2.00 per mile.

E. ☐ A store sells oranges at $2.40 per pound.

2. Does the table show a proportional relationship between x and y?

x	y
1	5
2	10
4	23
5	25

A. Yes, because $y \div x$ is the same for every pair of values.

B. No, because the x-values do not increase by the same amount.

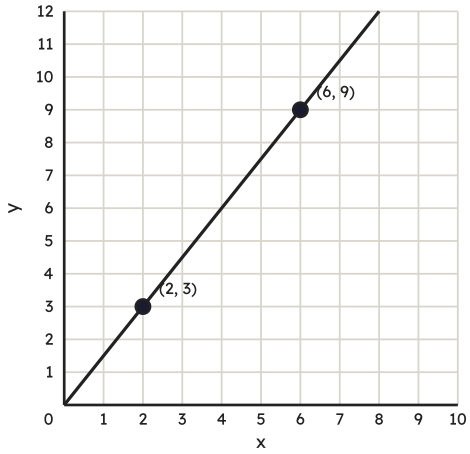
C. No, because $y \div x$ is not the same for every pair of values.

D. Yes, because the first two pairs of values have the same ratio.

Turn over →

Benchmark: **MA.7.AR.4.1** · Is it proportional? (table, graph, description) Student code: _____

3. Does the graph show a proportional relationship between x and y?



A. Yes, because the line is straight and passes through the origin.

B. Yes, because y increases whenever x increases.

C. No, because the line does not reach the edge of the grid.

D. No, because the line does not pass through the origin.

4. A machine takes 15 minutes plus 10 minutes for each box packed.

Is the relationship between the boxes packed and the total minutes proportional?

A. Yes, because there is no setup time at the start, only 10 minutes each time.

B. Yes, because the minutes added for each box packed never change.

C. No, because there is a setup time of 15 minutes before the first box packed.

D. No, because the minutes and the boxes packed are measured in different units.

5. The table shows values of x and y.

x	y
1.5	1.8
2	2.4
2.5	3
5	6

Noor says the relationship cannot be proportional because the table does not include $x = 0$.

Is Noor's reason correct? Is the relationship proportional? If it is, give the constant of proportionality; if it is not, name a pair whose ratio is different.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.4.1 · MA.7.AR.4.1 · Is it proportional? (table, graph, description)

Unit 11: Representing Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. A printer prints 30 pages every minute; E. A store sells oranges at $2.40 per pound.**

A relationship is proportional only when the two quantities have a constant ratio and both start at 0.
Proportional: "A store sells oranges at $2.40 per pound." — the same price for every pound, and 0 pounds cost $0; "A printer prints 30 pages every minute." — the same pages every minute, and 0 minutes print 0 pages.
Not proportional: "A phone plan costs $36 plus $0.10 for each text message." — the $36 is charged even with 0 texts; "A candle is 12 inches tall and burns down 1 inch each hour (its height compared with the hours burned)." — the candle is 12 inches tall at hour 0; "A taxi charges $3.00 plus $2.00 per mile." — the $3.00 charge is paid before the first mile.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

2. Regular **C. No, because $y \div x$ is not the same for every pair of values.**

Check $y \div x$ for each pair: $5 \div 1 = 5$, $10 \div 2 = 5$, $23 \div 4 = 5.75$, $25 \div 5 = 5$.
The ratios are not all the same, so the relationship is not proportional.

3. Regular **A. Yes, because the line is straight and passes through the origin.**

A graph shows a proportional relationship when it is a straight line through the origin $(0, 0)$.
This line passes through $(0, 0)$, so it is proportional.

4. Regular **C. No, because there is a setup time of 15 minutes before the first box packed.**

A proportional relationship must pass through $(0, 0)$ and have a constant ratio.
The starting setup time of 15 means 0 boxes packed already gives 15 minutes, so it is not proportional.

5. Rigor **Noor's reason is not correct; the relationship is proportional, with constant of proportionality 1.2. Score: 1 point for all three parts (the reason judged, the proportional decision, and the constant or a pair with a different ratio, in any equivalent form); two parts earn 0.**

A table does not have to show $(0, 0)$; test the ratio $y \div x$ for every pair.
$1.8 \div 1.5 = 1.2$, $2.4 \div 2 = 1.2$, $3 \div 2.5 = 1.2$, $6 \div 5 = 1.2$.
Every ratio is 1.2, so the relationship is proportional and the constant of proportionality is 1.2.

Re-check strip (3 items, C-AR.4.1-R)

1. Regular **B. A store sells grapes at $1.85 per pound; C. A recipe uses 7 cups of flour for every 4 loaves of bread.**

A relationship is proportional only when the two quantities have a constant ratio and both start at 0.
Proportional: "A store sells grapes at $1.85 per pound." — the same price for every pound, and 0 pounds cost $0; "A recipe uses 7 cups of flour for every 4 loaves of bread." — the same flour for every loaf, and 0 loaves need 0 cups.
Not proportional: "A gym charges $38 to join plus $11 each month." — the $38 joining fee is paid before the first month; "A phone plan costs $34 plus $0.15 for each text message." — the $34 is charged even with 0 texts; "A candle is 9 inches tall and burns down 1 inch each hour (its height compared with the hours burned)." — the candle is 9 inches tall at hour 0.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

2. Regular **D. Yes, because $y \div x$ is the same for every pair of values.**

Check $y \div x$ for each pair: $3 \div 1 = 3$, $9 \div 3 = 3$, $12 \div 4 = 3$, $15 \div 5 = 3$.
Every ratio is the same, so the relationship is proportional.

3. Regular **C. No, because there is a charge of 15 dollars before the first pound of apples.**

A proportional relationship must pass through $(0, 0)$ and have a constant ratio.
The starting charge of 15 means 0 pounds of apples already gives 15 dollars, so it is not proportional.

Comparison strip (2 items, C-AR.4.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **A. An empty tank fills with 4 liters of water each minute; C. A worker earns $22 for each hour worked; E. A store sells oranges at $1.60 per pound.**

A relationship is proportional only when the two quantities have a constant ratio and both start at 0.
Proportional: "A store sells oranges at $1.60 per pound." — the same price for every pound, and 0 pounds cost $0; "A worker earns $22 for each hour worked." — the same pay every hour, and 0 hours earn $0; "An empty tank fills with 4 liters of water each minute." — the same liters every minute, and the tank starts empty.
Not proportional: "A taxi charges $2.00 plus $3.25 per mile." — the $2.00 charge is paid before the first mile; "A plant is 7 inches tall and grows 1 inch each week (its height compared with the number of weeks)." — the plant is already 7 inches tall at week 0.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

2. Regular **86**

Every other pair gives $y \div x = 11$.
For $x = 8$, a proportional relationship needs $y = 88$, not 86.

Reteach plan — C-AR.4.1

MA.7.AR.4.1 · Is it proportional? (table, graph, description)

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.4.1-R

Benchmark: **MA.7.AR.4.1** · Is it proportional? (table, graph, description) Student code: _____ Calculator: yes

1. Select **all** the situations that describe a proportional relationship.

Select **all** the correct answers.

A. ☐ A gym charges $38 to join plus $11 each month.

B. ☐ A store sells grapes at $1.85 per pound.

C. ☐ A recipe uses 7 cups of flour for every 4 loaves of bread.

D. ☐ A candle is 9 inches tall and burns down 1 inch each hour (its height compared with the hours burned).

E. ☐ A phone plan costs $34 plus $0.15 for each text message.

2. Does the table show a proportional relationship between x and y?

x	y
1	3
3	9
4	12
5	15

A. Yes, because y increases whenever x increases.

B. No, because $y \div x$ is not the same for every pair of values.

C. No, because the x-values do not increase by the same amount.

D. Yes, because $y \div x$ is the same for every pair of values.

3. A stand charges 15 dollars plus 12 dollars for each pound of apples.

Is the relationship between the pounds of apples and the total dollars proportional?

A. Yes, because the dollars added for each pound of apples never change.

B. No, because the dollars and the pounds of apples are measured in different units.

C. No, because there is a charge of 15 dollars before the first pound of apples.

D. Yes, because there is no charge at the start, only 12 dollars each time.

Score: ____ / 3 → copy to your progress strip

Benchmark: **MA.7.AR.4.1** · Is it proportional? (table, graph, description) Student code: _____ Calculator: yes

1. Select **all** the situations that describe a proportional relationship.

Select **all** the correct answers.

A. ☐ An empty tank fills with 4 liters of water each minute.

B. ☐ A plant is 7 inches tall and grows 1 inch each week (its height compared with the number of weeks).

C. ☐ A worker earns $22 for each hour worked.

D. ☐ A taxi charges $2.00 plus $3.25 per mile.

E. ☐ A store sells oranges at $1.60 per pound.

2. The table does **not** show a proportional relationship.

x	y
2	22
4	44
6	66
8	86

Which value of y would have to change for the relationship to be proportional?

Score: ____ / 2 → **copy to your progress strip**

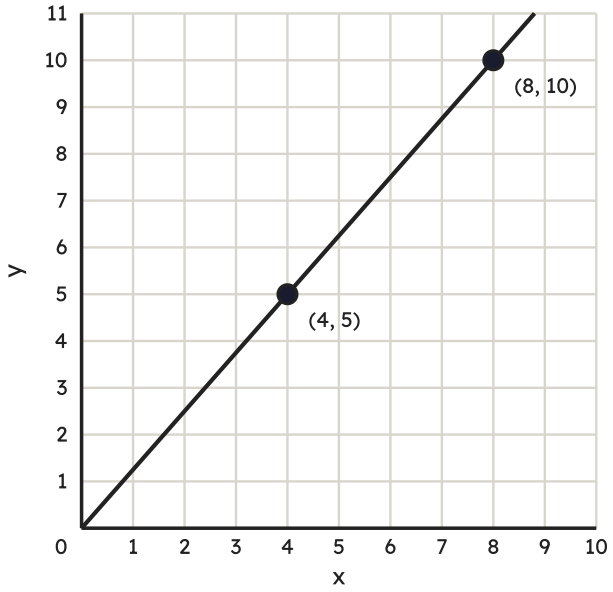
Benchmark: **MA.7.AR.4.4** · Translate between representations Student code: _____ Calculator: yes

Learning goal: I can write an equation for a proportional relationship. (circle your level after scoring)

4	I can translate between a table, a graph, an equation, and a description of the same relationship.
3	I can write the equation y = kx from a table, graph, or description and use it.
2	I can find the constant of proportionality.
1	I need help to start this benchmark.

1. The graph shows a proportional relationship between batches made (x) and cups of flour used (y).

Write an equation for this relationship.



2. A recipe uses $\frac{5}{4}$ cup of sugar for each batch of muffins. The relationship is proportional. Let x be the number of batches and let y be the number of cups of sugar.

Which equation represents this relationship?

A. $y = \left(\frac{5}{4}\right) x$

B. $x = \left(\frac{5}{4}\right) y$

C. $y = x + \frac{5}{4}$

D. $y = \left(\frac{4}{5}\right) x$

Turn over →

Benchmark: **MA.7.AR.4.4** · Translate between representations Student code: _____

3. The table shows a proportional relationship between x and y.

x	y
2	6
5	15
7	21

Write an equation for this relationship.

4. A machine fills 4.5 bottles each minute that it runs, on average. The relationship between the number of minutes and the number of bottles is proportional.

Number of minutes	Number of bottles
6	?
8	?
10	?

What are the three missing values in the table?

5. Diego earns \$11 per hour walking dogs. The relationship between the money earned, e, and the hours worked, h, is proportional.

Which equation gives the number of hours, h, that Diego must work to earn e dollars?

A. $h = 11e$

B. $h = e - 11$

C. $e = \left(\frac{1}{11}\right) h$

D. $h = \left(\frac{1}{11}\right) e$

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.4.4 · MA.7.AR.4.4 · Translate between representations

Unit 11: Representing Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $y = 1.25x$

Use a labeled point: $(4,\ 5)$. The constant of proportionality is $5 \div 4 = 1.25$.
The equation is $y = 1.25x$.

2. Regular **A.** $y = \left(\frac{5}{4} = 1\frac{1}{4}\right)x$

For each one of x, y increases by $\frac{5}{4}$, so the constant of proportionality is $\frac{5}{4}$ and $y = \left(\frac{5}{4}\right)x$.
$y = x + \frac{5}{4}$ would add $\frac{5}{4}$ once; $x = \left(\frac{5}{4}\right)y$ and $y = \left(\frac{4}{5}\right)x$ reverse the rate.

3. Regular $y = 3x$

The constant of proportionality is $y \div x = 6 \div 2 = 3$.
A proportional relationship has the form $y = kx$, so $y = 3x$.

4. Regular **27, 36, 45. Score: 1 point for all three values; two values earn 0.**

Multiply each number of minutes by 4.5, the number of bottles for one minute.
$4.5 \times 6 = 27$
$4.5 \times 8 = 36$
$4.5 \times 10 = 45$

5. Rigor **D.** $h = \left(\frac{1}{11}\right)e$

Each hour earns \$11, so each dollar takes $\frac{1}{11}$ of an hour.
$h = \left(\frac{1}{11}\right)e$. This is the same relationship as $e = 11h$, solved for h.

Re-check strip (3 items, C-AR.4.4-R)

1. Regular $y = 3.75x$

Use a labeled point: $(4,\ 15)$. The constant of proportionality is $15 \div 4 = 3.75$.
The equation is $y = 3.75x$.

2. Regular **A. The printer prints 18 pages each minute.**

In $y = 18x$, the constant 18 is the number of pages for each minute of printing: the printer prints 18 pages each minute.
The relationship is not additive (18 is not added on each time), 18 is not a total, and the rate is pages per minute of printing, not minutes of printing per page.

3. Regular **C.** $y = \left(\frac{1}{2}\right)x$

For each one of x, y increases by $\frac{1}{2}$, so the constant of proportionality is $\frac{1}{2}$ and $y = \left(\frac{1}{2}\right)x$.
$y = x + \frac{1}{2}$ would add $\frac{1}{2}$ once; $x = \left(\frac{1}{2}\right)y$ and $y = 2x$ reverse the rate.

Comparison strip (2 items, C-AR.4.4-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $y = 2.5x$

Use a labeled point: $(4,\ 10)$. The constant of proportionality is $10 \div 4 = 2.5$.
The equation is $y = 2.5x$.

2. Regular **A.** $y = \left(\frac{3}{2} = 1\frac{1}{2}\right)x$

For each one of x, y increases by $\frac{3}{2}$, so the constant of proportionality is $\frac{3}{2}$ and $y = \left(\frac{3}{2}\right)x$.
$y = x + \frac{3}{2}$ would add $\frac{3}{2}$ once; $x = \left(\frac{3}{2}\right)y$ and $y = \left(\frac{2}{3}\right)x$ reverse the rate.

Reteach plan — C-AR.4.4

MA.7.AR.4.4 · Translate between representations

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

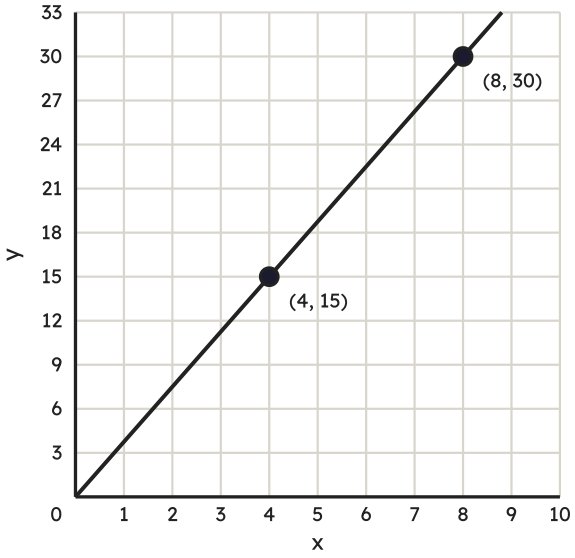
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.4.4-R

Benchmark: **MA.7.AR.4.4** · Translate between representations Student code: _____ Calculator: yes

1. The graph shows a proportional relationship between hours (x) and miles walked (y).

Write an equation for this relationship.



2. A printer prints at a constant rate. The proportional relationship is represented by $y = 18x$, where x is the number of minutes of printing and y is the number of pages.

Which description matches this relationship?

A. The printer prints 18 pages each minute.

B. The printer prints 18 more pages each minute than the minute before.

C. The printer takes 18 minutes for each page.

D. The printer prints 18 pages in all.

3. A gardener uses $\frac{1}{2}$ bag of soil for each planter. The relationship is proportional. Let x be the number of planters and let y be the number of bags of soil.

Which equation represents this relationship?

A. $y = 2x$

B. $x = \left(\frac{1}{2}\right) y$

C. $y = \left(\frac{1}{2}\right) x$

D. $y = x + \frac{1}{2}$

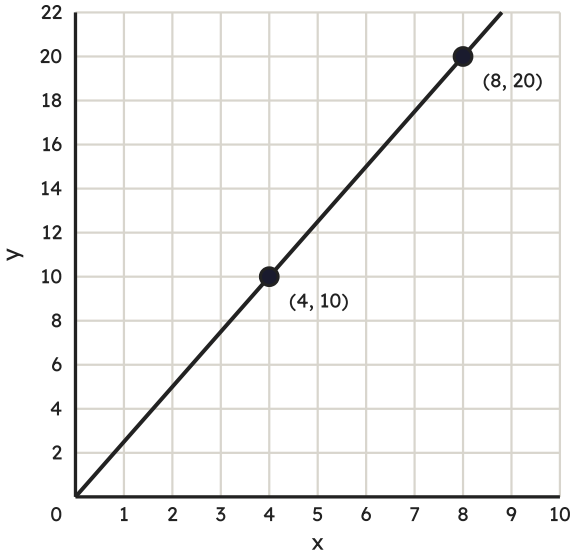
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-AR.4.4-X

Benchmark: **MA.7.AR.4.4** · Translate between representations Student code: _____ Calculator: yes

1. The graph shows a proportional relationship between minutes (x) and pages read (y).

Write an equation for this relationship.



2. A recipe uses $\frac{3}{2}$ cup of sugar for each batch of muffins. The relationship is proportional. Let x be the number of batches and let y be the number of cups of sugar.

Which equation represents this relationship?

A. $y = \left(\frac{3}{2}\right) x$

B. $y = \left(\frac{2}{3}\right) x$

C. $x = \left(\frac{3}{2}\right) y$

D. $y = x + \frac{3}{2}$

Score: ____ / 2 → copy to your progress strip

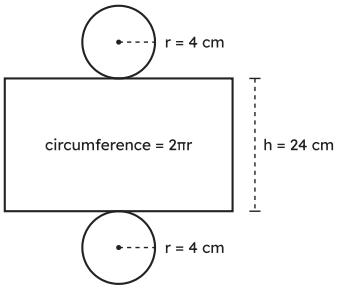
Benchmark: **MA.7.GR.2.1** · Surface area of a cylinder from its net Student code: _____ Calculator: yes

Learning goal: I can find the surface area of a right circular cylinder. (circle your level after scoring)

4	I can explain how the net gives the formula and find surface area with a missing dimension.
3	I can find the surface area of a cylinder from its radius and height.
2	I can identify the parts of a cylinder's net.
1	I need help to start this benchmark.

1. The net of a right circular cylinder is shown. The radius is 4 centimeters and the height is 24 centimeters.

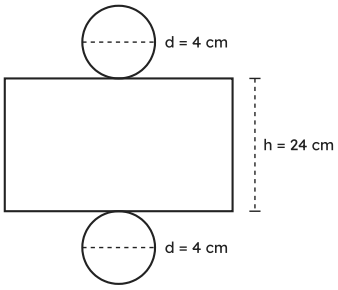
What is the surface area of the cylinder? Write your answer in terms of π.



Not drawn to scale

2. The net of a right circular cylinder is shown. Each circle has a diameter of 4 centimeters, and the rectangle is 24 centimeters tall.

What is the surface area of the cylinder? Write your answer in terms of π.

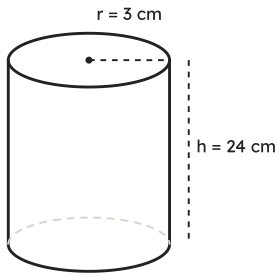


Not drawn to scale

Turn over →

3. The right circular cylinder shown has a radius of 3 centimeters and a height of 24 centimeters.

Which net, with its dimensions, matches the cylinder?



A. Two circles of radius 3 cm and a 3π cm by 24 cm rectangle

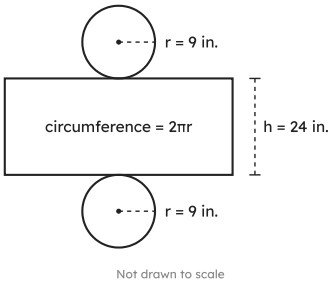
B. Two circles of radius 3 cm and a 6 cm by 24 cm rectangle

C. Two circles of radius 3 cm and a 6π cm by 24 cm rectangle

D. One circle of radius 3 cm and a 6π cm by 24 cm rectangle

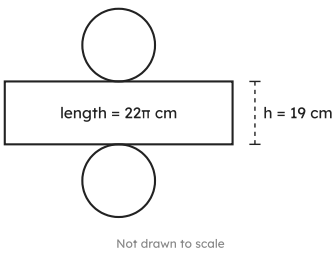
4. The net of a right circular cylinder is shown. The circles have a radius of 9 inches and the rectangle is 24 inches tall.

What is the area of the rectangle, in terms of π?



5. The net of a right circular cylinder is shown. The rectangle is 22π centimeters long and 19 centimeters tall.

What is the surface area of the cylinder? Write your answer in terms of π.



Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — C-GR.2.1 · MA.7.GR.2.1 · Surface area of a cylinder from its net

Unit 12: Surface Area and Volume. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **224π square centimeters**

Two bases: $2 \times \pi \times 4^2 = 32\pi$.
The rectangle: $2\pi \times 4 \times 24 = 192\pi$.
Total: $32\pi + 192\pi = 224\pi$ square centimeters.

2. Regular **104π square centimeters**

The radius is $4 \div 2 = 2$ centimeters.
Two circles: $2 \times \pi(2)^2 = 8\pi$.
The rectangle wraps around a circle, so its length is the circumference $2\pi(2) = 4\pi$ centimeters: $4\pi \times 24 = 96\pi$.
Surface area: $8\pi + 96\pi = 104\pi$ square centimeters.

3. Regular **C. Two circles of radius 3 cm and a 6π cm by 24 cm rectangle**

A cylinder has two circular bases, so the net has two circles of radius 3 cm.
The rectangle wraps around a base, so its length is the circumference, $2\pi(3) = 6\pi$ cm, and its width is the height, 24 cm.

4. Regular **432π square inches**

The rectangle wraps around the circle, so its width is the circumference: $2\pi r = 18\pi$ inches.
$\text{Area} = 18\pi \times 24 = 432\pi$ square inches.

5. Rigor **660π square centimeters**

The rectangle wraps around a base, so its length is the circumference of a circle: $2\pi r = 22\pi$, so $r = 22 \div 2 = 11$ centimeters.
Two circles: $2 \times \pi(11)^2 = 242\pi$.
Rectangle: $22\pi \times 19 = 418\pi$.
Surface area: $242\pi + 418\pi = 660\pi$ square centimeters.

Re-check strip (3 items, C-GR.2.1-R)

1. Regular **168π square centimeters**

Two bases: $2 \times \pi \times 6^2 = 72\pi$.
The rectangle: $2\pi \times 6 \times 8 = 96\pi$.
Total: $72\pi + 96\pi = 168\pi$ square centimeters.

2. Regular **160π square centimeters**

The radius is $8 \div 2 = 4$ centimeters.
Two circles: $2 \times \pi(4)^2 = 32\pi$.
The rectangle wraps around a circle, so its length is the circumference $2\pi(4) = 8\pi$ centimeters: $8\pi \times 16 = 128\pi$.
Surface area: $32\pi + 128\pi = 160\pi$ square centimeters.

3. Regular **84π square inches**

The rectangle wraps around the circle, so its width is the circumference: $2\pi r = 4\pi$ inches.
$\text{Area} = 4\pi \times 21 = 84\pi$ square inches.

Comparison strip (2 items, C-GR.2.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **182π square inches**

The rectangle wraps around the circle, so its width is the circumference: $2\pi r = 14\pi$ inches.
$\text{Area} = 14\pi \times 13 = 182\pi$ square inches.

2. Regular **414π square centimeters**

The radius is $18 \div 2 = 9$ centimeters.
Two circles: $2 \times \pi(9)^2 = 162\pi$.
The rectangle wraps around a circle, so its length is the circumference $2\pi(9) = 18\pi$ centimeters: $18\pi \times 14 = 252\pi$.
Surface area: $162\pi + 252\pi = 414\pi$ square centimeters.

Reteach plan — C-GR.2.1

MA.7.GR.2.1 · Surface area of a cylinder from its net

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

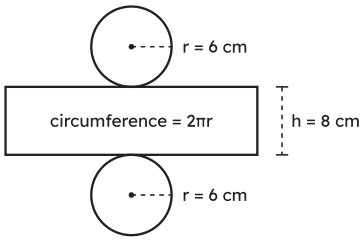
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.2.1-R

Benchmark: **MA.7.GR.2.1** · Surface area of a cylinder from its net Student code: _____ Calculator: yes

1. The net of a right circular cylinder is shown. The radius is 6 centimeters and the height is 8 centimeters.

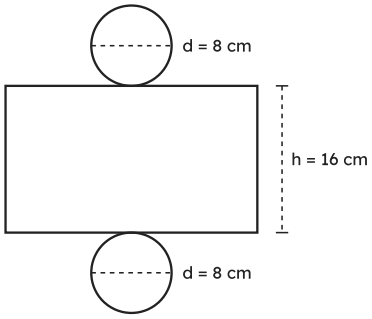
What is the surface area of the cylinder? Write your answer in terms of π.



Not drawn to scale

2. The net of a right circular cylinder is shown. Each circle has a diameter of 8 centimeters, and the rectangle is 16 centimeters tall.

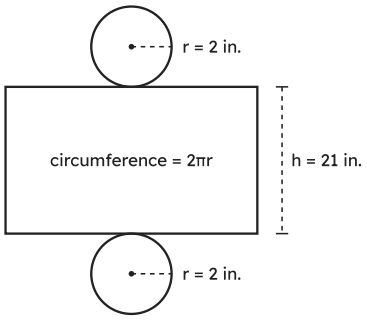
What is the surface area of the cylinder? Write your answer in terms of π.



Not drawn to scale

3. The net of a right circular cylinder is shown. The circles have a radius of 2 inches and the rectangle is 21 inches tall.

What is the area of the rectangle, in terms of π?



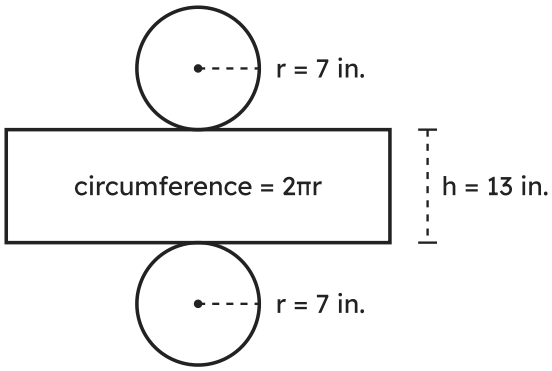
Not drawn to scale

Score: ____ / 3 → **copy to your progress strip**

Benchmark: **MA.7.GR.2.1** · Surface area of a cylinder from its net Student code: _____ Calculator: yes

1. The net of a right circular cylinder is shown. The circles have a radius of 7 inches and the rectangle is 13 inches tall.

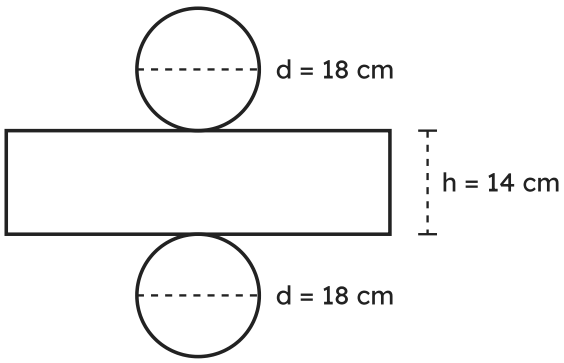
What is the area of the rectangle, in terms of π?



Not drawn to scale

2. The net of a right circular cylinder is shown. Each circle has a diameter of 18 centimeters, and the rectangle is 14 centimeters tall.

What is the surface area of the cylinder? Write your answer in terms of π.



Not drawn to scale

Score: ____ / 2 **→ copy to your progress strip**

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____ Calculator: yes

Learning goal: I can solve problems involving surface area of cylinders. (circle your level after scoring)

4	I can solve a multi-step surface-area problem, such as a label or an open-top can.
3	I can solve a real-world problem with the surface area of a cylinder.
2	I can find the area of a circle and a rectangle.
1	I need help to start this benchmark.

1. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical water tank has a radius of 9 feet and a height of 17 feet. Its whole outside surface, including the top and the bottom, will be covered with sheet metal.

What is the area that the sheet metal must cover? Write your answer in terms of π.

2. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical container is made of plastic. It has a radius of 3 inches and a height of 14 inches, and it has a bottom but no top.

What is the area of the plastic needed to make the container? Write your answer in terms of π.

Turn over →

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____

3. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical gift tin with a lid has a radius of 3 inches and a height of 4 inches. The whole tin, including the lid and the bottom, is covered with decorative paper.

What is the area the paper covers, in square inches?

A. 42π

B. 30π

C. 33π

D. 24π

4. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical oatmeal container has a radius of 5 inches and a height of 11 inches. A label wraps all the way around its curved side, with no overlap; the top and the bottom are not covered.

What is the area of the label? Round your answer to the nearest hundredth. Use the π key on your calculator.

5. Surface area of a right circular cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical water tank stands on the ground. It has a radius of 7 feet and a height of 17 feet. The curved side and the top will be painted, but not the bottom. One gallon of paint covers 400 square feet, and paint is sold only in whole gallons.

How many gallons of paint must be bought? Use the π key on your calculator.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.2.2 · MA.7.GR.2.2 · Real-world cylinder surface area

Unit 12: Surface Area and Volume. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **468π square feet**

The sheet metal covers both bases and the curved surface: the whole surface area.

$\text{SA} = 2\pi(9)^2 + 2\pi(9)(17) = 162\pi + 306\pi = 468\pi$ square feet.

2. Regular **93π square inches**

With no top, the container has one base and the curved surface.

One base: $\pi(3)^2 = 9\pi$.

Curved surface: $2\pi(3)(14) = 84\pi$.

Total: $9\pi + 84\pi = 93\pi$ square inches.

3. Regular **A.** 42π

The paper covers both circular ends and the curved side: the whole surface area.

$\text{SA} = 2\pi(3)^2 + 2\pi(3)(4) = 18\pi + 24\pi = 42\pi$.

4. Regular **345.58 square inches**

The label covers only the curved side, the $2\pi rh$ part of the formula; the $2\pi r^2$ part (the top and the bottom) is not included: $2\pi(5)(11) = 110\pi$.

With the π key: $110 \times \pi \approx 345.58$ square inches.

5. Rigor **3 gallons**

The painted surface is one base and the curved surface: $\pi(7)^2 + 2\pi(7)(17) = 49\pi + 238\pi = 287\pi$ square feet.

$287\pi \approx 901.64$ square feet.

$901.64 \div 400 \approx 2.25$, so 2 gallons would not be enough.

3 gallons must be bought.

Re-check strip (3 items, C-GR.2.2-R)

1. Regular **455.53 square inches**

The planter has one circular base and the curved side, so only one πr^2 is used: $\pi(5)^2 + 2\pi(5)(12) = 25\pi + 120\pi = 145\pi$.

With the π key: $145 \times \pi \approx 455.53$ square inches.

2. Regular **112π square centimeters**

The label covers only the curved side, the $2\pi rh$ part of the formula; the $2\pi r^2$ part is the top and the bottom.

$2\pi(7)(8) = 112\pi$ square centimeters.

3. Regular **462π square feet**

The sheet metal covers both bases and the curved surface: the whole surface area.

$\text{SA} = 2\pi(11)^2 + 2\pi(11)(10) = 242\pi + 220\pi = 462\pi$ square feet.

Comparison strip (2 items, C-GR.2.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **407π square inches**

With no top, the container has one base and the curved surface.

One base: $\pi(11)^2 = 121\pi$.

Curved surface: $2\pi(11)(13) = 286\pi$.

Total: $121\pi + 286\pi = 407\pi$ square inches.

2. Regular **188.40 square inches**

The label covers only the curved side, the $2\pi rh$ part of the formula; the $2\pi r^2$ part (the top and the bottom) is not included: $2\pi(3)(10) = 60\pi$.

With 3.14 for π: $60 \times 3.14 = 188.40$ square inches.

Reteach plan — C-GR.2.2

MA.7.GR.2.2 · Real-world cylinder surface area

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.2.2-R

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____ Calculator: yes

1. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical planter has no top. It has a radius of 5 inches and a height of 12 inches. The outside of the planter, including its bottom, will be painted.

What is the area that will be painted? Round your answer to the nearest hundredth. Use the π key on your calculator.

2. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical can has a radius of 7 centimeters and a height of 8 centimeters. A paper label wraps around the side of the can. It does not cover the top or the bottom.

What is the area of the label? Write your answer in terms of π.

3. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical water tank has a radius of 11 feet and a height of 10 feet. Its whole outside surface, including the top and the bottom, will be covered with sheet metal.

What is the area that the sheet metal must cover? Write your answer in terms of π.

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-GR.2.2-X

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____ Calculator: yes

1. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical container is made of plastic. It has a radius of 11 inches and a height of 13 inches, and it has a bottom but no top.

What is the area of the plastic needed to make the container? Write your answer in terms of π.

2. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical oatmeal container has a radius of 3 inches and a height of 10 inches. A label wraps all the way around its curved side, with no overlap; the top and the bottom are not covered.

What is the area of the label? Round your answer to the nearest hundredth. Use 3.14 for π.

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-GR.2.2-X

Benchmark: **MA.7.GR.2.2** · Real-world cylinder surface area Student code: _____ Calculator: yes

1. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical container is made of plastic. It has a radius of 11 inches and a height of 13 inches, and it has a bottom but no top.

What is the area of the plastic needed to make the container? Write your answer in terms of π.

2. Surface area of a right circular cylinder: $\mathrm{SA} = 2\pi r^2 + 2\pi rh$

A cylindrical oatmeal container has a radius of 3 inches and a height of 10 inches. A label wraps all the way around its curved side, with no overlap; the top and the bottom are not covered.

What is the area of the label? Round your answer to the nearest hundredth. Use 3.14 for π.

Score: ____ **/ 2** → **copy to your progress strip**

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder Student code: _____ Calculator: yes

Learning goal: I can find the volume of a right circular cylinder. (circle your level after scoring)

4	I can solve a real-world volume problem, including comparing two cylinders.
3	I can find the volume of a cylinder from its radius or diameter and height.
2	I can find the area of the base.
1	I need help to start this benchmark.

1. A cylindrical tank has a radius of 8 feet and a height of 12 feet.

What is the volume of the tank? Round your answer to the nearest hundredth. Use 3.14 for π.

2. A cylindrical container has a diameter of 16 inches and a height of 24 inches.

What is the volume of the container? Write your answer in terms of π.

Turn over →

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder Student code: _____

3. A right circular cylinder has a radius of 6 units and a height of 16 units.

What is the volume of the cylinder, in cubic units?

A. 192π

B. 1152π

C. 576π

D. 96π

4. A cylindrical trash can has a diameter of 20 inches and a height of 26 inches.

What is the volume of the trash can? Round your answer to the nearest hundredth. Use 3.14 for π.

5. A cylindrical storage tank has a volume of 478 cubic feet and a radius of 3.2 feet.

What is the height of the storage tank? Round your answer to the nearest hundredth. Use the π key on your calculator.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.2.3 · MA.7.GR.2.3 · Volume of a cylinder

Unit 12: Surface Area and Volume. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **2411.52 cubic feet**

$V = \pi r^2 h = \pi(8)^2(12) = 768\pi$.

With 3.14 for π: $768 \times 3.14 = 2411.52$ cubic feet.

2. Regular **1536π cubic inches**

The radius is $16 \div 2 = 8$ inches.

$V = \pi r^2 h = \pi(8)^2(24) = 1536\pi$ cubic inches.

3. Regular **C. 576π**

$V = \pi r^2 h = \pi(6)^2(16) = 576\pi$.

The value 192π would be the curved surface area instead.

4. Regular **8164 cubic inches (8164.00 also accepted)**

The radius is $20 \div 2 = 10$ inches.

$V = \pi r^2 h = \pi(10)^2(26) = 2600\pi$.

With 3.14 for π: $2600 \times 3.14 = 8164.00$ cubic inches.

5. Rigor **14.86 feet**

$V = \pi r^2 h$, so $478 = \pi(3.2)^2 h = 10.24\pi h$.

$h = 478 \div (10.24\pi) \approx 14.86$ feet with the π key.

Re-check strip (3 items, C-GR.2.3-R)

1. Regular **C. 256π**

$V = \pi r^2 h = \pi(8)^2(4) = 256\pi$.

The value 64π would be the curved surface area instead.

2. Regular **339.29 cubic feet**

The radius is $12 \div 2 = 6$ feet.

$V = \pi r^2 h = \pi(6)^2(3) = 108\pi$.

With the π key: $108 \times \pi \approx 339.29$ cubic feet.

3. Regular **384π cubic inches**

The radius is $8 \div 2 = 4$ inches.

$V = \pi r^2 h = \pi(4)^2(24) = 384\pi$ cubic inches.

Comparison strip (2 items, C-GR.2.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **1005.31 cubic feet**

$V = \pi r^2 h = \pi(8)^2(5) = 320\pi$.

With the π key: $320 \times \pi \approx 1005.31$ cubic feet.

2. Regular **64π cubic inches**

The radius is $4 \div 2 = 2$ inches.

$V = \pi r^2 h = \pi(2)^2(16) = 64\pi$ cubic inches.

Reteach plan — C-GR.2.3

MA.7.GR.2.3 · Volume of a cylinder

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items)

C-GR.2.3-R

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder

Student code: _____ Calculator: yes

1. A right circular cylinder has a radius of 8 units and a height of 4 units.

What is the volume of the cylinder, in cubic units?

A. 64π

B. 32π

C. 256π

D. 512π

2. A cylindrical swimming pool has a diameter of 12 feet and a height of 3 feet.

What is the volume of the pool? Round your answer to the nearest hundredth. Use the π key on your calculator.

3. A cylindrical container has a diameter of 8 inches and a height of 24 inches.

What is the volume of the container? Write your answer in terms of π.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items)

C-GR.2.3-R

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder

Student code: _____ Calculator: yes

1. A right circular cylinder has a radius of 8 units and a height of 4 units.

What is the volume of the cylinder, in cubic units?

A. 64π

B. 32π

C. 256π

D. 512π

2. A cylindrical swimming pool has a diameter of 12 feet and a height of 3 feet.

What is the volume of the pool? Round your answer to the nearest hundredth. Use the π key on your calculator.

3. A cylindrical container has a diameter of 8 inches and a height of 24 inches.

What is the volume of the container? Write your answer in terms of π.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-GR.2.3-X

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder

Student code: _____ Calculator: yes

1. A cylindrical tank has a radius of 8 feet and a height of 5 feet.

What is the volume of the tank? Round your answer to the nearest hundredth. Use the π key on your calculator.

2. A cylindrical container has a diameter of 4 inches and a height of 16 inches.

What is the volume of the container? Write your answer in terms of π.

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-GR.2.3-X

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder

Student code: _____ Calculator: yes

1. A cylindrical tank has a radius of 8 feet and a height of 5 feet.

What is the volume of the tank? Round your answer to the nearest hundredth. Use the π key on your calculator.

2. A cylindrical container has a diameter of 4 inches and a height of 16 inches.

What is the volume of the container? Write your answer in terms of π.

Score: ____ / 2 → **copy to your progress strip**

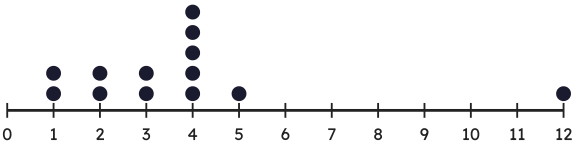
Benchmark: **MA.7.DP.1.1** · Choose a measure of center or variation Student code: _____ Calculator: yes

Learning goal: I can choose and justify measures of center and spread. (circle your level after scoring)

4	I can explain how an outlier changes the mean, median, and range.
3	I can choose the best measure of center or spread for a data set and explain why.
2	I can find the mean, median, and range of a data set.
1	I need help to start this benchmark.

1. The line plot shows the number of pets in each household surveyed.

Which measure of center best describes a typical value in this data set, and why?



A. the median, because one value is far from the rest and pulls the mean away

B. the mean, because every value is close to the rest, with no outlier

C. the range, because it shows how spread out the data is

D. the mean, because it is easier to find on a line plot than the median

2. The numbers of minutes a student practiced piano on six days are 32, 41, 50, 57, 64, 186.

Which measure of variation better describes the spread of the data, and why?

A. the range, because there is no outlier to stretch it

B. the mean, because it uses every value in the data set

C. the interquartile range, because an outlier stretches the range

D. the range, because an outlier makes the spread easier to see

3. The prices of the cars in a lot, in thousands of dollars, are 25, 29, 39, 42, 53, 243.

Which measure of center best describes a typical value in this data set?

A. the mean, because no value is far enough away to pull it off center

B. the range, because it shows how spread out the data is

C. the mean, because only the middle value affects it

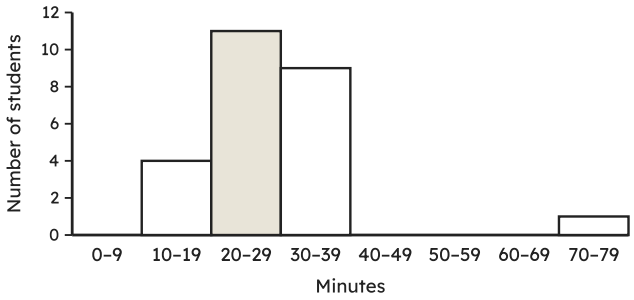
D. the median, because an outlier pulls the mean away from the rest

Turn over →

Benchmark: **MA.7.DP.1.1** · Choose a measure of center or variation Student code: _____

4. The histogram shows the numbers of minutes 25 students spent on homework last night.

Which measure of center best describes a typical value in this data set, and why?



A. the range, because one value is far from the rest of the data

B. the mean, because no value is far from the rest of the data

C. the median, because one value is far from the rest of the data

D. the mean, because one value is far from the rest of the data

5. The prices, in dollars, of eight used bikes at a yard sale are 90, 30, 34, 26, 28, 22, 32, 30.

One value in this data set is an outlier. Select **all** the statements that are true.

Select **all** the correct answers.

A. ☐ Removing 90 from the data set would change the mean more than it would change the median.

B. ☐ Removing 90 from the data set would not change the range.

C. ☐ The range is more than twice the interquartile range.

D. ☐ The mean is greater than the median.

E. ☐ The interquartile range is more than half of the range.

F. ☐ Removing 90 from the data set would change the median more than it would change the mean.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.1 · MA.7.DP.1.1 · Choose a measure of center or variation

Unit 13: Numerical and Categorical Data. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. the median, because one value is far from the rest and pulls the mean away**

One dot sits at 12, far to the right of the cluster between 1 and 5.
That one value pulls the mean toward it, but it does not move the middle value, so the median describes a typical value better.

2. Regular **C. the interquartile range, because an outlier stretches the range**

186 is far from every other value, so it is an outlier.
The range, $186 - 32 = 154$ minutes, is stretched by the outlier. The interquartile range uses only the middle half of the ordered data, $Q_3 - Q_1 = 64 - 41 = 23$ minutes, so the outlier does not stretch it.
The interquartile range describes the spread better; it goes with the median as the measure of center.

3. Regular **D. the median, because an outlier pulls the mean away from the rest**

243 is far greater than every other value, so it is an outlier.
An outlier pulls the mean toward it, but the median, the middle of the ordered data, stays with the other values. The median describes a typical value better.

4. Regular **C. the median, because one value is far from the rest of the data**

One value, in the 70–79 interval, is separated from all the others by three empty intervals, so it is an outlier.
An outlier pulls the mean toward it, but the median, the middle of the ordered data, stays with the other values. The median describes a typical value better.

5. Rigor **A. Removing 90 from the data set would change the mean more than it would change the median; C. The range is more than twice the interquartile range; D. The mean is greater than the median.**

In order: 22, 26, 28, 30, 30, 32, 34, 90. The outlier is 90.
$\text{Mean} = 292 \div 8 = 36.5$; $\text{median} = (30 + 30) \div 2 = 30$; $Q_1 = 27$ and $Q_3 = 33$, so the interquartile range is 6; $\text{range} = 90 - 22 = 68$.
Without 90: $\text{mean} = 202 \div 7 \approx 28.86$, $\text{median} = 30$, $\text{range} = 12$. The mean moves by about 7.64 and the median by 0.
True: The mean is greater than the median. Removing 90 from the data set would change the mean more than it would change the median. The range is more than twice the interquartile range.
Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Re-check strip (3 items, C-DP.1.1-R)

1. Regular **B. the median, because an outlier pulls the mean away from the rest**

210 is far greater than every other value, so it is an outlier.
An outlier pulls the mean toward it, but the median, the middle of the ordered data, stays with the other values. The median describes a typical value better.

2. Regular **D. the mean, because no value is far from the rest of the data**

The bars form one unbroken group, and no value is set apart from the rest, so the data set has no outlier.
With no outlier to pull it off center, the mean uses every value and describes a typical value well.

3. Regular **A. the interquartile range, because one long whisker stretches the range**

The box holds the middle half of the data: $Q_3 - Q_1 = 30 - 20 = 10$ minutes.
The right whisker reaches 65, which is 35 minutes from the box, more than three times the width of the box. The highest data lie far from the middle half of the data.
Those data stretch the range, $65 - 15 = 50$ minutes, but the interquartile range describes only the middle half and is not affected by them, so the interquartile range describes the spread better.

Comparison strip (2 items, C-DP.1.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **A. the interquartile range, because an outlier stretches the range**

153 is far from every other value, so it is an outlier.
The range, $153 - 33 = 120$ customers, is stretched by the outlier. The interquartile range uses only the middle half of the ordered data, $Q_3 - Q_1 = 59 - 43 = 16$ customers, so the outlier does not stretch it.
The interquartile range describes the spread better; it goes with the median as the measure of center.

2. Regular **B. the median, because one value is far from the rest of the data**

One value, in the 70–79 interval, is separated from all the others by three empty intervals, so it is an outlier.
An outlier pulls the mean toward it, but the median, the middle of the ordered data, stays with the other values. The median describes a typical value better.

Reteach plan — C-DP.1.1

MA.7.DP.1.1 · Choose a measure of center or variation

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.1.1-R

Benchmark: **MA.7.DP.1.1** · Choose a measure of center or variation Student code: _____ Calculator: yes

1. The yearly salaries at a small shop, in thousands of dollars, are 26, 28, 34, 46, 52, 210.

Which measure of center best describes a typical value in this data set?

A. the mean, because no value is far enough away to pull it off center

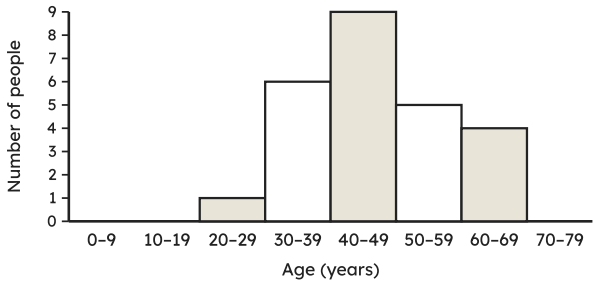
B. the median, because an outlier pulls the mean away from the rest

C. the median, because it is easier to calculate than the mean

D. the mean, because only the middle value affects it

2. The histogram shows the ages of the 25 people at a family picnic.

Which measure of center best describes a typical value in this data set, and why?



A. the mean, because one value is far from the rest of the data

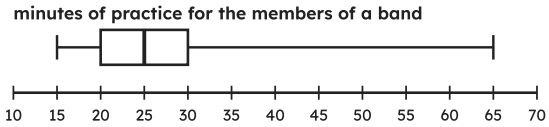
B. the range, because no value is far from the rest of the data

C. the median, because one value is far from the rest of the data

D. the mean, because no value is far from the rest of the data

3. The box plot shows the minutes of practice for the members of a band.

Which measure of variation better describes the spread of the data, and why?



A. the interquartile range, because one long whisker stretches the range

B. the median, because it is in the middle of the box

C. the range, because both whiskers are short and nothing stretches it

D. the interquartile range, because it uses every value in the data set

Score: ____ / 3 → copy to your progress strip

Benchmark: **MA.7.DP.1.1** · Choose a measure of center or variation Student code: _____ Calculator: yes

1. The numbers of customers at a food truck on six days are 33, 43, 50, 55, 59, 153.

Which measure of variation better describes the spread of the data, and why?

A. the interquartile range, because an outlier stretches the range

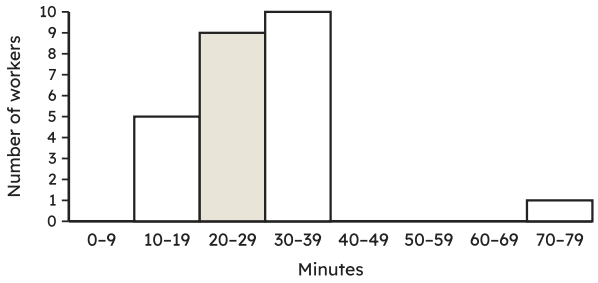
B. the range, because an outlier makes the spread easier to see

C. the range, because there is no outlier to stretch it

D. the median, because it is the middle value of the ordered data

2. The histogram shows the commute times, in minutes, of 25 workers at an office.

Which measure of center best describes a typical value in this data set, and why?



A. the range, because one value is far from the rest of the data

B. the median, because one value is far from the rest of the data

C. the mean, because no value is far from the rest of the data

D. the mean, because one value is far from the rest of the data

Score: ____ / 2 → copy to your progress strip

Benchmark: **MA.7.DP.1.2** · Compare two data sets Student code: _____ Calculator: yes

Learning goal: I can compare two data sets using their displays. (circle your level after scoring)

4	I can make and justify a claim about two populations using their data displays.
3	I can compare two data sets using measures of center and spread from their displays.
2	I can read values from a box plot or histogram.
1	I need help to start this benchmark.

1. Route A: 30, 28, 26, 30, 26
Route B: 21, 20, 24, 20, 25

The lists show the times, in minutes, of five trips on each of two bus routes. Which conclusion do the means of the two data sets support?

A. Trips on Route A typically took about 6 more minutes than trips on Route B.

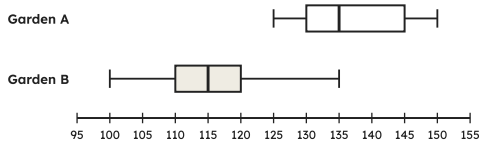
B. Trips on Route A typically took about 30 more minutes than trips on Route B.

C. Trips on Route A typically took about 5 more minutes than trips on Route B.

D. Trips on Route B typically took about 6 more minutes than trips on Route A.

2. The box plots show the heights, in centimeters, of the sunflowers grown in two gardens. The two gardens have the same number of sunflowers.

A sunflower from one of the two gardens is 132 centimeters tall. Which garden did the sunflower more likely come from, and why?



A. Garden A, because 132 is in the middle half for Garden A but in the top fourth for Garden B

B. Garden B, because the median of the data for Garden B is closer to 132 than the median for Garden A

C. Garden B, because 132 is between the least and the greatest values of the data for Garden B

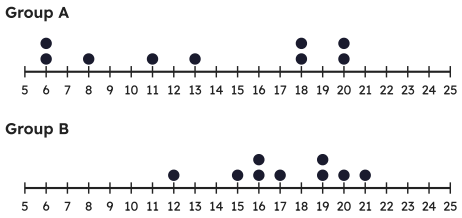
D. Either one is equally likely, because 132 is within the range of the data for both gardens

Turn over →

Benchmark: **MA.7.DP.1.2** · Compare two data sets Student code: _____

3. The line plots show the numbers of pages read last night by the 9 students in each of two reading groups.

Which conclusion do the medians of the two groups support?



A. Students in Group B typically read about 1 more pages than students in Group A.

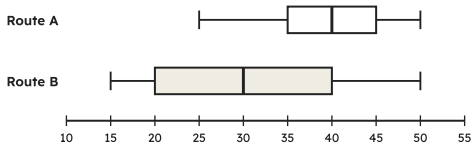
B. Students in Group B typically read about 6 more pages than students in Group A.

C. Students in Group A typically read about 4 more pages than students in Group B.

D. Students in Group B typically read about 4 more pages than students in Group A.

4. The box plots show the times, in minutes, of trips on two bus routes.

Which conclusion about the two routes is supported by the box plots?



A. Trips on Route A typically took longer, and the trip times on Route B varied more.

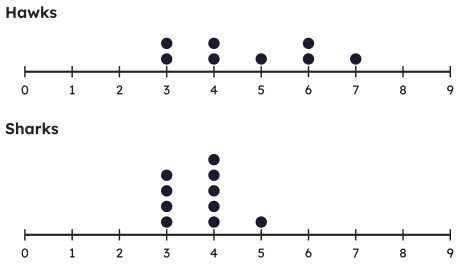
B. Trips on Route A typically took longer, and the trip times on Route A varied more.

C. Trips on the two routes typically took the same amount of time, and the trip times on Route A varied more.

D. Trips on Route B typically took longer, and the trip times on Route B varied more.

5. The line plots show the numbers of goals two soccer teams scored in their games.

Use the mean and the range of each data set. Which conclusion is supported by the data?



A. The Sharks typically scored more goals, and the Hawks' numbers of goals varied more.

B. The Sharks typically scored more goals, and the Sharks' numbers of goals varied more.

C. The Hawks typically scored more goals, and the Sharks' numbers of goals varied more.

D. The Hawks typically scored more goals, and the Hawks' numbers of goals varied more.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.2 · MA.7.DP.1.2 · Compare two data sets

Unit 13: Numerical and Categorical Data. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. Trips on Route A typically took about 6 more minutes than trips on Route B.**

Route A: $\text{mean} = 140 \div 5 = 28$.
Route B: $\text{mean} = 110 \div 5 = 22$.
$28 - 22 = 6$, so: Trips on Route A typically took about 6 more minutes than trips on Route B.

2. Regular **A. Garden A, because 132 is in the middle half for Garden A but in the top fourth for Garden B**

Garden A: the box runs from 130 to 145, so 132 is among the middle half of its values.
Garden B: the upper quartile is 120, so only about one-fourth of its values are greater than 120, and they are spread out from 120 to 135.
The gardens are the same size, and far more values of Garden A lie near 132, so the sunflower is more likely from Garden A.

3. Regular **D. Students in Group B typically read about 4 more pages than students in Group A.**

Each plot has 9 dots, so the median is the 5th dot counting from either end.
Group A: in order, 6, 6, 8, 11, 13, 18, 18, 20, 20; the median is 13.
Group B: in order, 12, 15, 16, 16, 17, 19, 19, 20, 21; the median is 17.
$17 - 13 = 4$, so: Students in Group B typically read about 4 more pages than students in Group A.

4. Regular **A. Trips on Route A typically took longer, and the trip times on Route B varied more.**

Route A: median 40 minutes; interquartile range $45 - 35 = 10$; range $50 - 25 = 25$.
Route B: median 30 minutes; interquartile range $40 - 20 = 20$; range $50 - 15 = 35$.
The median shows the typical value, and the interquartile range and the range show how much the values vary. Trips on Route A typically took longer, and the trip times on Route B varied more.

5. Rigor **D. The Hawks typically scored more goals, and the Hawks' numbers of goals varied more.**

Hawks: $\text{mean} = 38 \div 8 = 4.75$; $\text{range} = 7 - 3 = 4$.
Sharks: $\text{mean} = 37 \div 10 = 3.7$; $\text{range} = 5 - 3 = 2$.
The greater mean shows the greater typical value, and the greater range shows more variation. The Hawks typically scored more goals, and the Hawks' numbers of goals varied more.

Re-check strip (3 items, C-DP.1.2-R)

1. Regular **C. Team A typically scored about 5 more points per game than Team B.**

Team A: $\text{mean} = 250 \div 5 = 50$.
Team B: $\text{mean} = 225 \div 5 = 45$.
$50 - 45 = 5$, so: Team A typically scored about 5 more points per game than Team B.

2. Regular **A. Class A, because 42 is in the middle half for Class A but in the top fourth for Class B**

Class A: the box runs from 35 to 50, so 42 is among the middle half of its values.
Class B: the upper quartile is 25, so only about one-fourth of its values are greater than 25, and they are spread out from 25 to 45.
The classes are the same size, and far more values of Class A lie near 42, so the student is more likely from Class A.

3. Regular **D. Students in the two classes typically scored the same, and the scores in Class A varied more.**

Class A: median 25 points; interquartile range $35 - 15 = 20$; range $40 - 10 = 30$.
Class B: median 25 points; interquartile range $30 - 20 = 10$; range $35 - 10 = 25$.
The median shows the typical value, and the interquartile range and the range show how much the values vary. Students in the two classes typically scored the same, and the scores in Class A varied more.

Comparison strip (2 items, C-DP.1.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **A. Students in Class B typically did about 6 more push-ups than students in Class A.**

Each plot has 9 dots, so the median is the 5th dot counting from either end.
Class A: in order, 5, 5, 7, 9, 12, 13, 13, 15, 19; the median is 12.
Class B: in order, 12, 13, 15, 16, 18, 19, 19, 23, 23; the median is 18.
$18 - 12 = 6$, so: Students in Class B typically did about 6 more push-ups than students in Class A.

2. Regular **C. Garden B, because 139 is in the middle half for Garden B but in the top fourth for Garden A**

Garden B: the box runs from 130 to 145, so 139 is among the middle half of its values.
Garden A: the upper quartile is 125, so only about one-fourth of its values are greater than 125, and they are spread out from 125 to 140.
The gardens are the same size, and far more values of Garden B lie near 139, so the sunflower is more likely from Garden B.

Reteach plan — C-DP.1.2

MA.7.DP.1.2 · Compare two data sets

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.1.2-R

Benchmark: **MA.7.DP.1.2** · Compare two data sets Student code: _____ Calculator: yes

1. Team A: 48, 52, 55, 48, 47
Team B: 49, 43, 45, 45, 43

The lists show the points scored in five games by each of two basketball teams. Which conclusion do the means of the two data sets support?

A. Team A typically scored about 25 more points per game than Team B.

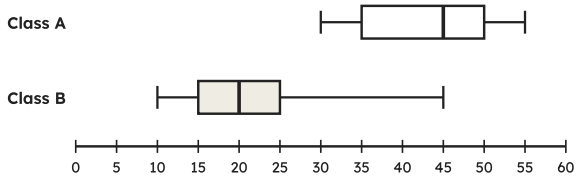
B. Team A typically scored about 6 more points per game than Team B.

C. Team A typically scored about 5 more points per game than Team B.

D. Team B typically scored about 5 more points per game than Team A.

2. The box plots show the numbers of books read over the summer by the students in two classes. The two classes have the same number of students.

A student from one of the two classes read 42 books over the summer. Which class is the student more likely in, and why?



A. Class A, because 42 is in the middle half for Class A but in the top fourth for Class B

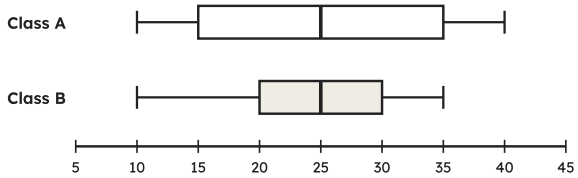
B. Class B, because 42 is between the least and the greatest values of the data for Class B

C. Class B, because the median of the data for Class B is closer to 42 than the median for Class A

D. Either one is equally likely, because 42 is within the range of the data for both classes

3. The box plots show the quiz scores of the students in two classes.

Which conclusion about the two classes is supported by the box plots?



A. Students in Class A typically scored higher, and the scores in Class A varied more.

B. Students in Class B typically scored higher, and the scores in Class A varied more.

C. Students in Class A typically scored higher, and the scores in Class B varied more.

D. Students in the two classes typically scored the same, and the scores in Class A varied more.

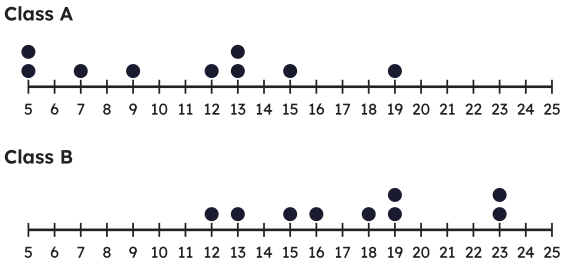
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.2-X

Benchmark: **MA.7.DP.1.2** · Compare two data sets Student code: _____ Calculator: yes

1. The line plots show the numbers of push-ups done in one minute by the 9 students in each of two classes.

Which conclusion do the medians of the two groups support?



A. Students in Class B typically did about 6 more push-ups than students in Class A.

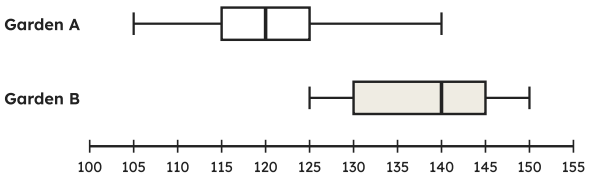
B. Students in Class A typically did about 6 more push-ups than students in Class B.

C. Students in Class B typically did about 7 more push-ups than students in Class A.

D. Students in Class B typically did about 4 more push-ups than students in Class A.

2. The box plots show the heights, in centimeters, of the sunflowers grown in two gardens. The two gardens have the same number of sunflowers.

A sunflower from one of the two gardens is 139 centimeters tall. Which garden did the sunflower more likely come from, and why?



A. Garden A, because 139 is between the least and the greatest values of the data for Garden A

B. Either one is equally likely, because 139 is within the range of the data for both gardens

C. Garden B, because 139 is in the middle half for Garden B but in the top fourth for Garden A

D. Garden A, because the median of the data for Garden A is closer to 139 than the median for Garden B

Score: ____ / 2 → copy to your progress strip

Benchmark: **MA.7.DP.1.5** · Choose an appropriate graph Student code: _____ Calculator: yes

Learning goal: I can choose and create an appropriate data display. (circle your level after scoring)

4	I can explain why one display is better than another for a given question.
3	I can choose the right display for numerical or categorical data and create it accurately.
2	I can make a bar graph or dot plot from data.
1	I need help to start this benchmark.

1. A student recorded the 280 kilograms of material collected at a recycling drive.

Paper	Plastic	Glass	Metal
126	70	56	28

The student wants a display that answers this question at a glance: is Paper more than half of the total? Which display is the most appropriate?

A. a bar chart
B. a histogram
C. a line plot
D. a circle graph

2. A class surveyed 140 students about the lunch they like best.
Pizza: 21
Tacos: 35
Salad: 63
Pasta: 21

The class wants a display that answers this question at a glance: did about one-fourth of the students choose Pizza? Which display is the most appropriate?

A. a histogram
B. a box plot
C. a circle graph
D. a bar chart

3. A gardener recorded the heights of 120 sunflowers, measured to the nearest centimeter. Almost every value is different. The table summarizes the data.

Height (cm)	Number
90–109	11
110–129	23
130–149	45
150–169	37
170–189	4

The gardener wants a display that shows the shape of the distribution. Which display is the most appropriate for this data set?

A. a circle graph
B. a histogram
C. a line plot
D. a stem-and-leaf plot

Turn over →

Benchmark: **MA.7.DP.1.5** · Choose an appropriate graph Student code: _____

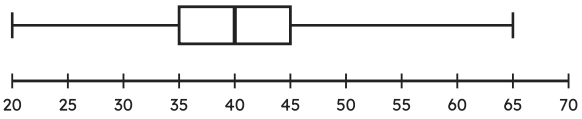
4. The data shows the number of pets for 16 students:
1, 0, 3, 1, 1, 2, 1, 0, 1, 2, 0, 1, 0, 0, 1, 2

Choose the display that shows every value and how often it occurs. Explain why it fits these data, and describe what it would look like.

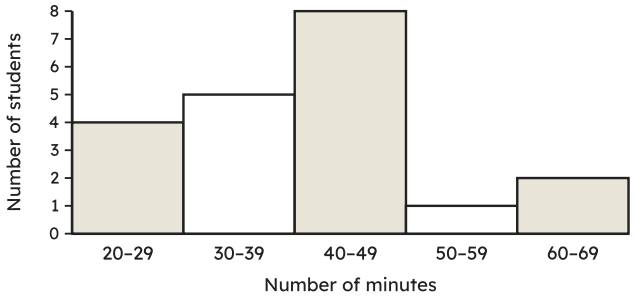
5. The two displays show the same data: the numbers of minutes 20 students spent reading over the weekend.

Which display can be used to find the interquartile range of the data? Explain why the other display cannot, and then use the display to find the interquartile range of the data.

Display A



Display B



Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.5 · MA.7.DP.1.5 · Choose an appropriate graph

Unit 13: Numerical and Categorical Data. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **D. a circle graph**

The question compares one category with the whole. In a circle graph each section is the same fraction of the circle as its category is of the total, so a section can be compared with half of the circle at a glance (Paper is 45%).

A bar chart shows the amounts but not how each compares with the total; a histogram and a line plot are for numerical data.

2. Regular **C. a circle graph**

The question compares one category with the whole class. In a circle graph each section is the same fraction of the circle as its category is of the class, so half, one-third or one-fourth of the circle can be seen at a glance.

A bar chart shows the counts, but not how a count compares with the total; a box plot and a histogram are for numerical data.

3. Regular **B. a histogram**

There are 120 values, almost all different, so a line plot or a stem-and-leaf plot would need a mark or a leaf for nearly every value, and the table gives only counts for intervals.

The data are numerical, so a circle graph (parts of a whole) does not fit. A histogram uses the equal intervals in the table as its bars and shows the shape of the distribution.

4. Regular **A line plot, because the data are numbers with only a few different values, so every value can be shown. It would have an X above each value for each time it appears: 0 → 5 Xs, 1 → 7 Xs, 2 → 3 Xs, 3 → 1 X. Score: 1 point — a line plot or dot plot (a bar graph of the counts also accepted) with a reason that mentions the few different values, plus the description; the name alone earns 0.**

The data are numbers with only a few different values $(0,\ 1,\ 2,\ 3)$, so a line plot keeps every value and shows the shape; a stem-and-leaf plot would put every value on one stem, and a histogram would hide the individual values.

Count how many times each value occurs, then stack that many Xs above the value on a number line.

0: 5; 1: 7; 2: 3; 3: 1.

5. Rigor **Display A, the box plot, because it marks the median and the quartiles; the histogram groups the values into intervals, so it shows only which interval holds the quartiles, not the exact value. From it, the interquartile range is $45 - 35 = 10$. Score: 1 point — the display named, why the other cannot be used, and the value; the value alone earns 0.**

A box plot marks the minimum, Q_1, the median, Q_3 and the maximum, so the interquartile range is $Q_3 - Q_1 = 45 - 35 = 10$.

A histogram groups the values into intervals, so it shows only which interval holds the quartiles, not the exact value.

Re-check strip (3 items, C-DP.1.5-R)

1. Regular **D. a bar chart**

The question compares two counts. A bar chart shows one bar for each category, with heights read on a scale, so the two counts and their difference can be read.

A circle graph shows shares of the whole, not the counts; a box plot and a histogram are for numerical data.

2. Regular **A line plot, because the data are numbers with only a few different values, so every value can be shown. It would have an X above each value for each time it appears: 0 → 4 Xs, 1 → 3 Xs, 2 → 3 Xs, 3 → 2 Xs, 4 → 1 X. Score: 1 point — a line plot or dot plot (a bar graph of the counts also accepted) with a reason that mentions the few different values, plus the description; the name alone earns 0.**

The data are numbers with only a few different values $(0,\ 1,\ 2,\ 3,\ 4)$, so a line plot keeps every value and shows the shape; a stem-and-leaf plot would put every value on one stem, and a histogram would hide the individual values.

Count how many times each value occurs, then stack that many Xs above the value on a number line.

0: 4; 1: 3; 2: 3; 3: 2; 4: 1.

3. Regular **A. a stem-and-leaf plot**

The scores are numerical. A stem-and-leaf plot keeps every value (each leaf is one score), and the lengths of its leaf rows show the shape.

A box plot shows the shape and spread but not every value.

Comparison strip (2 items, C-DP.1.5-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B. a bar chart**

The question compares two counts. A bar chart shows one bar for each category, with heights read on a scale, so the two counts and their difference can be read.

A circle graph shows shares of the whole, not the counts; a box plot and a histogram are for numerical data.

2. Regular **A. a circle graph**

The question compares one category with the whole. In a circle graph each section is the same fraction of the circle as its category is of the total, so a section can be compared with half of the circle at a glance (Rent is 45%).

A bar chart shows the amounts but not how each compares with the total; a histogram and a line plot are for numerical data.

Reteach plan — C-DP.1.5

MA.7.DP.1.5 · Choose an appropriate graph

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.1.5-R

Benchmark: **MA.7.DP.1.5** · Choose an appropriate graph Student code: _____ Calculator: yes

1. A class surveyed 260 students about the way they get to school.
Bus: 52
Car: 39
Walk: 104
Bike: 26
Train: 39

The class wants a display that answers this question at a glance: how many more students chose Bus than chose Train? Which display is the most appropriate?

A. a histogram
B. a circle graph
C. a box plot
D. a bar chart

2. The data shows the number of pets for 13 students:
0, 4, 0, 2, 2, 1, 1, 3, 1, 0, 3, 2, 0

Choose the display that shows every value and how often it occurs. Explain why it fits these data, and describe what it would look like.

3. A teacher recorded the test scores of 15 students:
97, 97, 79, 95, 97, 90, 79, 87, 89, 72, 72, 72, 91, 85, 98

The teacher wants a display that keeps every value and shows the shape of the data. Which display is the most appropriate for this data set?

A. a stem-and-leaf plot
B. a box plot
C. a bar chart of categories
D. a circle graph

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.5-X

Benchmark: **MA.7.DP.1.5** · Choose an appropriate graph Student code: _____ Calculator: yes

1. A class surveyed 200 students about the way they get to school.
Bus: 60
Car: 20
Walk: 40
Bike: 50
Train: 30

The class wants a display that answers this question at a glance: how many more students chose Bus than chose Car? Which display is the most appropriate?

A. a box plot
B. a bar chart
C. a circle graph
D. a histogram

2. A student recorded how a family divides its monthly income of $4,800.

Rent	Food	Transportation	Savings
$2,160	$1,200	$960	$480

The student wants a display that answers this question at a glance: is Rent more than half of the total? Which display is the most appropriate?

A. a circle graph
B. a line plot
C. a histogram
D. a bar chart

Score: ____ / 2 → copy to your progress strip

Mixed benchmarks from earlier units Student code: _____ Calculator: yes

1. A probability is given for each event.

Describe each event as likely, unlikely or neither.

Check one box in each row.

	Likely	Unlikely	Neither likely nor unlikely
The probability that a tossed coin lands on heads is 50%.	☐	☐	☐
The probability that it snows in Orlando, Florida, this winter is 0.01.	☐	☐	☐
The probability that a thunderstorm is within 5 miles of your house tomorrow is $\frac{7}{10}$.	☐	☐	☐

2. Evaluate. Write your answer in simplest form.

$$\frac{1}{4} - \frac{3}{4}$$

3. A group of 4 friends owes \$148.80 for a group gift. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Maya pays $\frac{5}{6}$ of her share.

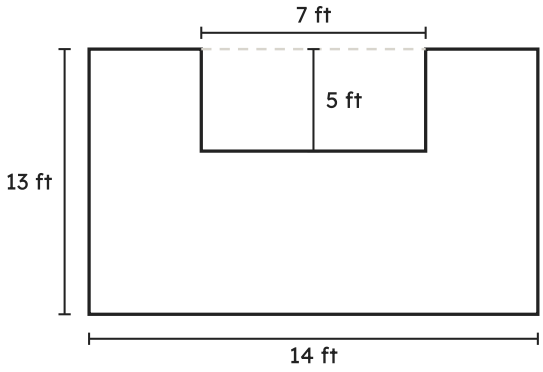
What is Maya's balance now?

Turn over →

Mixed benchmarks from earlier units Student code: _____

4. A patio is shaped like the figure shown: a 14 ft by 13 ft rectangle with a 7 ft by 5 ft rectangular notch removed from the middle of one edge. The dashed lines show the part that was removed.

What is the area of the patio, in square feet?



Not drawn to scale

5. The circumference of a round pool is 125.66 feet.

What is the diameter of the round pool? Round your answer to the nearest hundredth. Use the π key on your calculator.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — BACKSTOP-Q2

Given when no new check is ready by Thursday. Logged as a real check (Backstop = y in CHECKS).

1. MA.7.DP.2.2 Regular **The probability that a tossed coin lands on heads is 50%. → Neither likely nor unlikely; The probability that it snows in Orlando, Florida, this winter is 0.01. → Unlikely; The probability that a thunderstorm is within 5 miles of your house tomorrow is $\frac{7}{10}$. → Likely**

Write each probability as a decimal and compare it with 0.5: an event with a probability greater than 0.5 is likely, less than 0.5 is unlikely, and exactly 0.5 is neither.

$50\% = 0.5$: exactly 0.5, so neither.

0.01: less than 0.5, so unlikely.

$\frac{7}{10} = 0.7$: greater than 0.5, so likely.

Scoring: 1 point only if every match is correct; anything else earns 0.

---

2. MA.7.NSO.2.2 Regular $-\frac{1}{2}$

Both fractions already have the denominator 4.

$\frac{1}{4} - \frac{3}{4} = -\frac{2}{4}$.

Simplify: $-\frac{1}{2}$.

---

3. MA.7.NSO.2.3 Regular **−$6.20**

Each share: $-\$148.80 \div 4 = -\$37.20$.

Paying $\frac{5}{6}$ of the share leaves $1 - \frac{5}{6} = \frac{1}{6}$ of it: $\frac{1}{6} \times (-\$37.20) = -\$6.20$.

Maya's balance is −$6.20.

---

4. MA.7.GR.1.2 Regular $147\,\text{ft}^2$

Whole rectangle: $14 \times 13 = 182\,\text{ft}^2$.

Removed notch: $7 \times 5 = 35\,\text{ft}^2$.

Area: $182 - 35 = 147\,\text{ft}^2$.

---

5. MA.7.GR.1.3 Regular **40 feet (40.00 also accepted)**

$C = \pi d$, so $d = C \div \pi$.

With the π key: $125.66 \div \pi \approx 40.00$ feet.

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Mixed benchmarks from earlier units Student code: _____ Calculator: yes

1. What is the value of the expression?

$$-1\frac{4}{5} \div \left(-\frac{4}{5}\right)$$

A. $\frac{4}{9}$

B. $2\frac{1}{4}$

C. $\frac{9}{20}$

D. $1\frac{11}{25}$

2. A photo is reduced by a scale factor of $\frac{1}{5}$. A side that measures 40 inches in the original is scaled down.

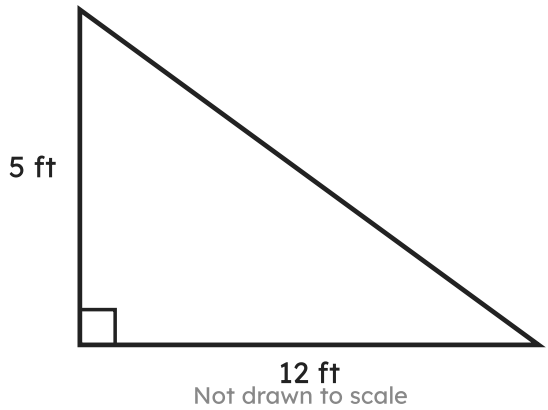
What is the length of the corresponding side in the reduction, in inches?

Turn over →

Mixed benchmarks from earlier units Student code: _____

3. A 5-foot person casts a 12-foot shadow. At the same time a tree casts a 39-foot shadow.

How tall is the tree, in feet?



Not drawn to scale

A. 7.80 ft

B. 16.25 ft

C. 93.60 ft

D. 32.00 ft

4. A student estimated the number of seats in an auditorium to be 774. The actual number was 750 seats.

What is the percent error of the estimate?

5. Write the expression using the fewest possible terms.

$$\left(\left(\frac{8}{3}\right)x - 4\right) - \left(\left(\frac{7}{3}\right)x - 5\right)$$

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — BACKSTOP-Q3

Given when no new check is ready by Thursday. Logged as a real check (Backstop = y in CHECKS).

1. MA.7.NSO.2.2 Regular **B.** $2\frac{1}{4} = \frac{9}{4}$

Dividing by a fraction means multiplying by its reciprocal.

Write the mixed number as a fraction: $-1\frac{4}{5} = -\frac{9}{5}$.

$-\frac{9}{5} \times \left(-\frac{5}{4}\right) = \frac{9}{4} = 2\frac{1}{4}$.

---

2. MA.7.GR.1.5 Regular **8 inches**

Multiply by the scale factor: $40 \times \frac{1}{5} = 8$ inches.

---

3. MA.7.AR.3.2 Regular **B. 16.25 ft**

The tree's shadow is $39 \div 12 = 3.25$ times as long as the person's.

The tree is 3.25 times as tall: $5 \times 3.25 = 16.25\,\text{ft}$.

---

4. MA.7.AR.3.1 Regular **3.2%**

Percent error compares the size of the error with the ACTUAL amount.

Error: $|774 - 750| = 24$.

$24 \div 750 = 0.032$.

$0.032 \times 100 = 3.2\%$.

---

5. MA.7.AR.1.1 Regular $\left(\frac{1}{3}\right)x + 1$

Subtracting the whole expression changes the sign of each of its terms: $\left(\frac{8}{3}\right)x - 4 - \left(\frac{7}{3}\right)x + 5$.

x-terms: $\left(\frac{8}{3}\right)x - \left(\frac{7}{3}\right)x = \left(\frac{1}{3}\right)x$. Constants: $-4 - (-5) = 1$.

$= \left(\frac{1}{3}\right)x + 1$

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Mixed benchmarks from earlier units Student code: _____ Calculator: yes

1. The price of one share of a company's stock changed by these amounts over 4 days.

−$0.60, +$1.55, −$1.95, −$0.40

What was the mean (average) daily change in the price?

2. A pair of shoes costs $27.11 and is on sale for 37.5% off.

What is the sale price, to the nearest cent?

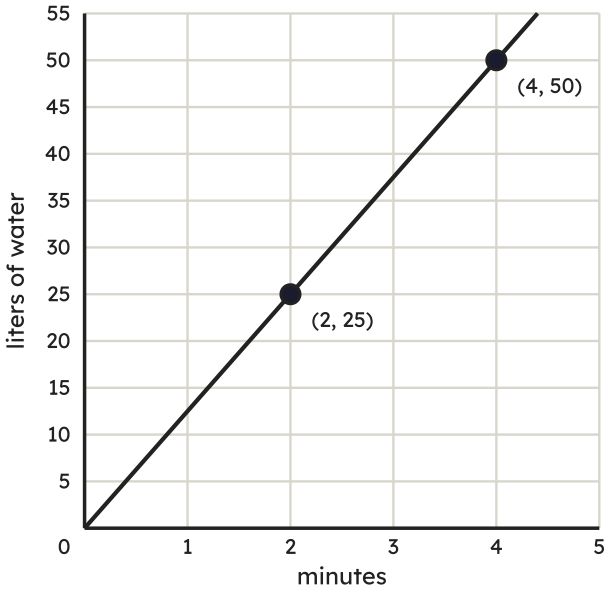
3. What is the exact value of m in the equation shown?

$$-3\frac{19}{25} = \left(\frac{3}{5}\right)m - 4$$

Turn over →

4. The graph shows the proportional relationship between minutes and liters of water.

What is the constant of proportionality, in liters per minute?



5. A right circular cylinder has a radius of 4 units and a height of 6 units.

What is the volume of the cylinder, in cubic units?

A. 48π

B. 24π

C. 192π

D. 96π

Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — BACKSTOP-Q4

Given when no new check is ready by Thursday. Logged as a real check (Backstop = y in CHECKS).

1. MA.7.NSO.2.3 Regular **−$0.35**

Add the 4 changes: the total change is −$1.40.

Divide by the number of days: $-\$1.40 \div 4 = -\$0.35$.

---

2. MA.7.AR.3.1 Regular **$16.94**

Discount: $\$27.11 \times 0.375 = \$10.16625 \approx \$10.17$.

Sale price: $\$27.11 - \$10.17 = \$16.94$.

37.5% must be written as 0.375 before multiplying.

---

3. MA.7.AR.2.2 Regular $m = \dfrac{2}{5}$

Add 4 to both sides: $-3\dfrac{19}{25} + 4 = \dfrac{6}{25}$, so $\dfrac{6}{25} = \left(\dfrac{3}{5}\right) m$.

Divide both sides by $\dfrac{3}{5}$, which means multiplying by its reciprocal $\dfrac{5}{3}$: $m = \dfrac{6}{25} \times \dfrac{5}{3} = \dfrac{2}{5}$.

---

4. MA.7.AR.4.2 Regular **12.5 liters per minute**

Use a labeled point: $(2,\ 25)$.

The constant of proportionality is $y \div x = 25 \div 2 = 12.5$ liters per minute.

It is not the y-value of a labeled point, because no labeled point has $x = 1$.

---

5. MA.7.GR.2.3 Regular **D.** 96π

$V = \pi r^2 h = \pi(4)^2(6) = 96\pi$.

The value 48π would be the curved surface area instead.

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